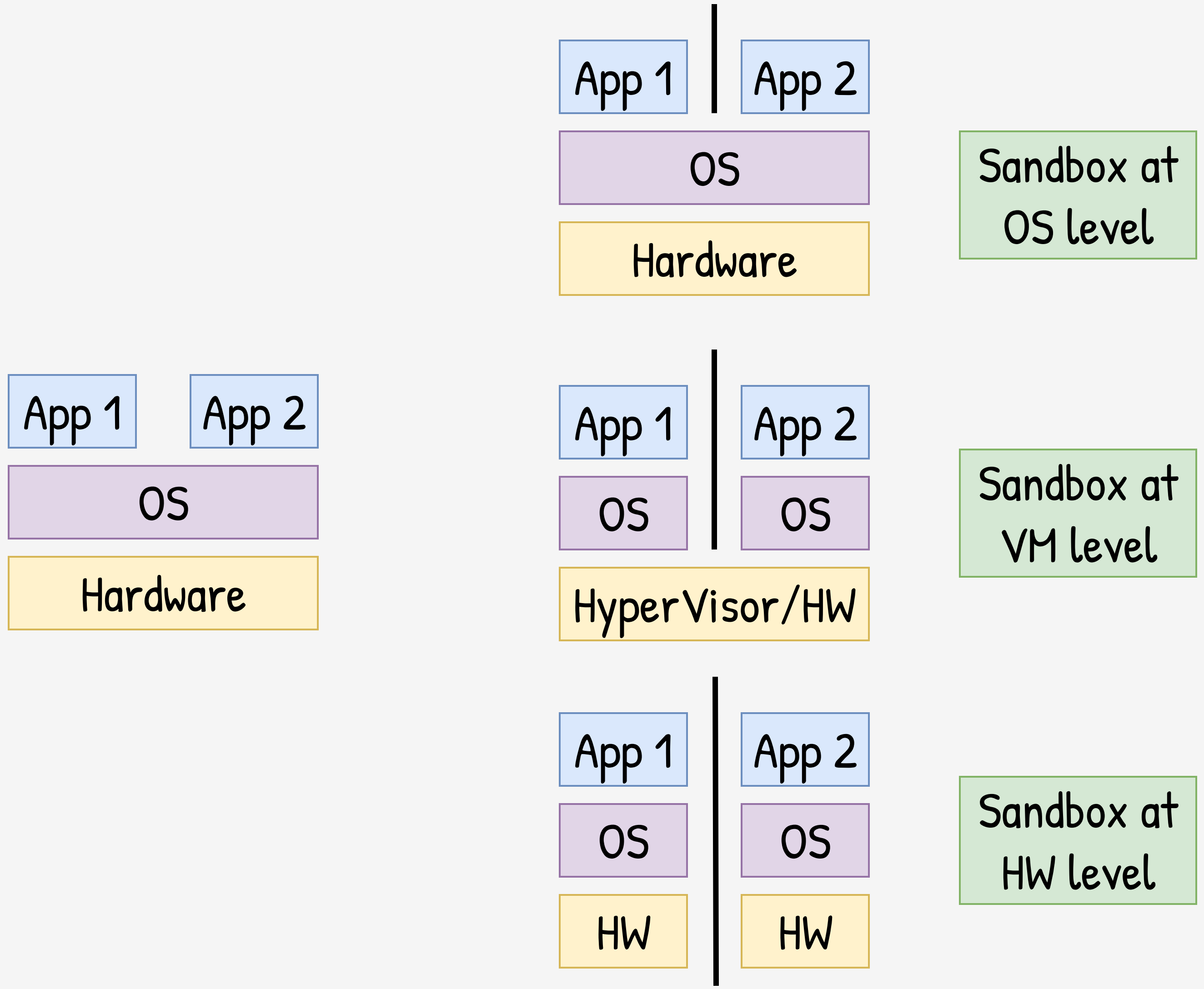
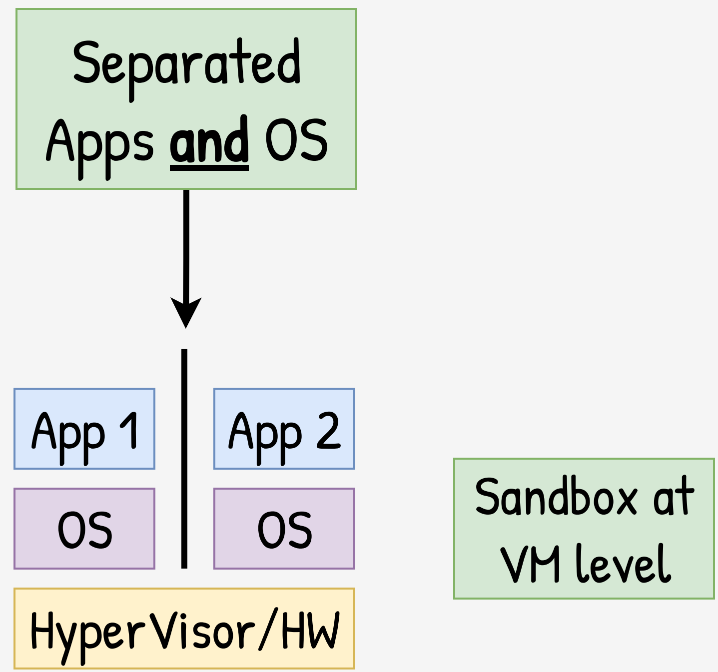
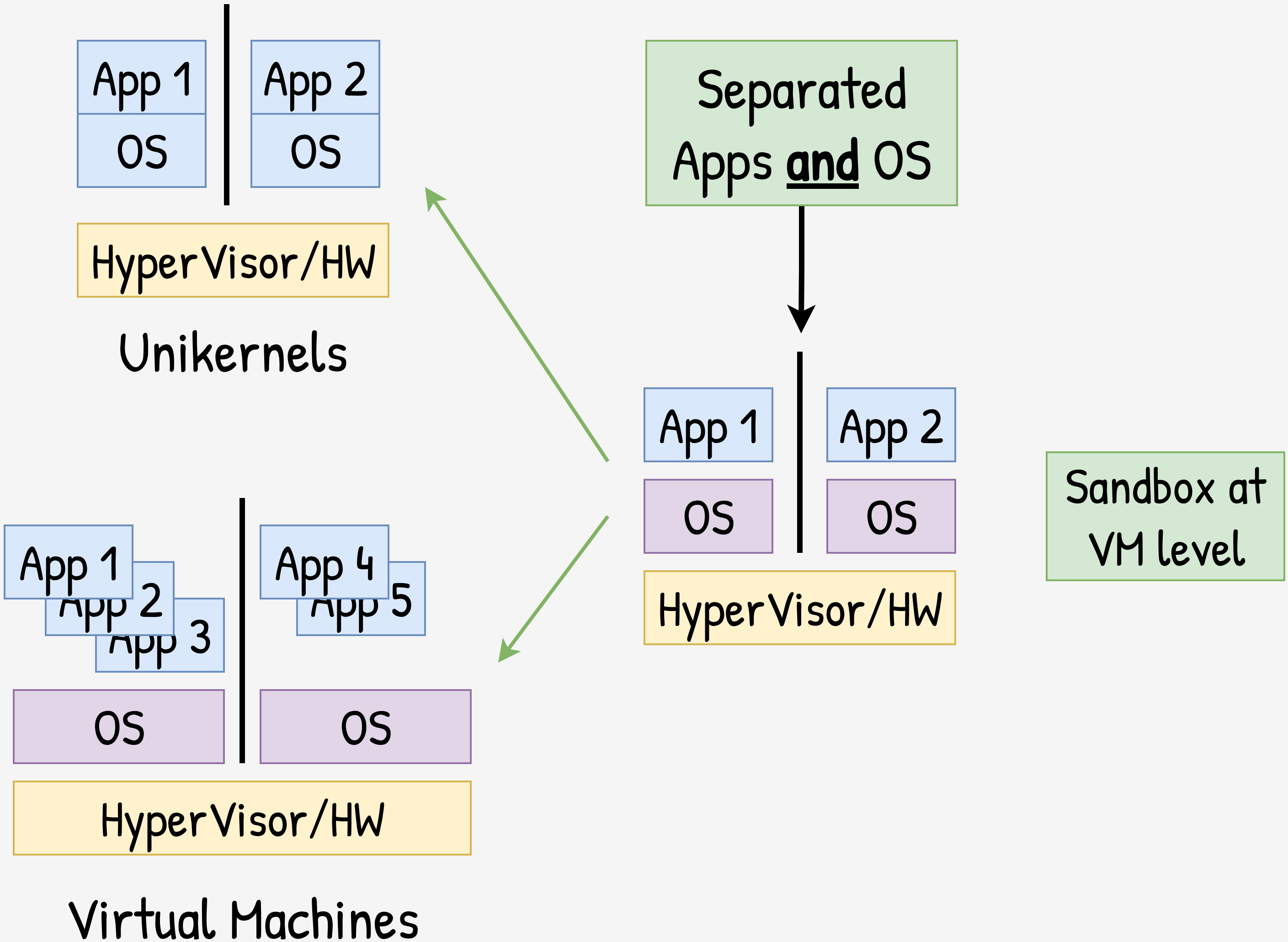
Sandbox at VM level
Virtual Machine, Virtualization, Emulation, Hypervisor, Row Hammering

Taller de Seguridad

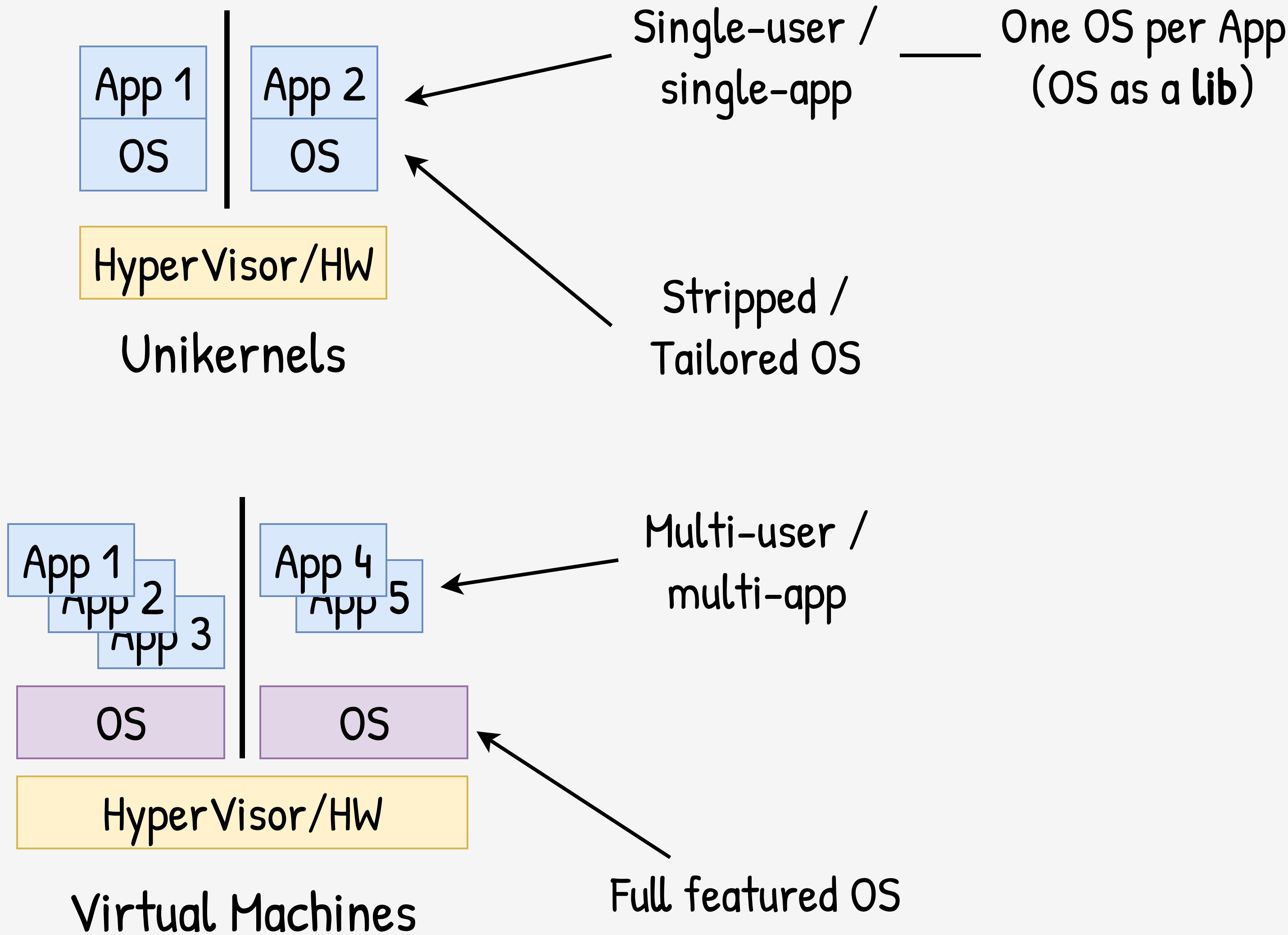
Ing. Martin Di Paola - Fiuba

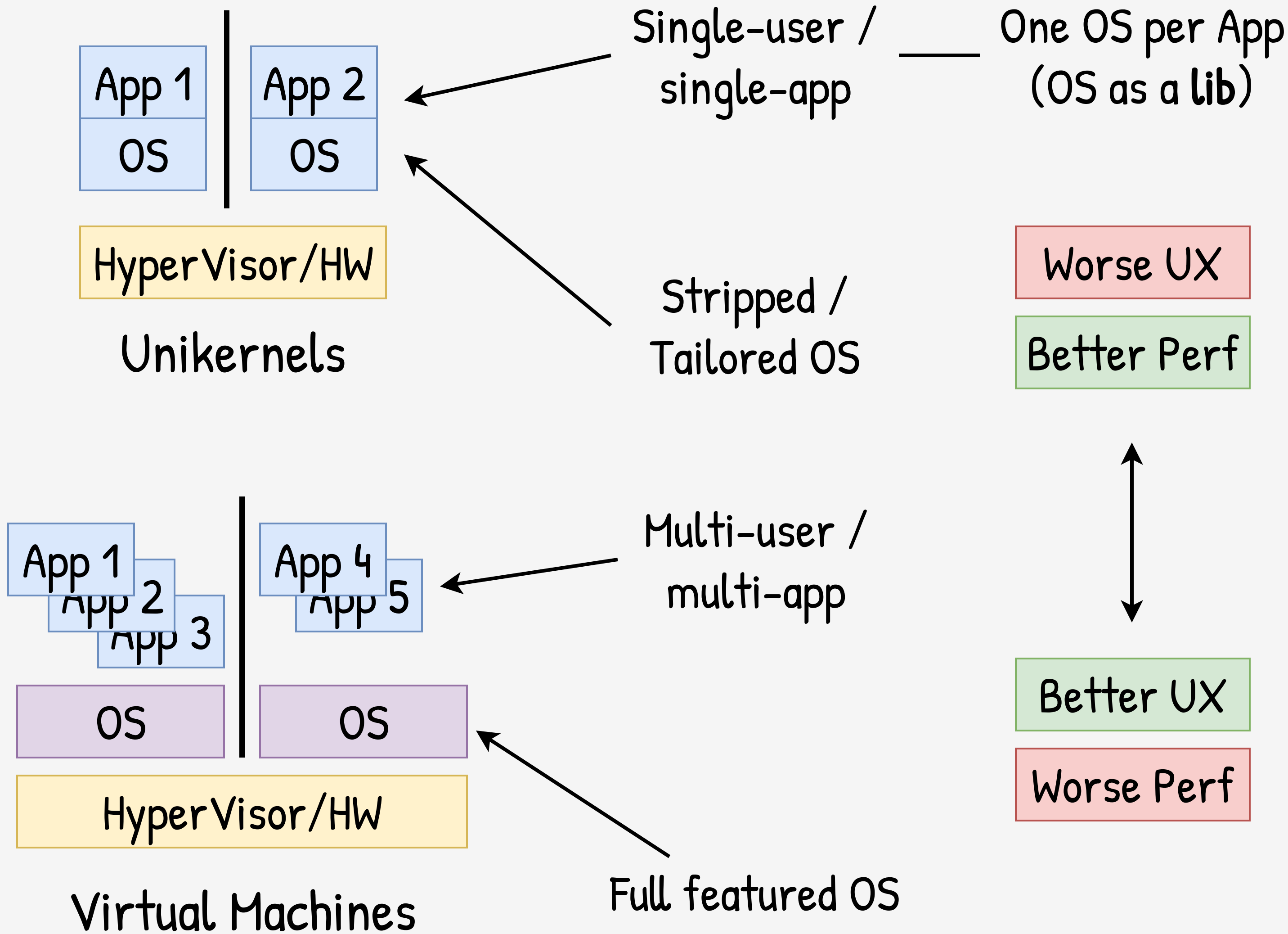

App 1
App 2
OS
Hardware
App 1
App 2
OS
Hardware
Sandbox at OS level
App 1
App 2
OS
OS
HyperVisor/HW
Sandbox at VM level
App 1
App 2
OS
OS
HW
HW
Sandbox at HW level


Separated
Apps **and** OS
App 1
App 2
OS
OS
HyperVisor/HW
Sandbox at
VM level

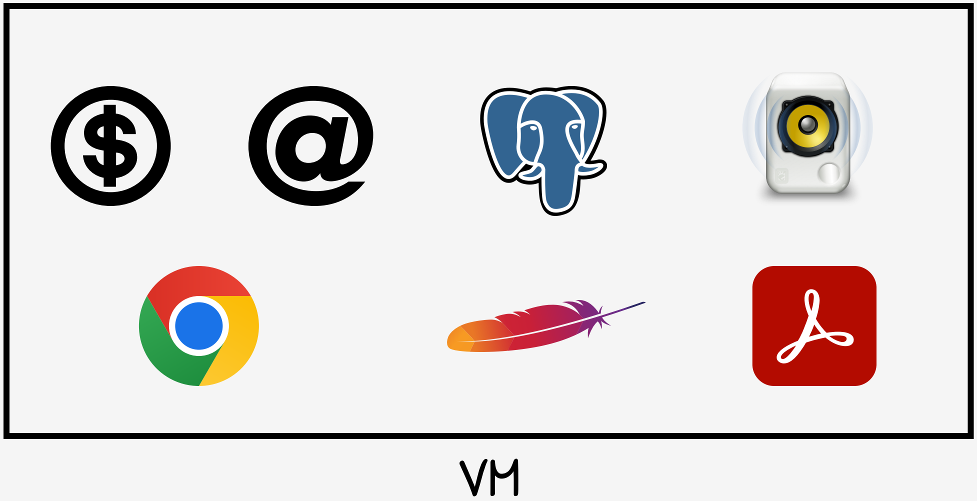


App 1
App 2
OS
OS
HyperVisor/HW
Unikernels
Separated
Apps and OS
App 1
App 2
OS
OS
HyperVisor/HW
Sandbox at
VM level
App 1
App 2
App 3
App 4
App 5
OS
OS
HyperVisor/HW
Virtual Machines

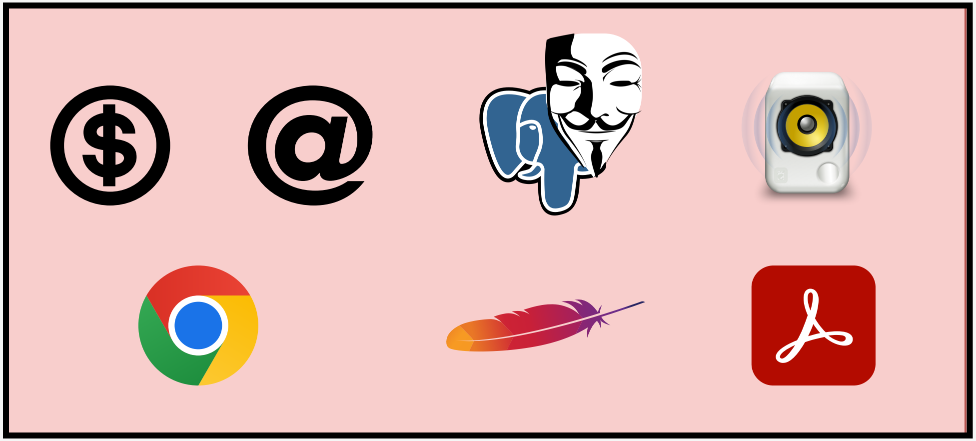

App 1
OS
App 2
OS
HyperVisor/HW
Unikernels
Single-user / single-app
One OS per App (OS as a **lib**)
Stripped / Tailored OS
App 1
App 2
App 3
App 4
App 5
OS
OS
HyperVisor/HW
Virtual Machines
Multi-user / multi-app
Full featured OS



App 1
OS
App 2
OS
HyperVisor/HW
Unikernels
Single-user /
single-app
One OS per App
(OS as a **lib**)
Stripped /
Tailored OS
Worse UX
Better Perf
App 1
App 2
App 3
App 4
App 5
OS
OS
HyperVisor/HW
Virtual Machines
Multi-user /
multi-app
Full featured OS
Better UX
Worse Perf



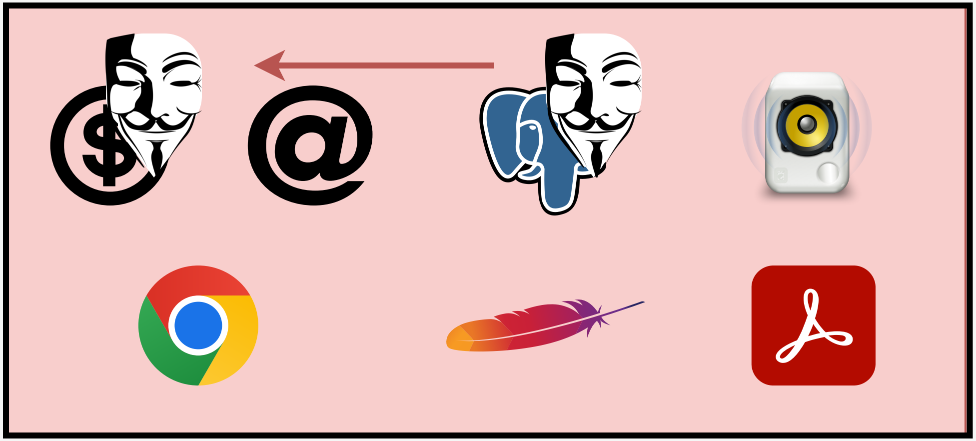
Mixed Concerns



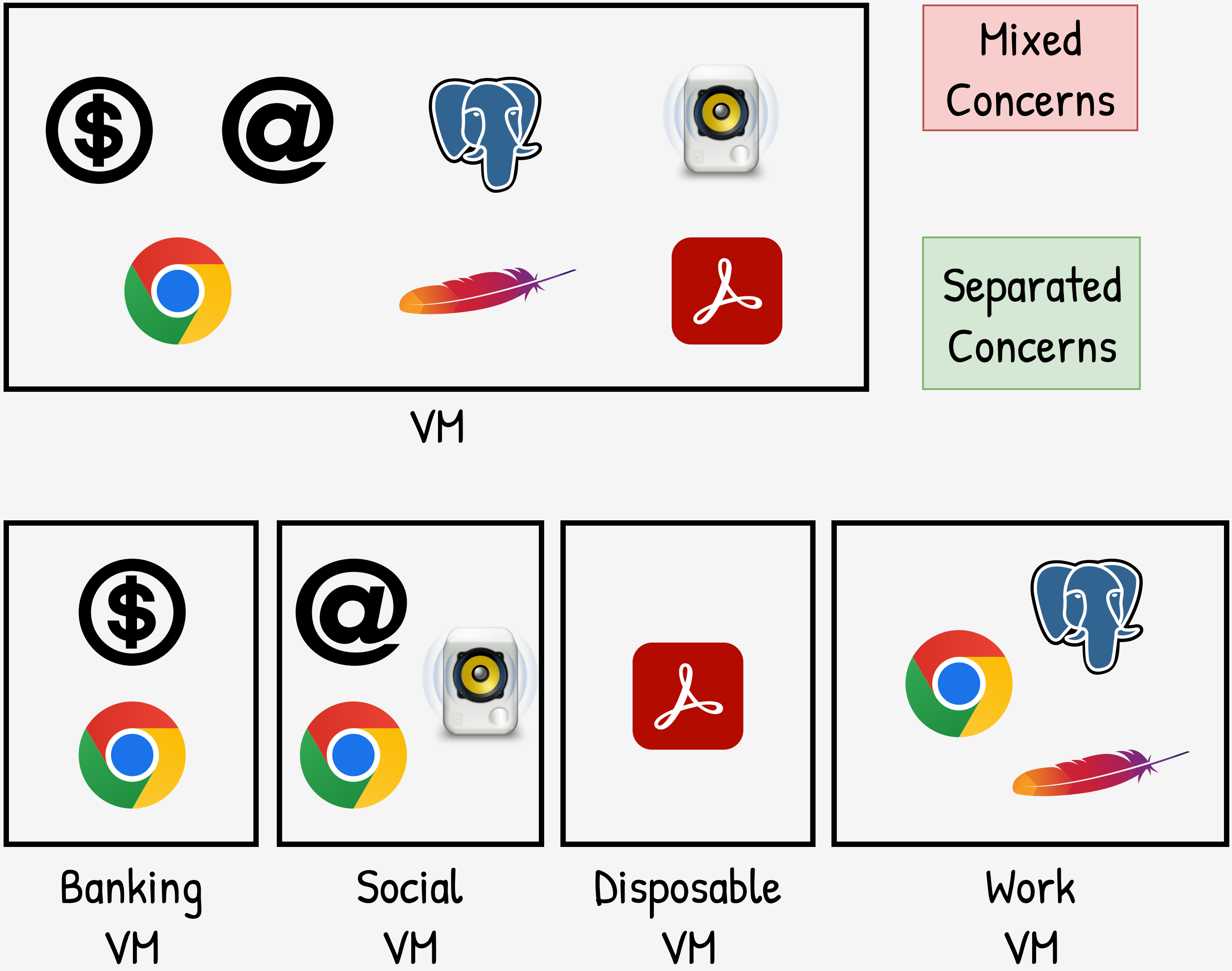
VM

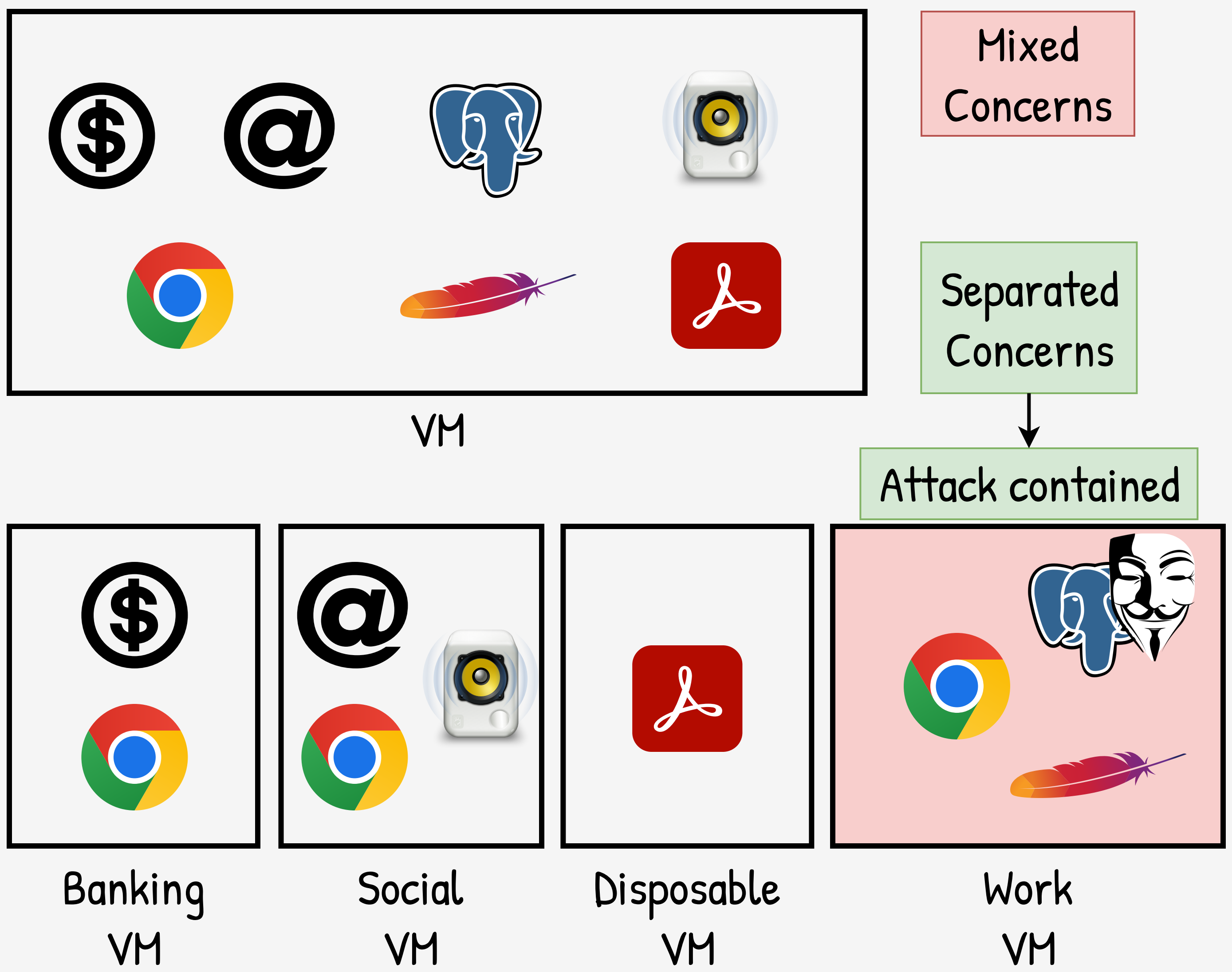
Mixed Concerns

Mixed Concerns

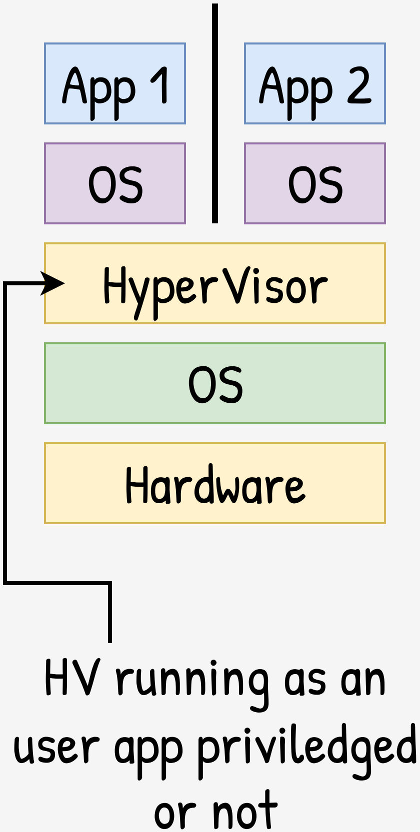


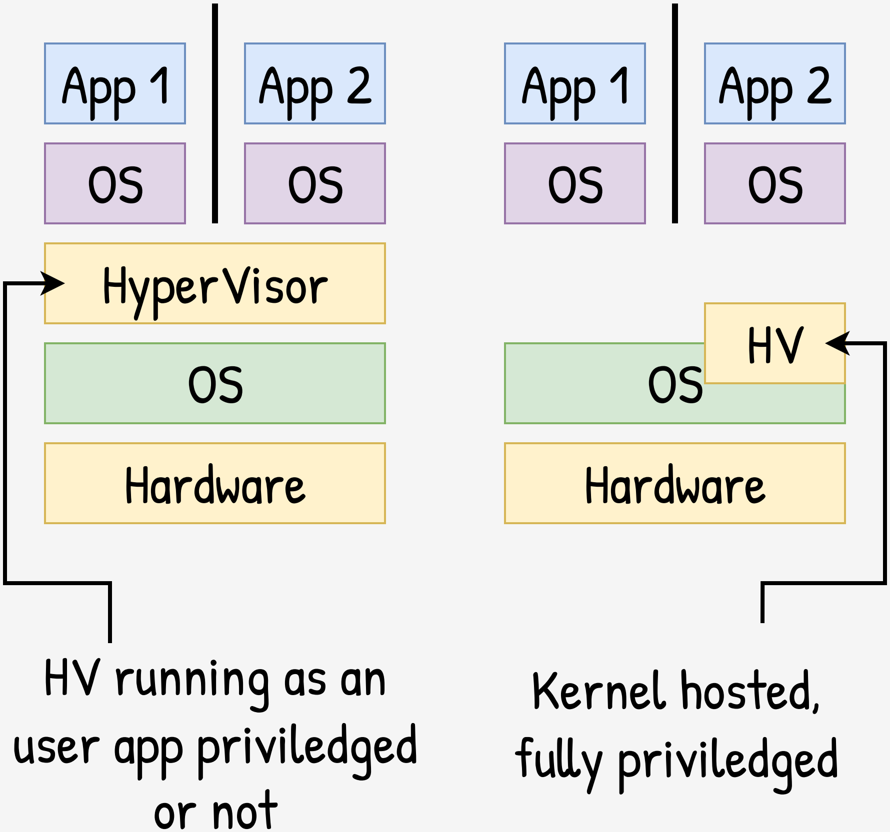
VM

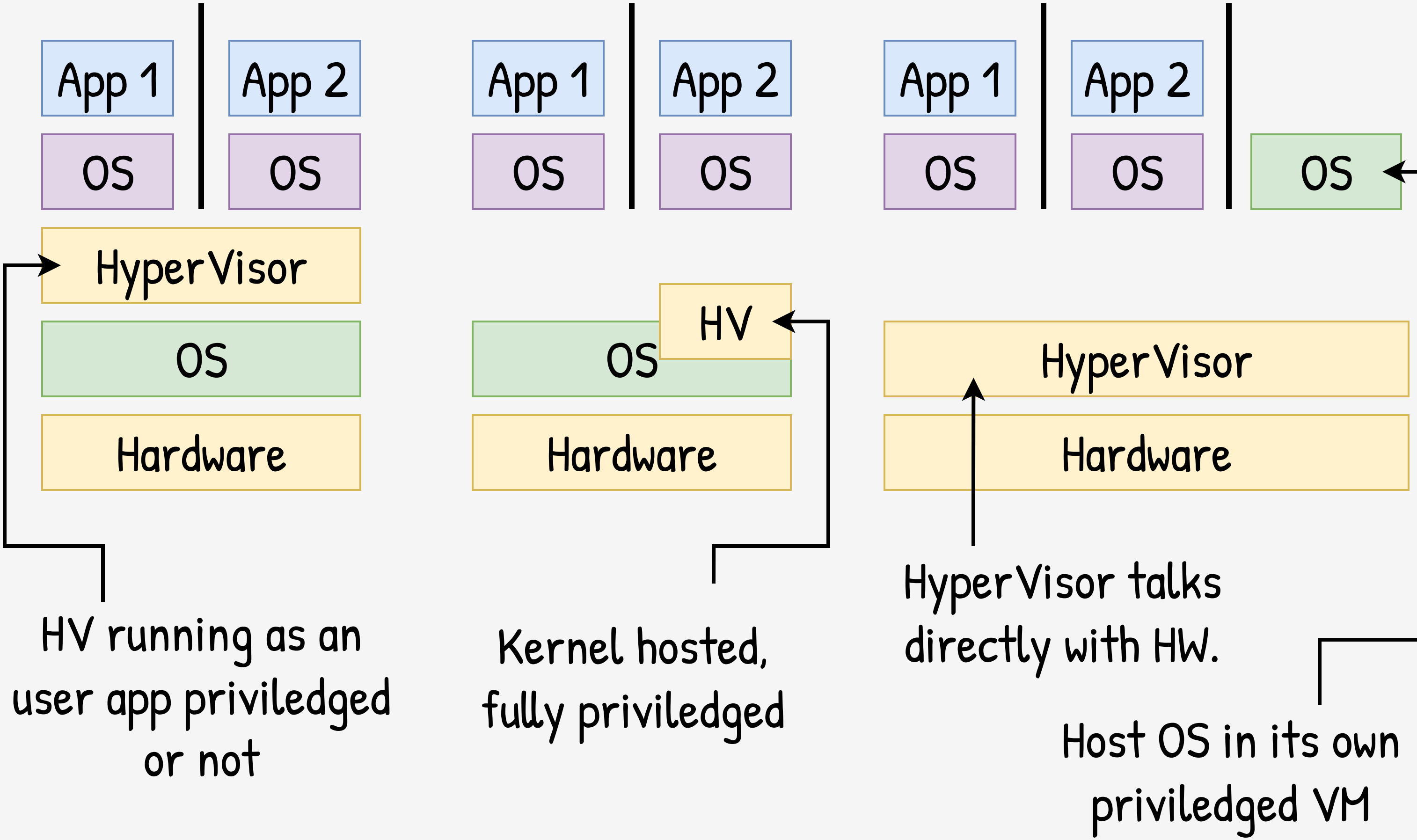

Mixed Concerns
Separated Concerns
VM
Banking VM
Social VM
Disposable VM
Work VM

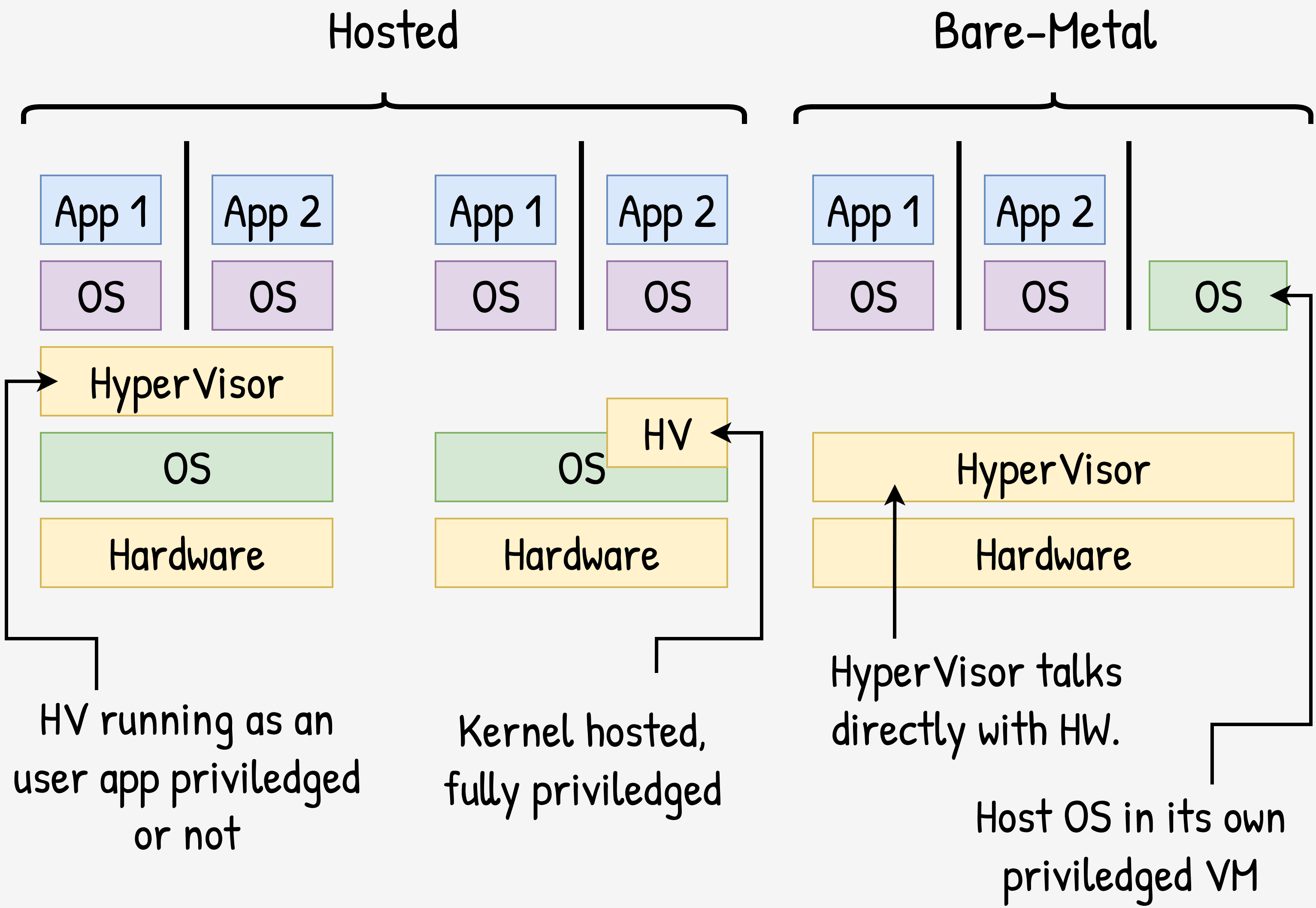


Mixed Concerns
VM
Separated Concerns
Attack contained
Banking VM
Social VM
Disposable VM
Work VM

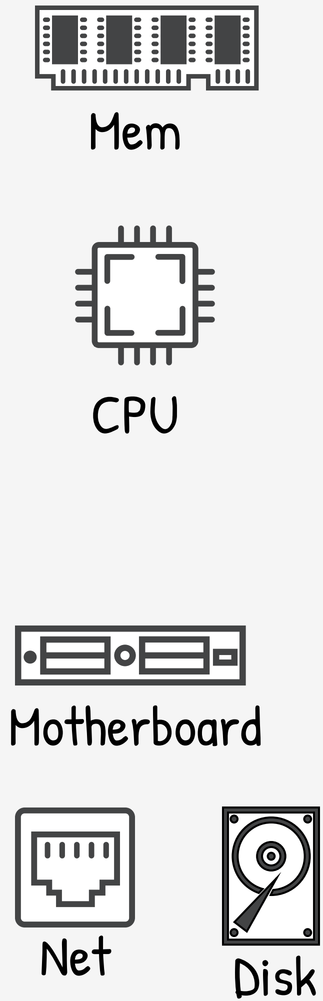

App 1
App 2
OS
OS
HyperVisor
OS
Hardware
HV running as an
user app priviledged
or not

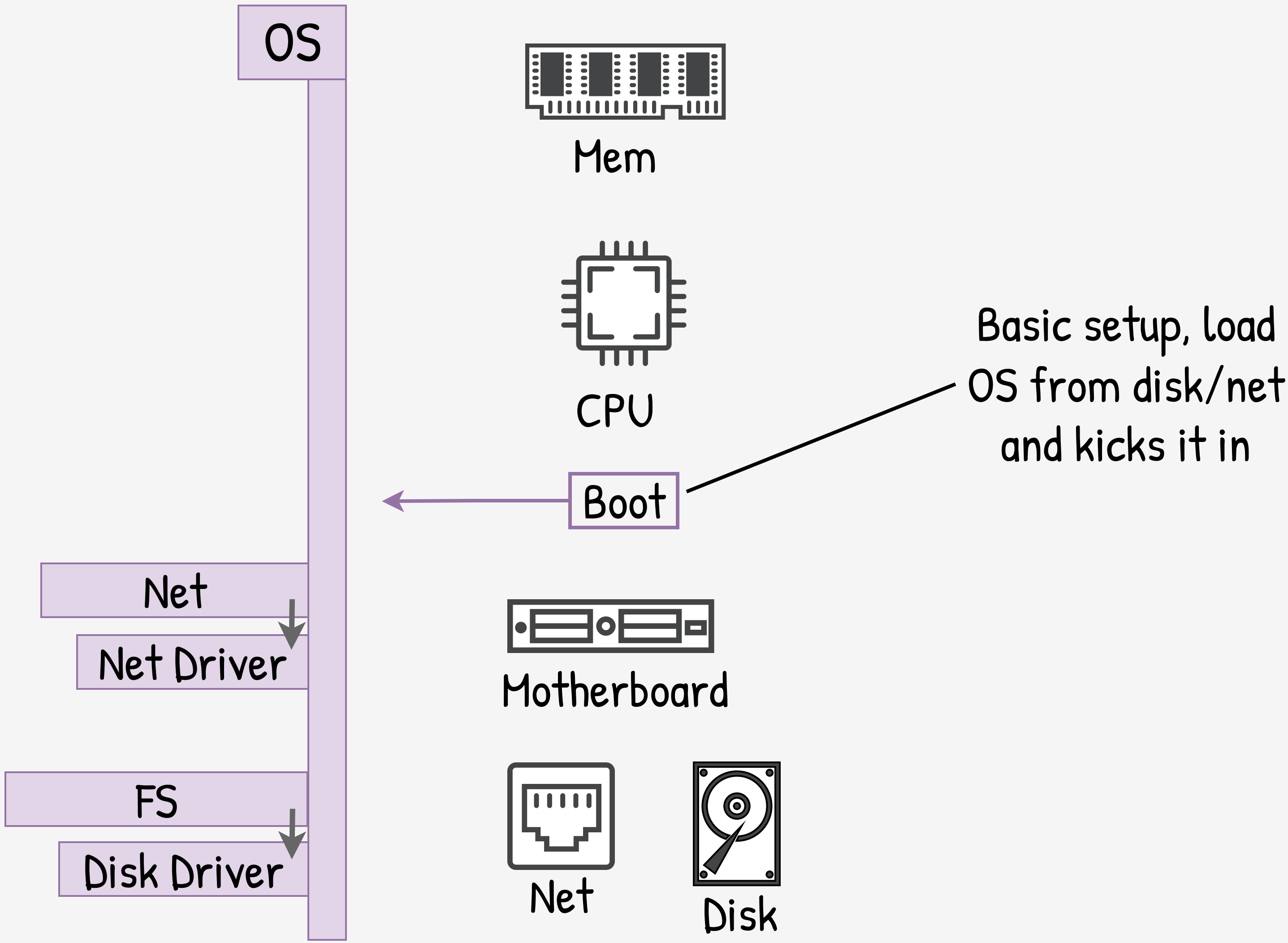

App 1
App 2
OS
OS
HyperVisor
OS
Hardware
App 1
App 2
OS
OS
HV
OS
Hardware
HV running as an user app priviledged or not
Kernel hosted, fully priviledged

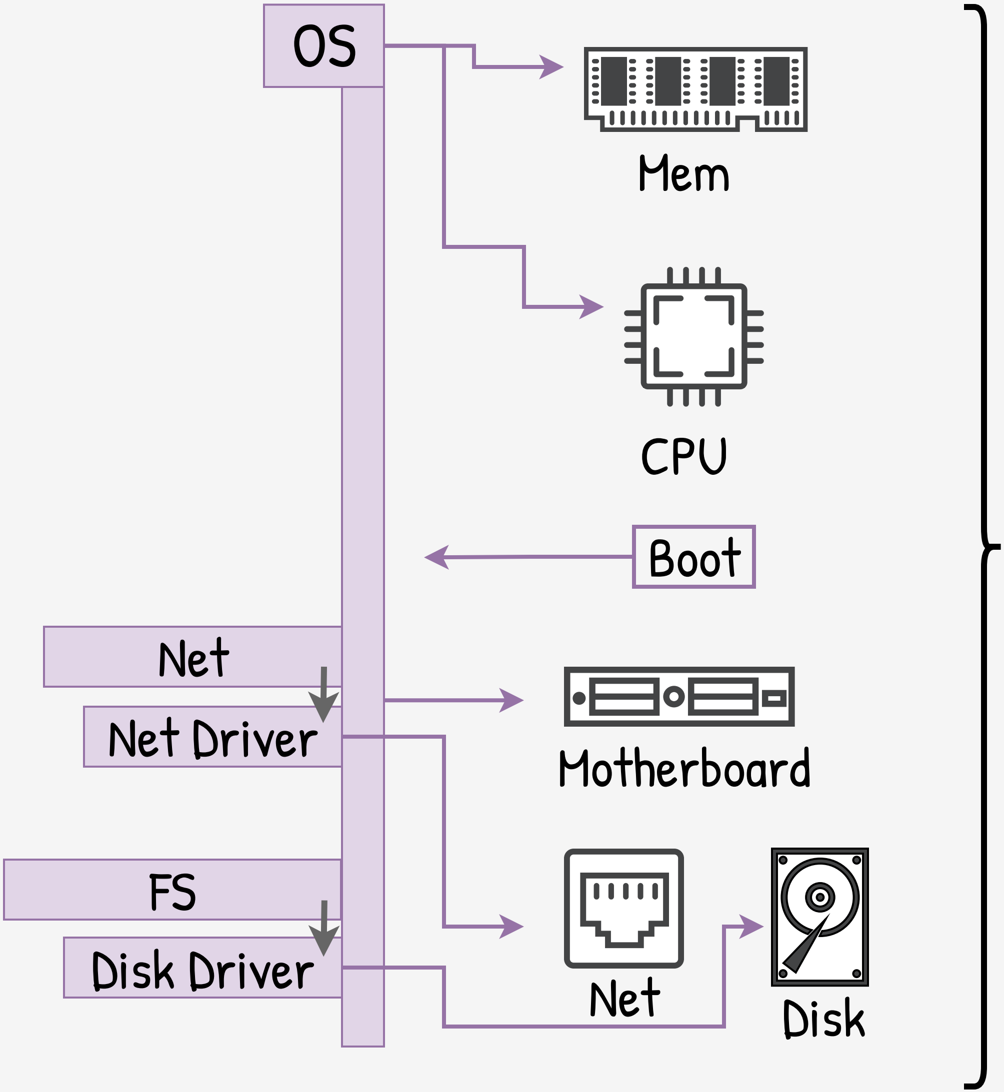

App 1
App 2
OS
OS
HyperVisor
OS
Hardware
App 1
App 2
OS
OS
HV
OS
Hardware
App 1
App 2
OS
OS
OS
HyperVisor
Hardware
HV running as an user app priviledged or not
Kernel hosted, fully priviledged
HyperVisor talks directly with HW.
Host OS in its own priviledged VM



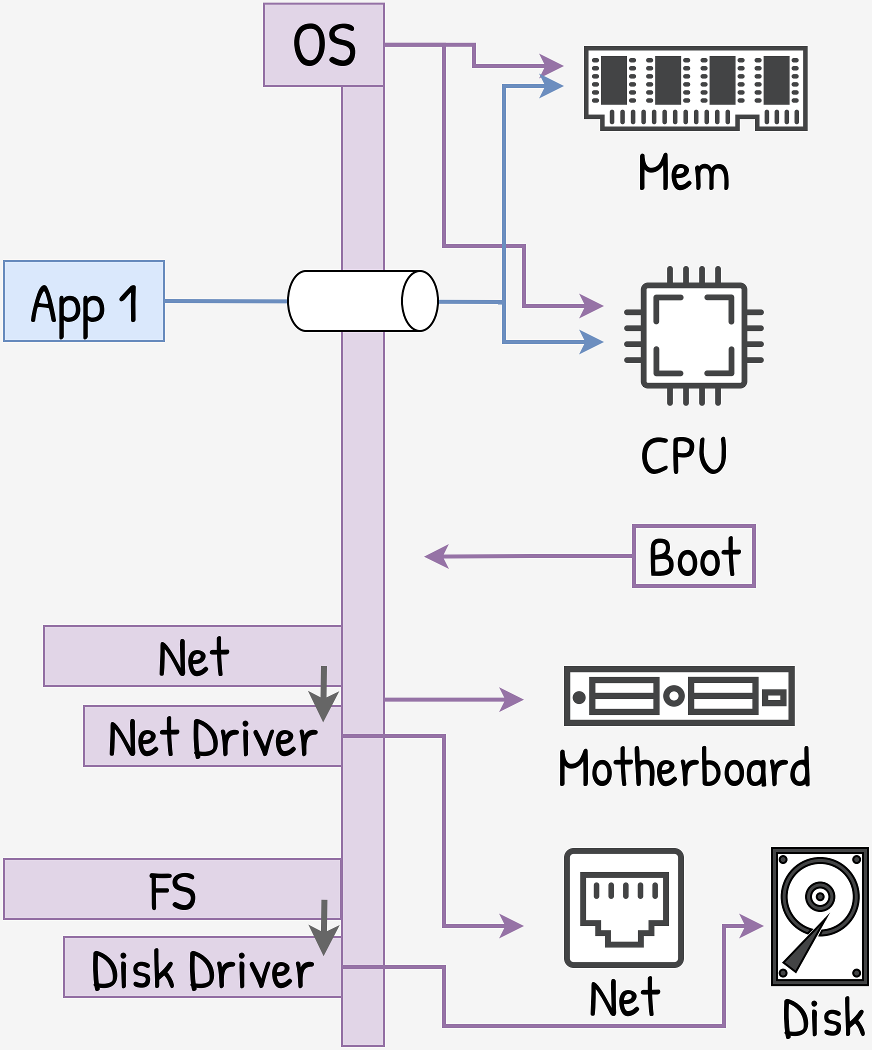
Hosted
Bare-Metal
App 1
App 2
OS
OS
HyperVisor
OS
Hardware
App 1
App 2
OS
OS
HV
OS
Hardware
App 1
App 2
OS
OS
OS
HyperVisor
Hardware
HV running as an user app priviledged or not
Kernel hosted, fully priviledged
HyperVisor talks directly with HW.
Host OS in its own priviledged VM


Mem
CPU
Motherboard
Net
Disk

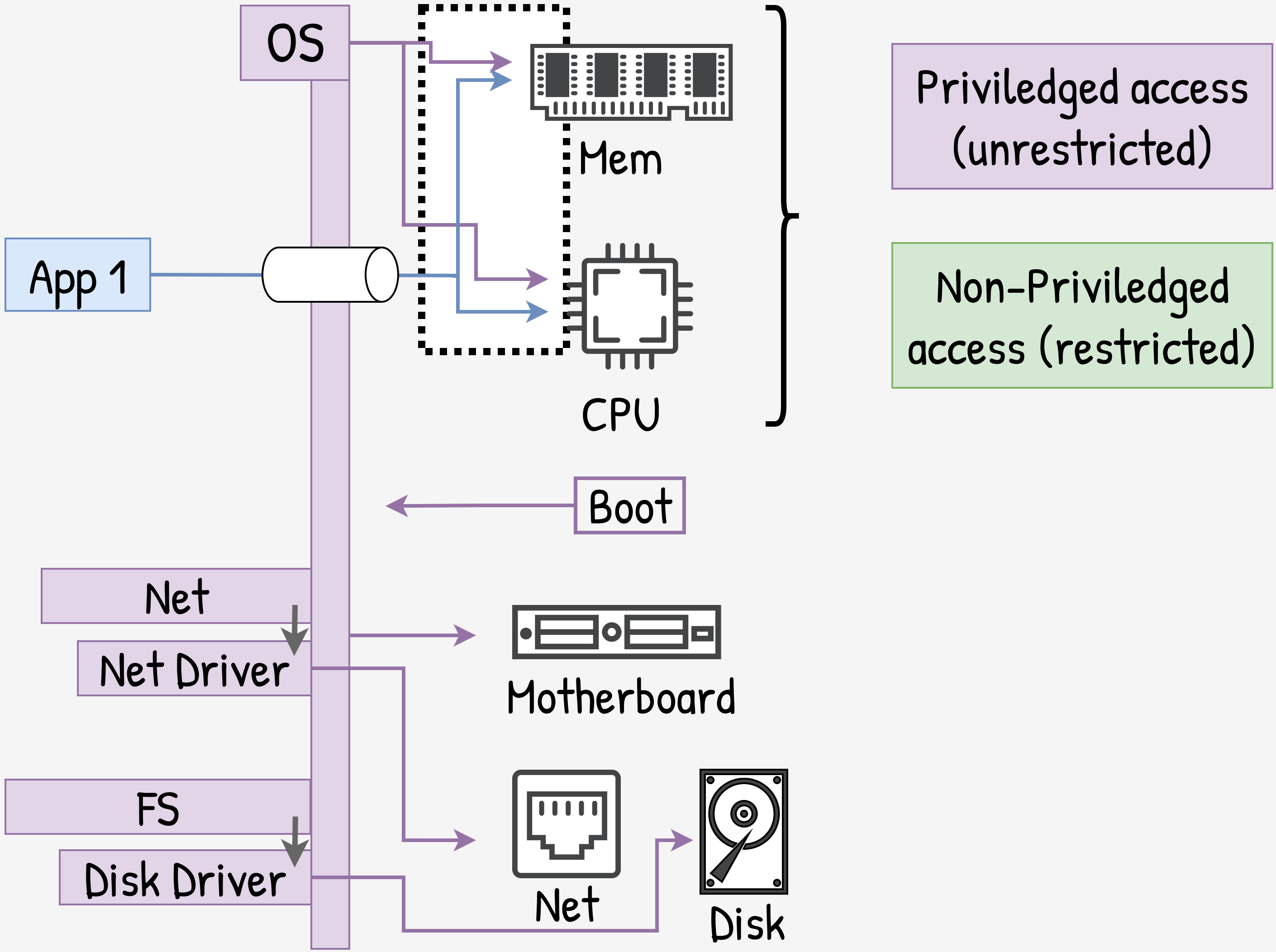

OS
Mem
CPU
Basic setup, load OS from disk/net and kicks it in
Boot
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk



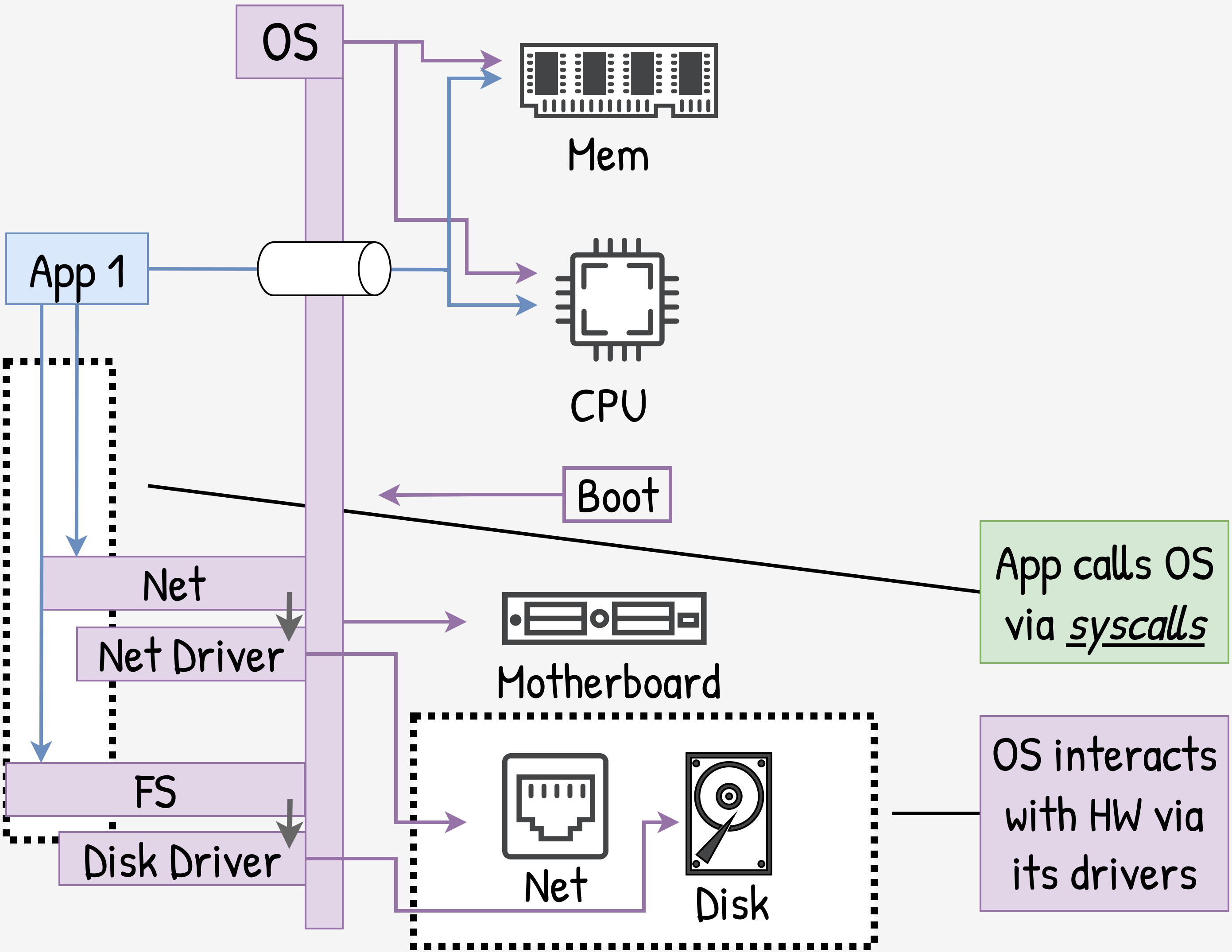
Then, OS takes control of the hardware

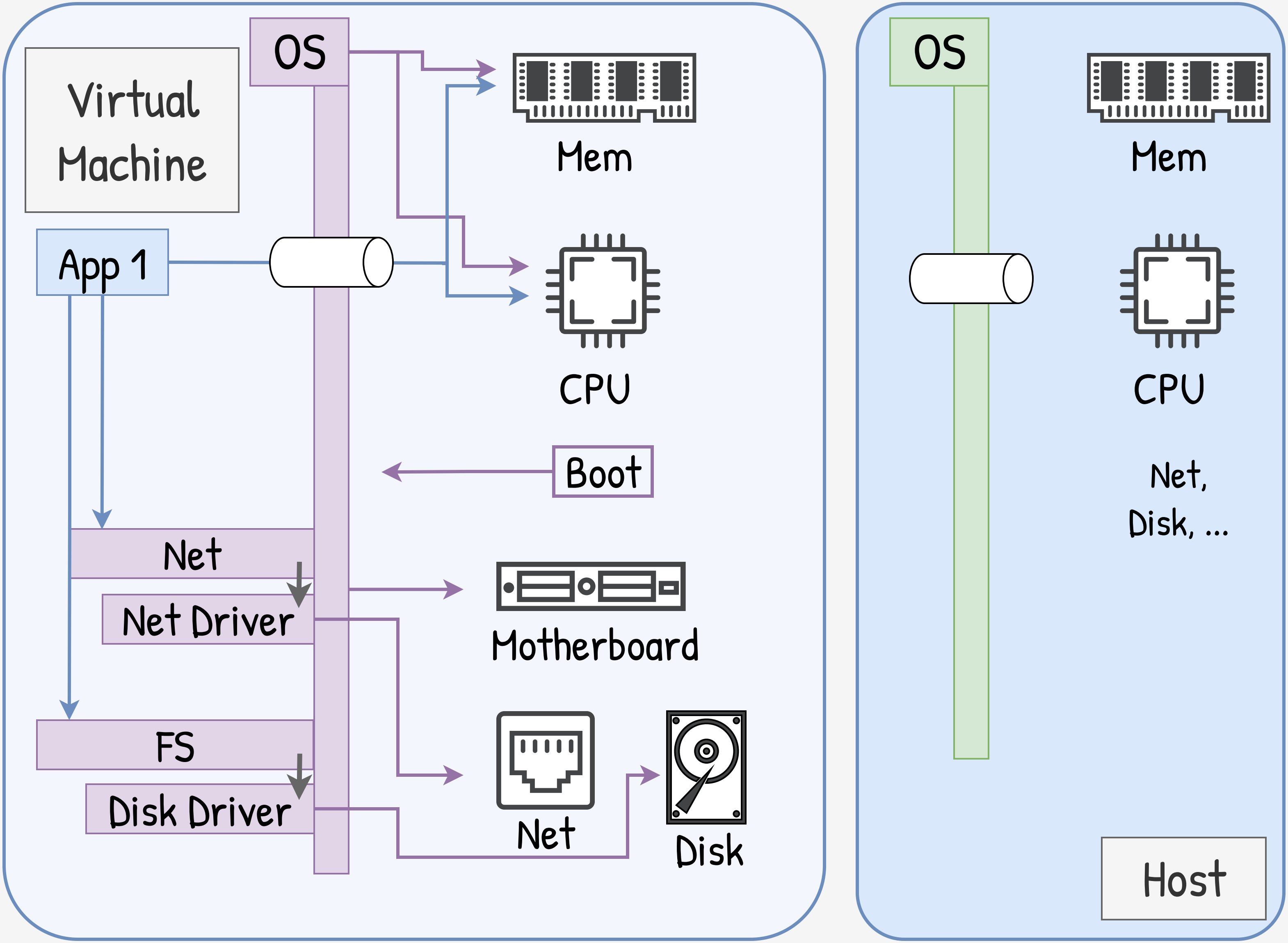


OS
Mem
App 1
CPU
Boot
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk

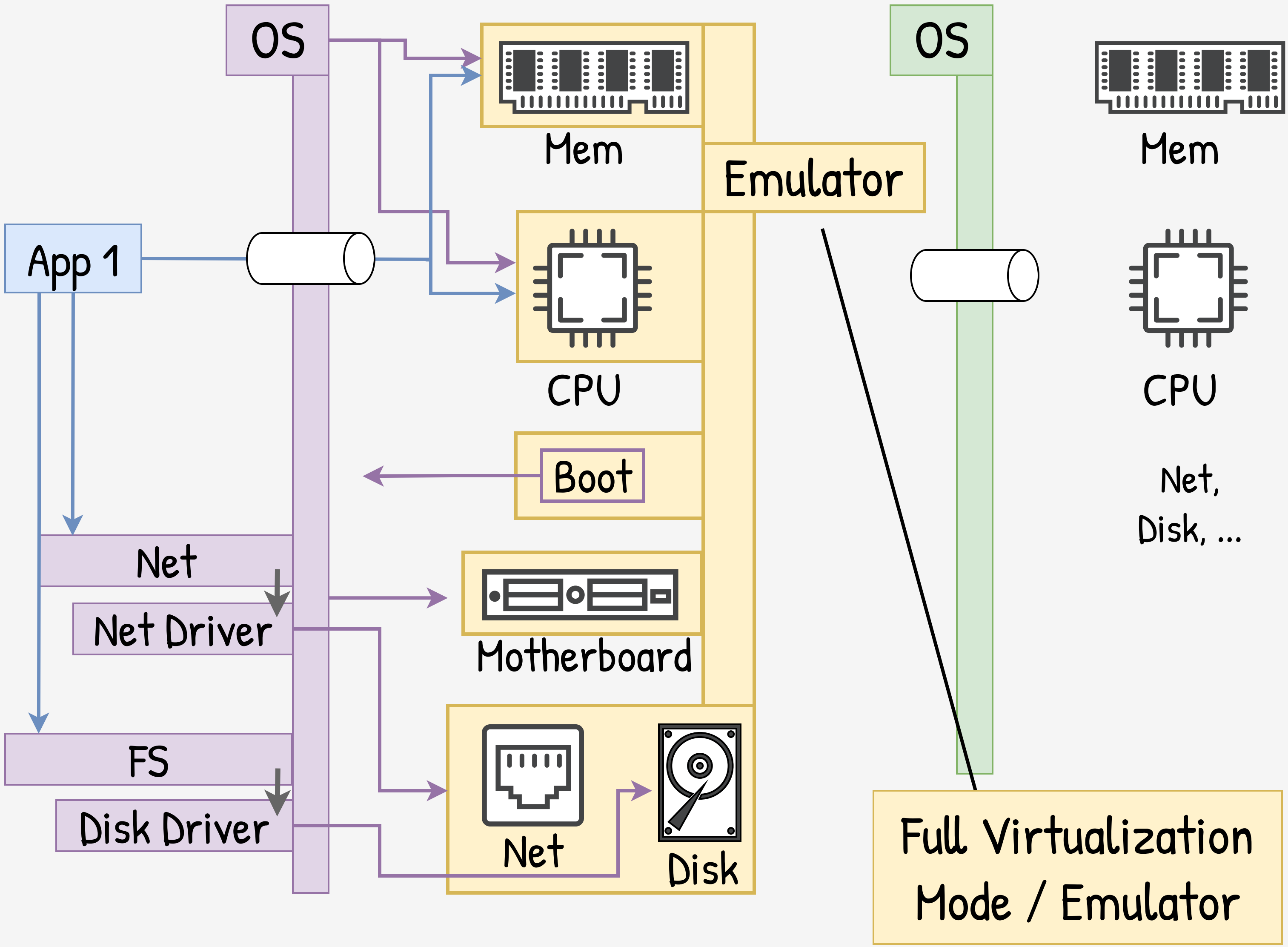


OS
Mem
App 1
CPU
Boot
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
Priviledged access (unrestricted)
Non-Priviledged access (restricted)

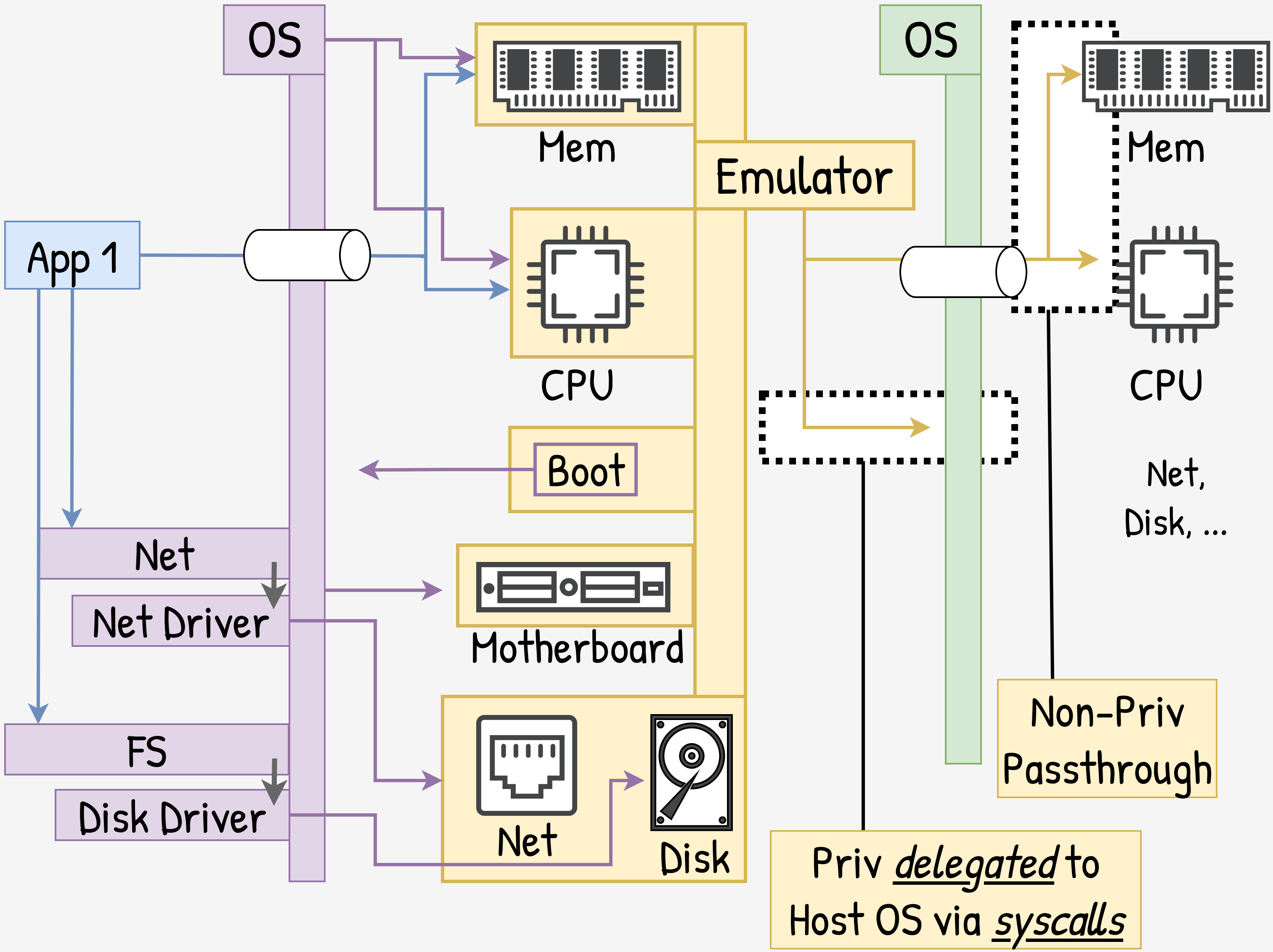

OS
Mem
App 1
CPU
Boot
App calls OS via syscalls
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
OS interacts with HW via its drivers



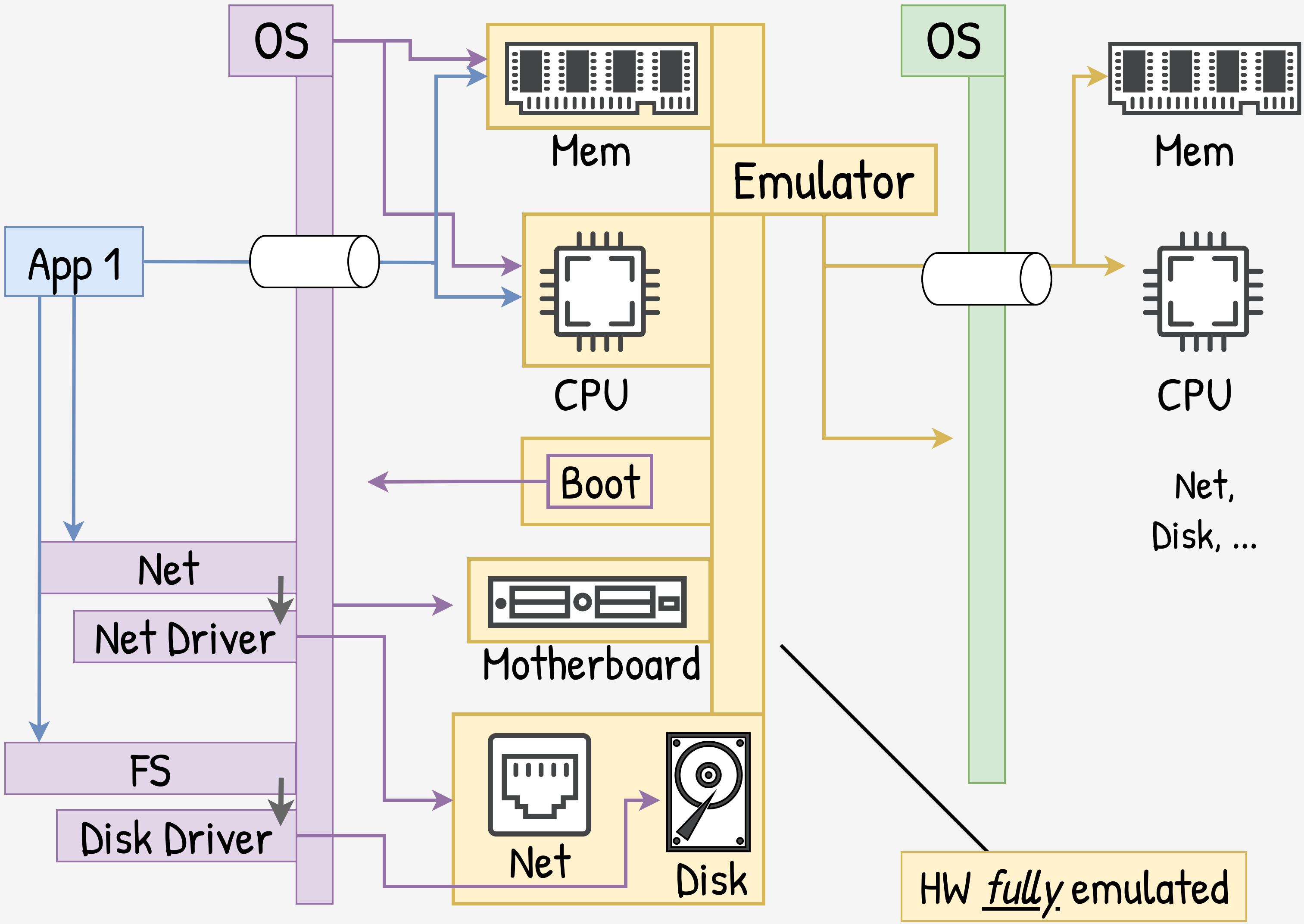
Virtual Machine
OS
Mem
App 1
CPU
Boot
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
OS
Mem
CPU
Net, Disk, ...
Host



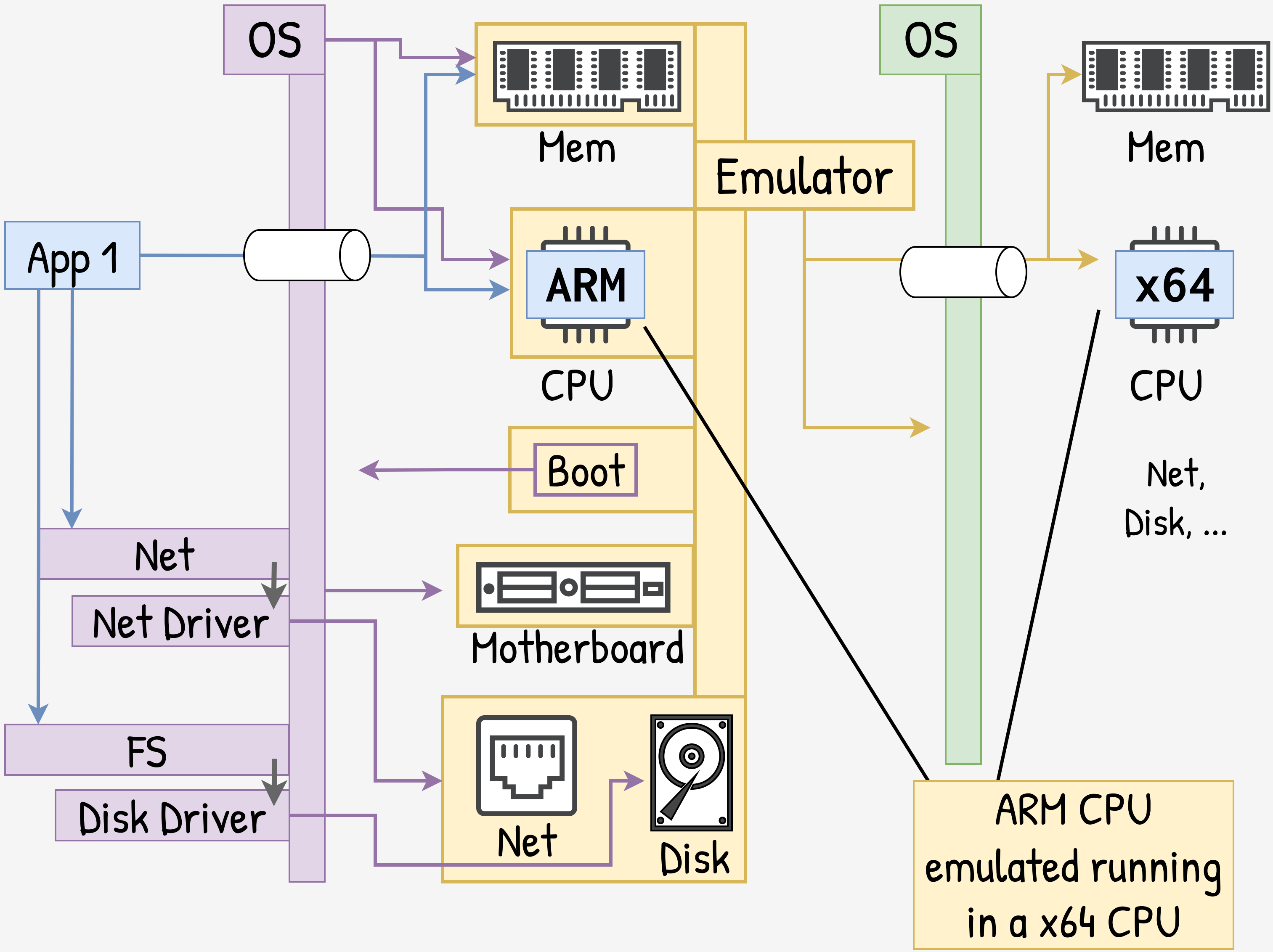
OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
Full Virtualization
Mode / Emulator



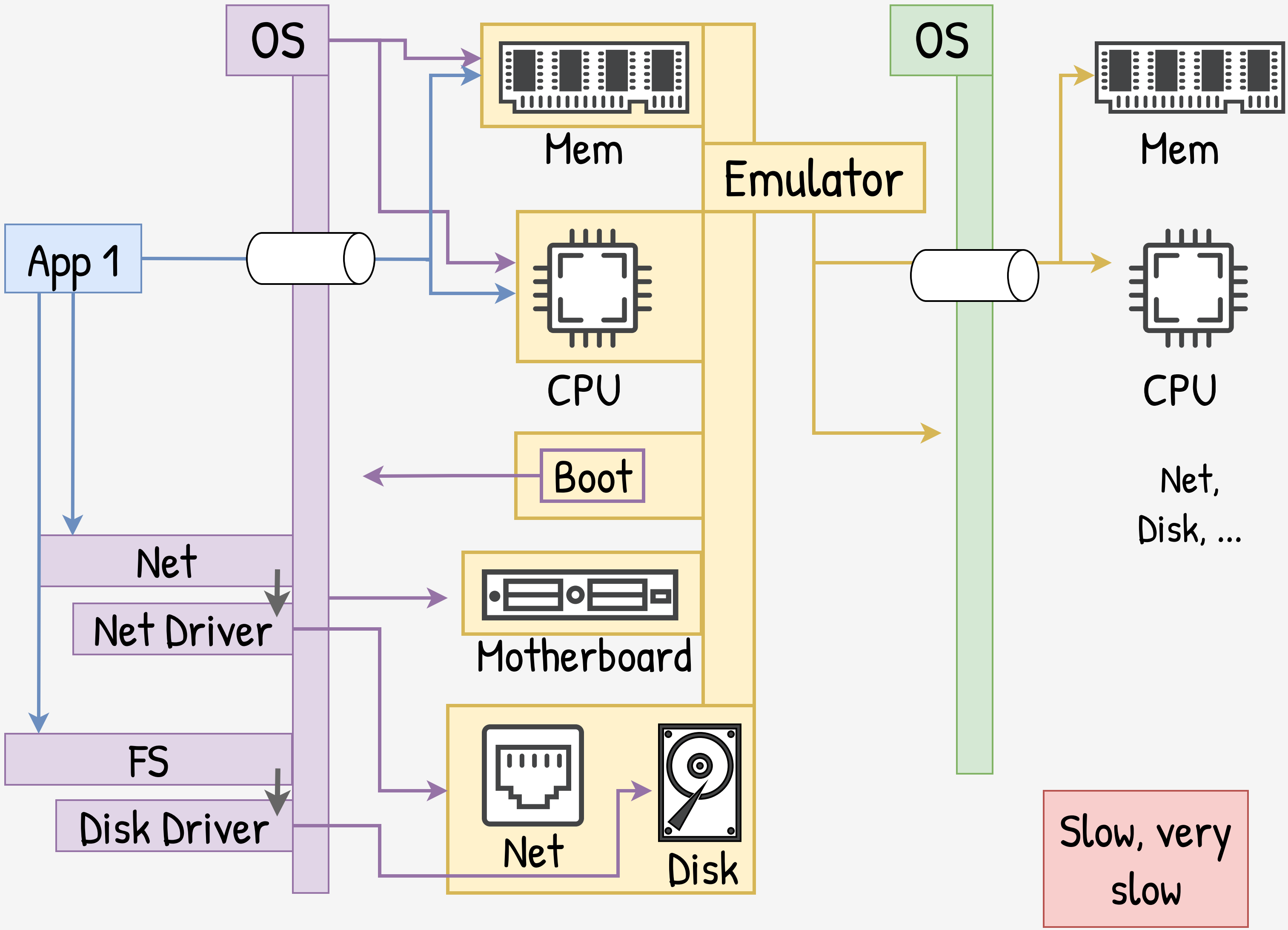
OS
Mem
Emulator
App 1
CPU
Boot
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
OS
Mem
CPU
Net,
Disk, ...
Non-Priv
Passthrough
Priv *delegated* to
Host OS via *syscalls*



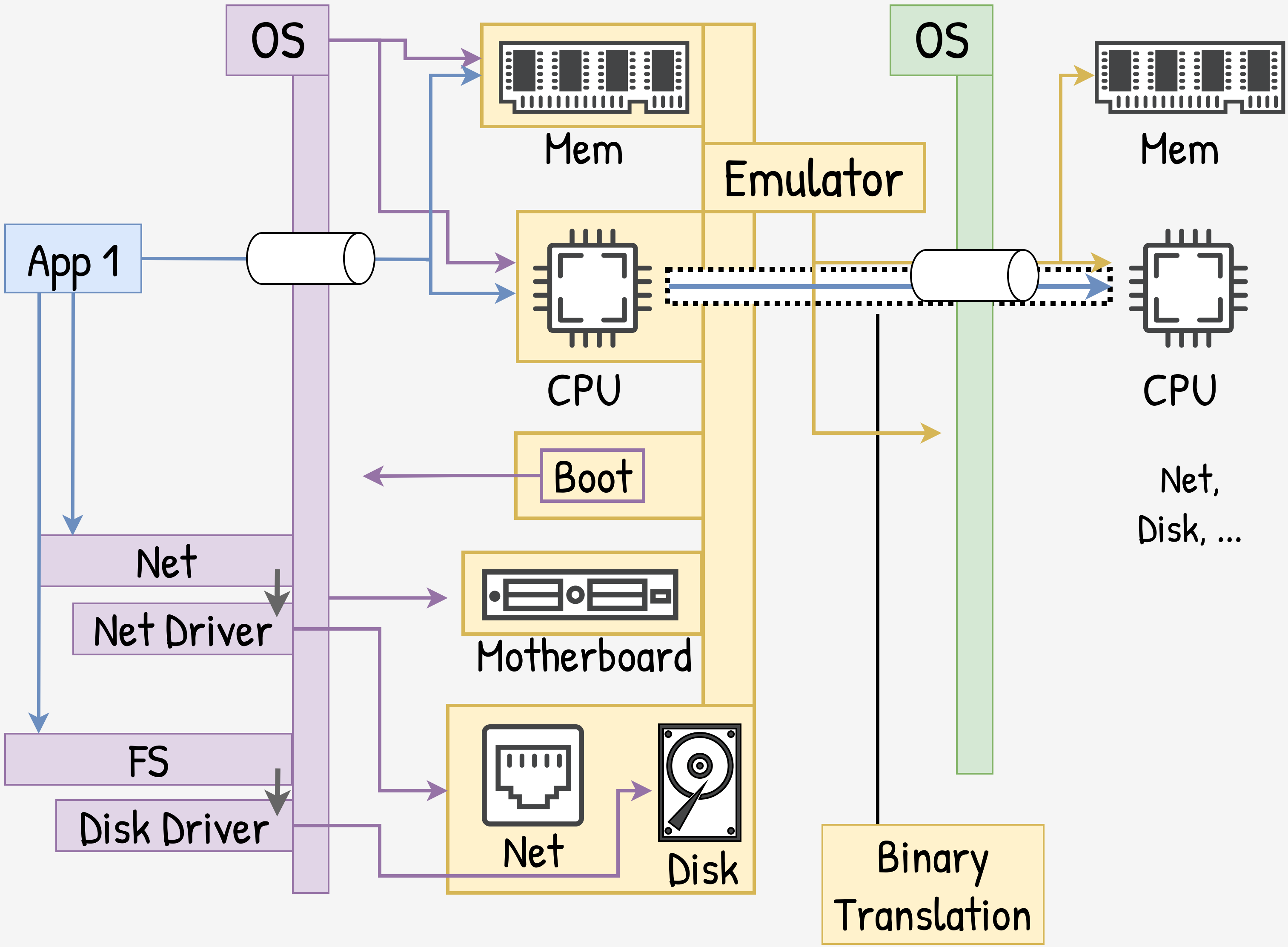
OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
HW fully emulated



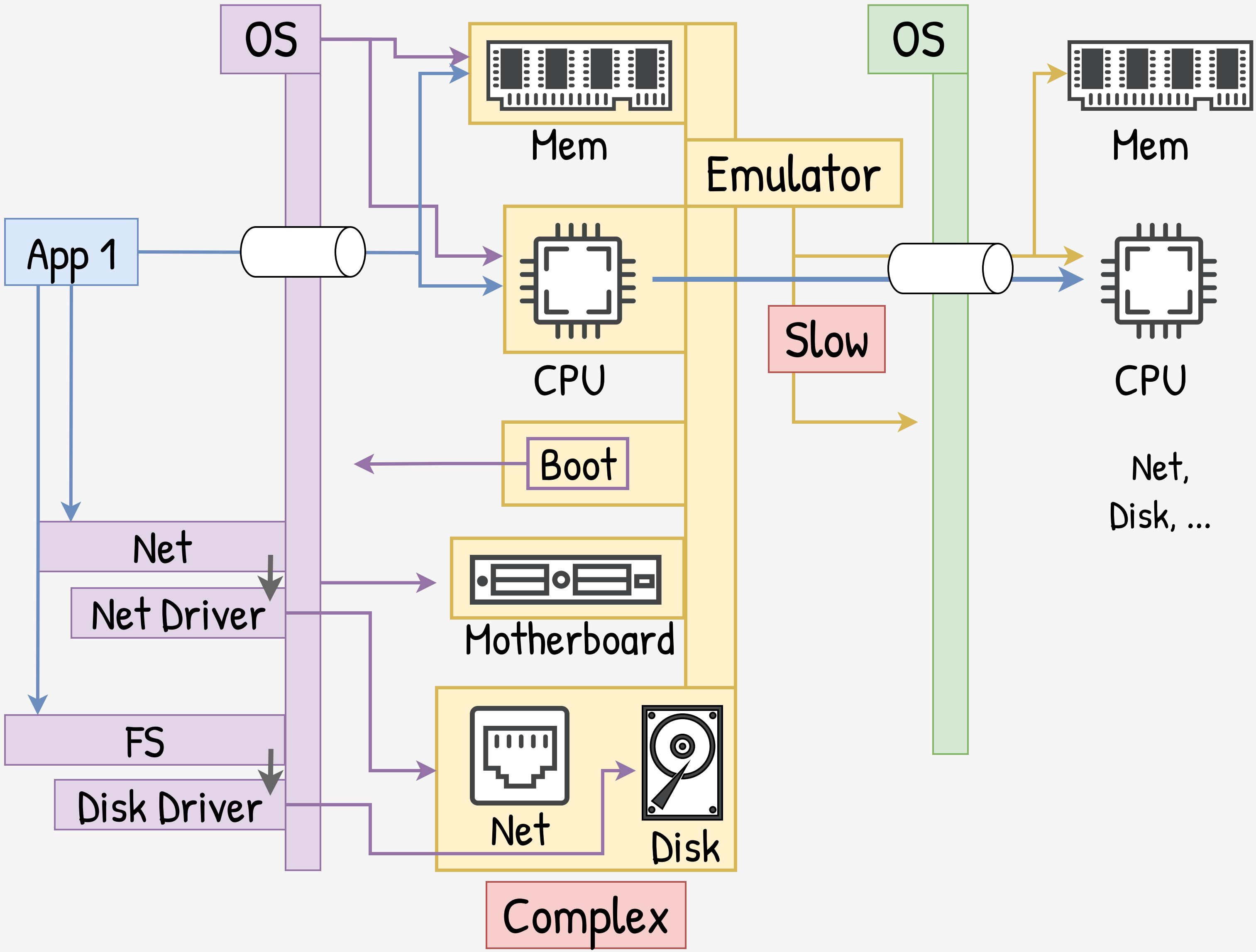
OS
Mem
Emulator
OS
Mem
App 1
ARM
x64
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
ARM CPU
emulated running
in a x64 CPU

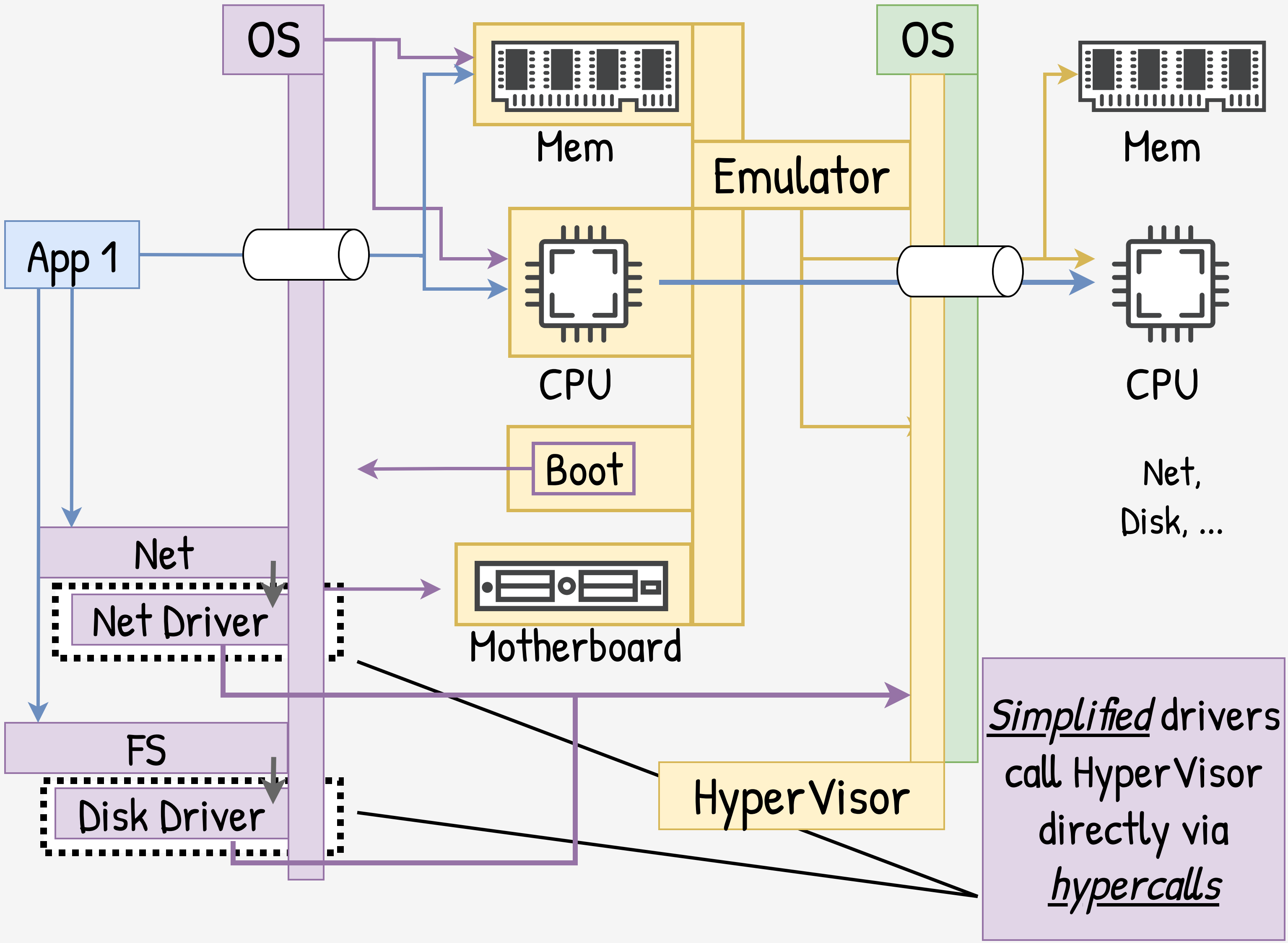


OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
Slow, very slow

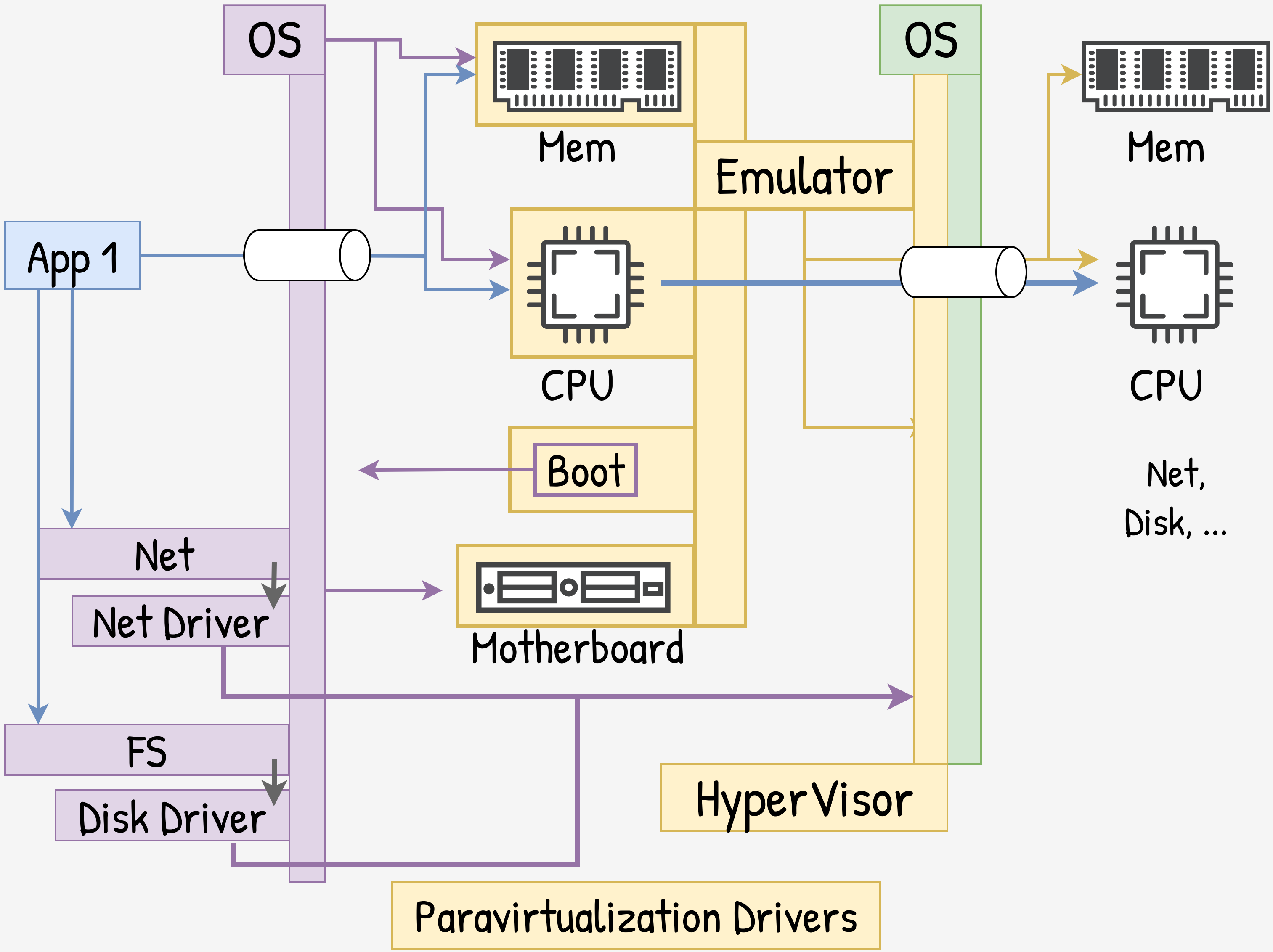


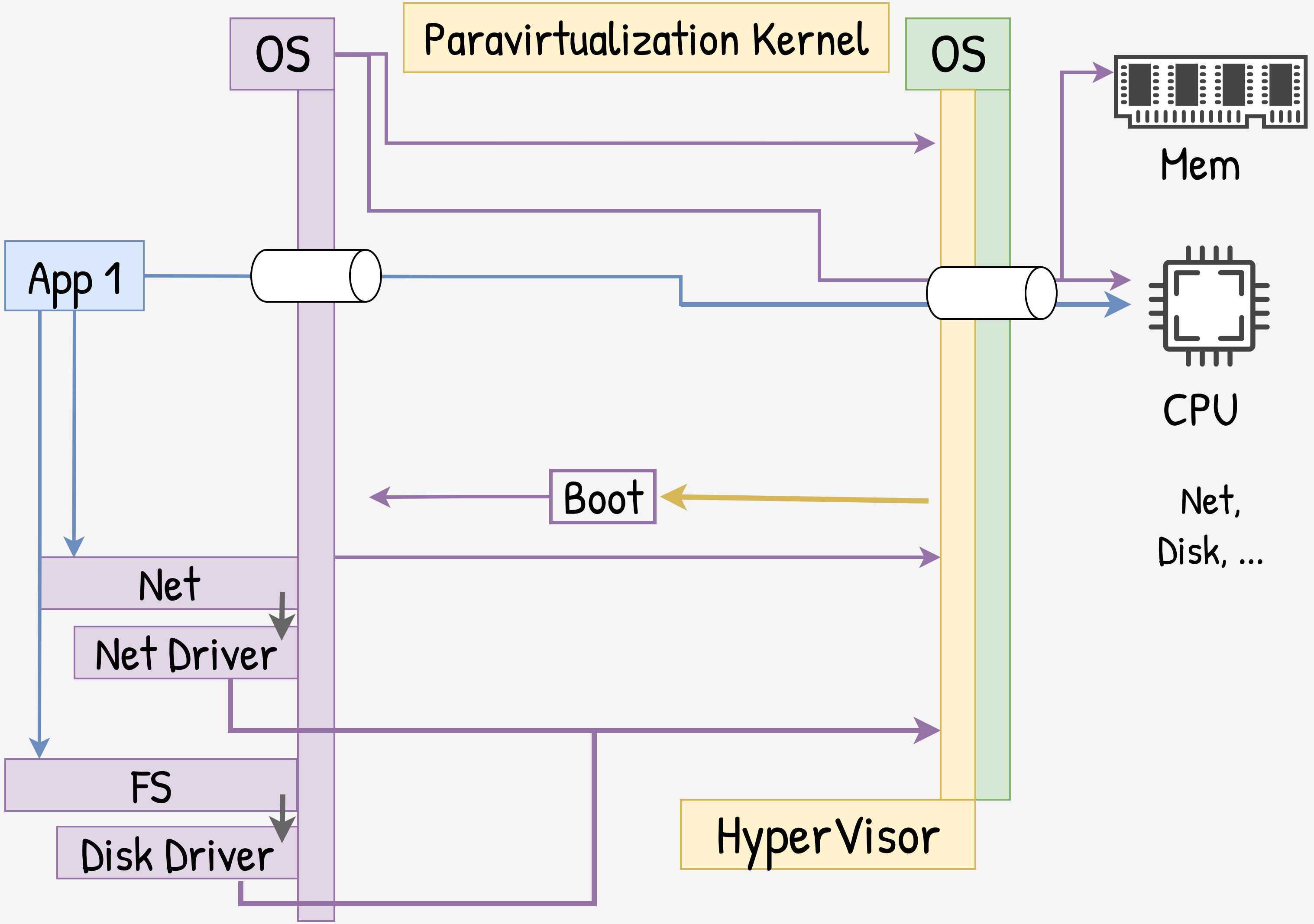
OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
Binary
Translation


OS
Mem
Emulator
OS
Mem
App 1
CPU
Slow
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
Net
Disk
Complex

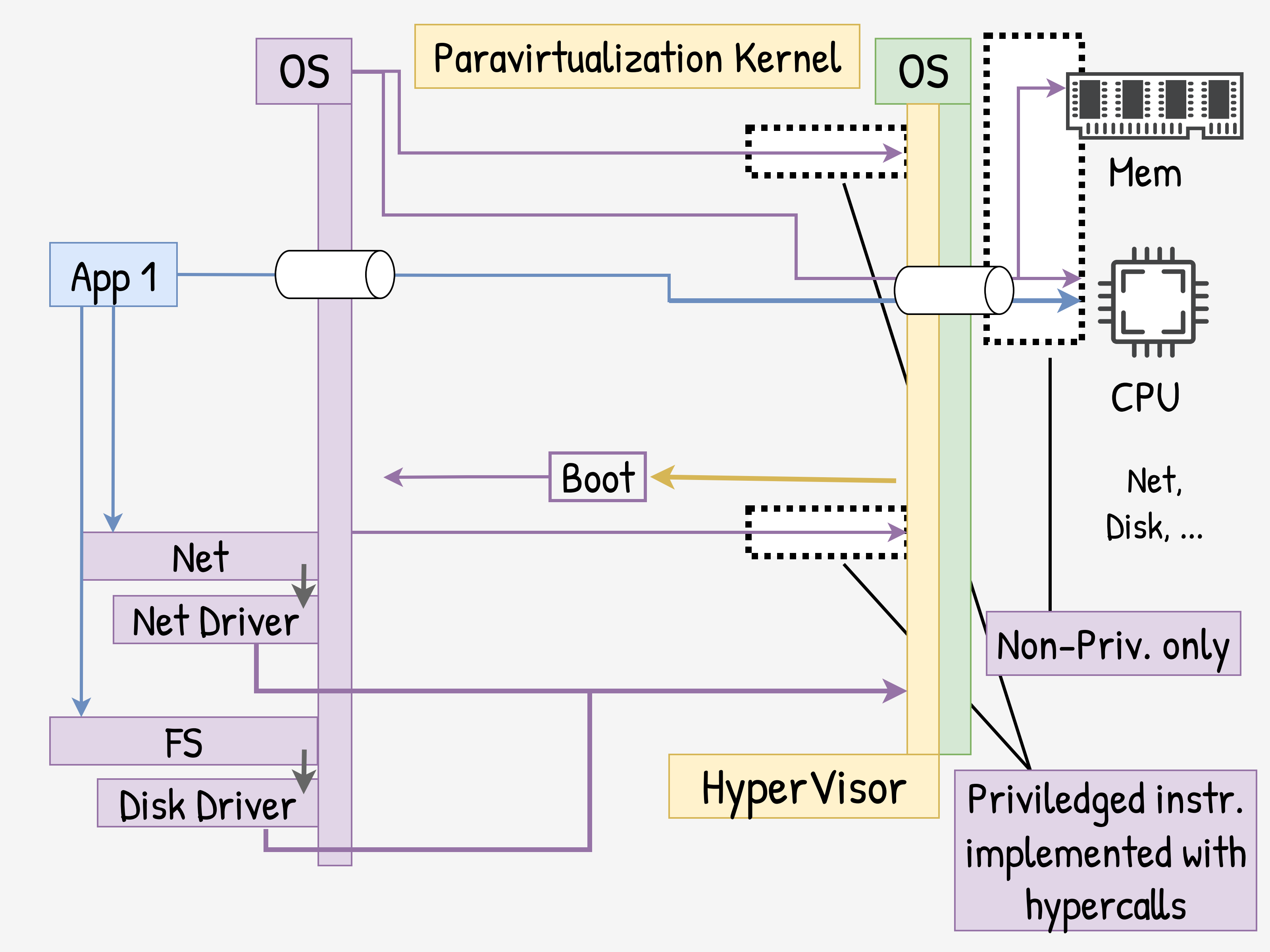


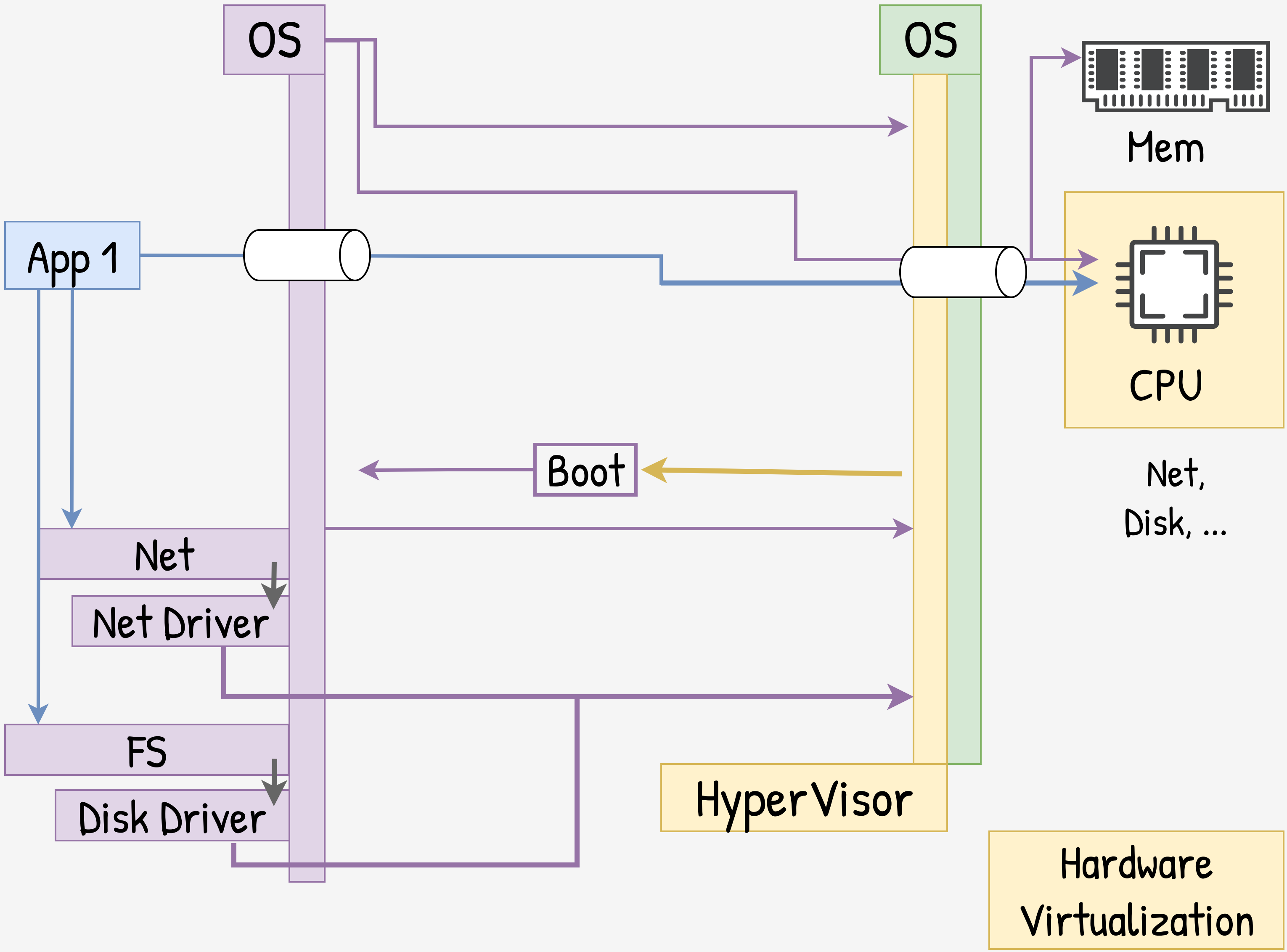
OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
HyperVisor
Simplified drivers call HyperVisor directly via *hypercalls*


OS
Mem
Emulator
OS
Mem
App 1
CPU
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Motherboard
FS
Disk Driver
HyperVisor
Paravirtualization Drivers

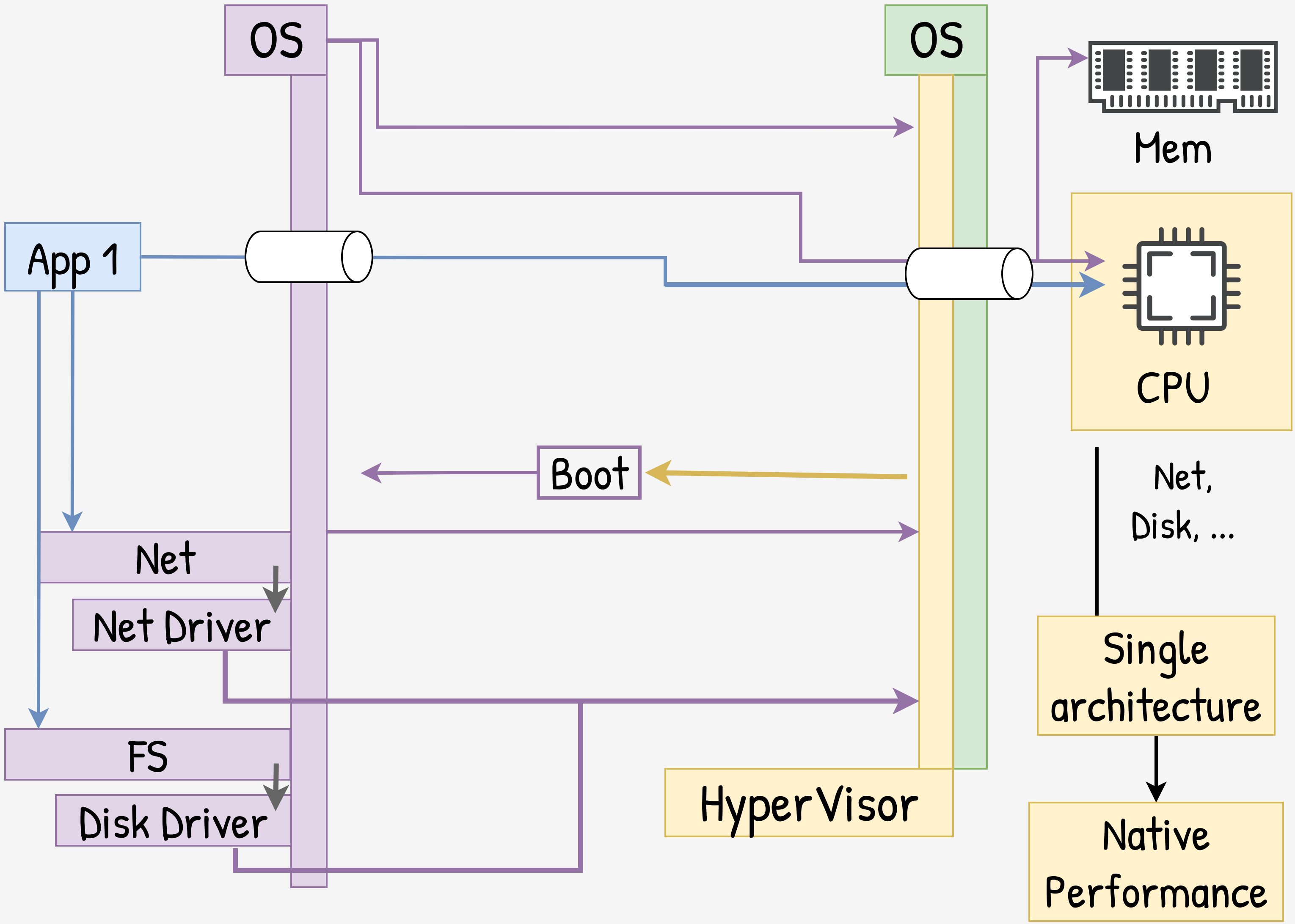


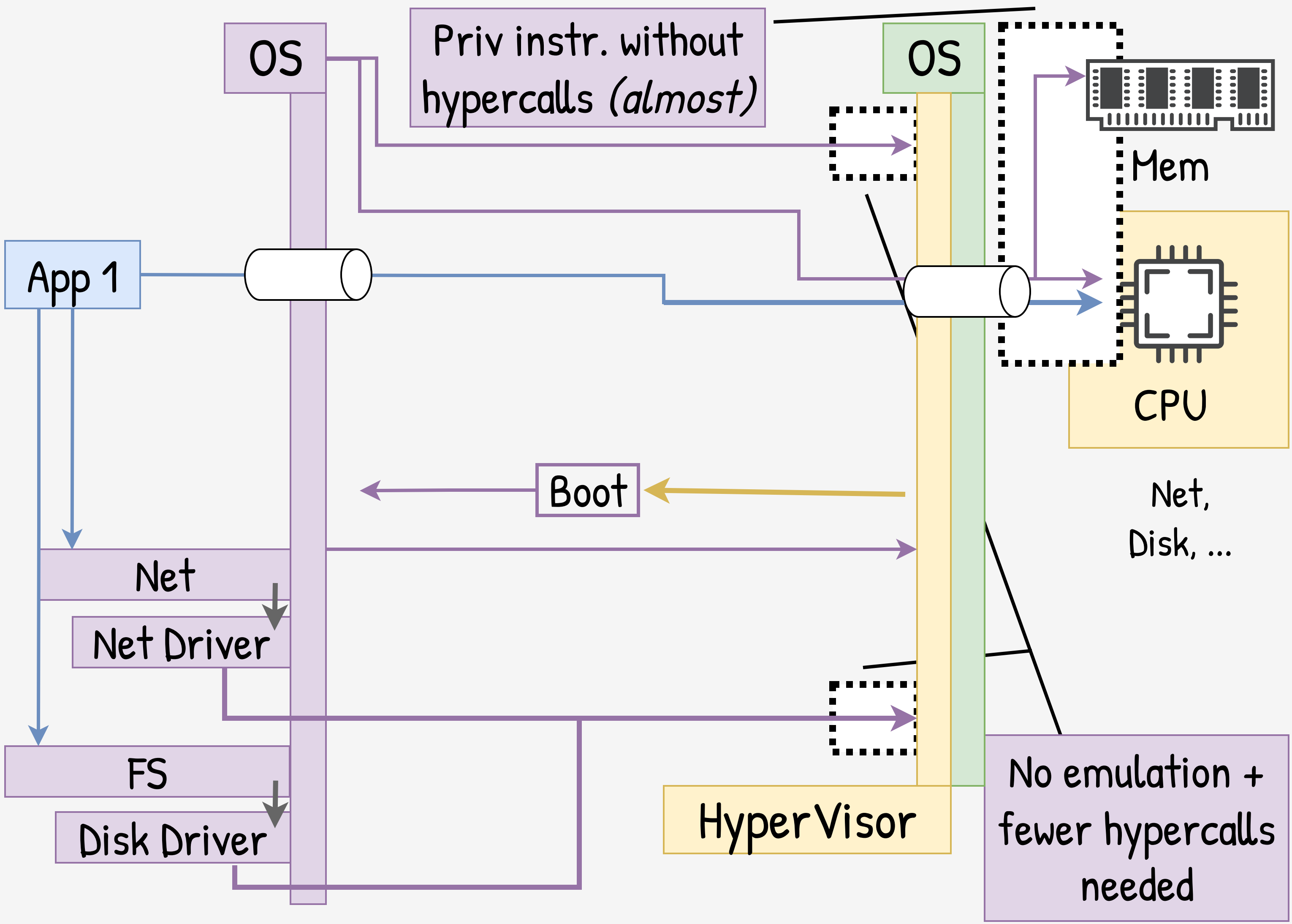
OS
Paravirtualization Kernel
OS
Mem
App 1
CPU
Boot
Net,
Disk, ...
Net
Net Driver
FS
Disk Driver
HyperVisor


OS
Paravirtualization Kernel
OS
Mem
App 1
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Non-Priv. only
FS
Disk Driver
HyperVisor
Priviledged instr.
implemented with
hypercalls

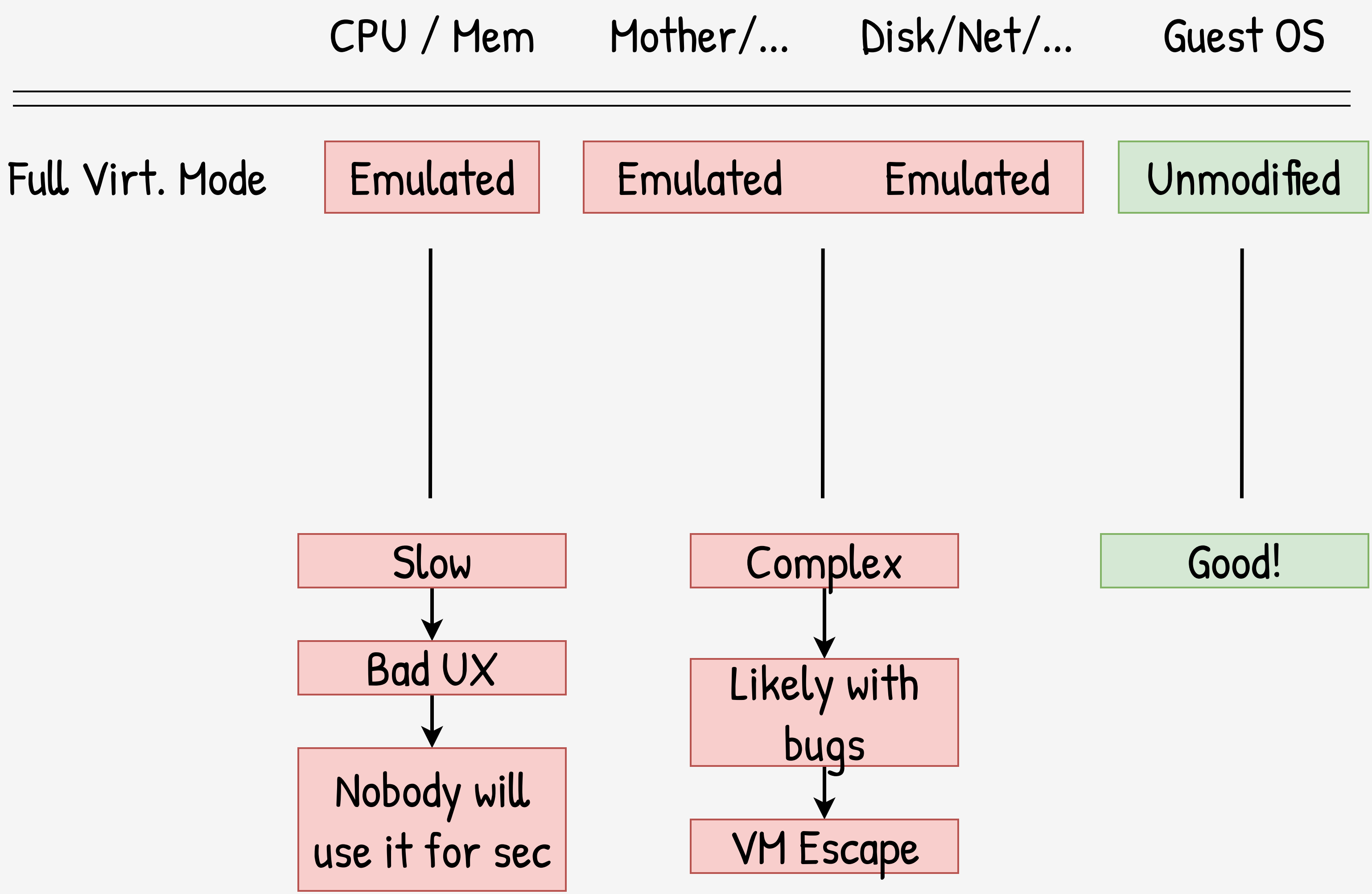


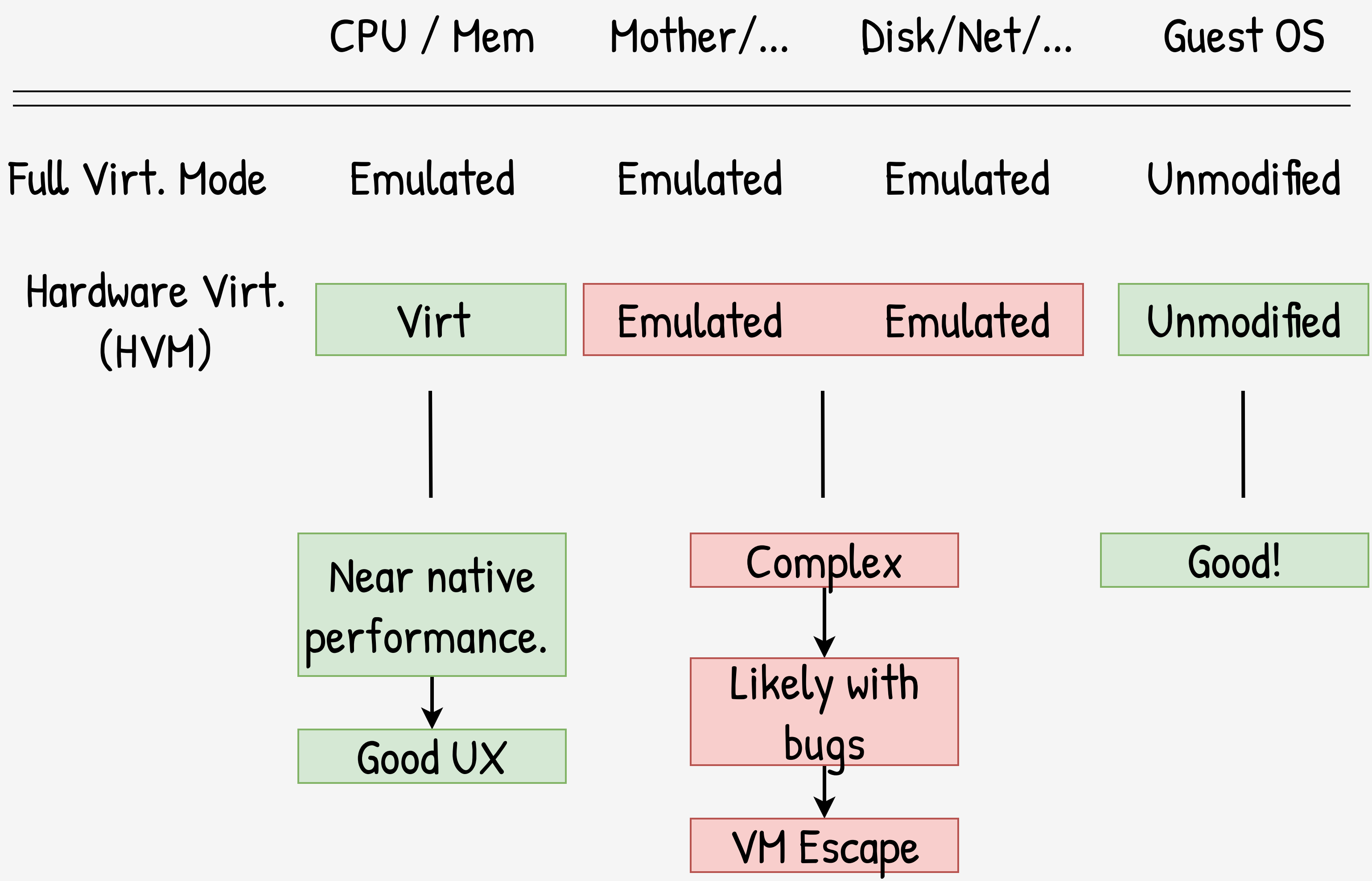
OS
OS
Mem
App 1
CPU
Boot
Net,
Disk, ...
Net
Net Driver
FS
Disk Driver
HyperVisor
Hardware
Virtualization


OS
OS
Mem
App 1
CPU
Boot
Net,
Disk, ...
Net
Net Driver
Single
architecture
FS
Disk Driver
HyperVisor
Native
Performance

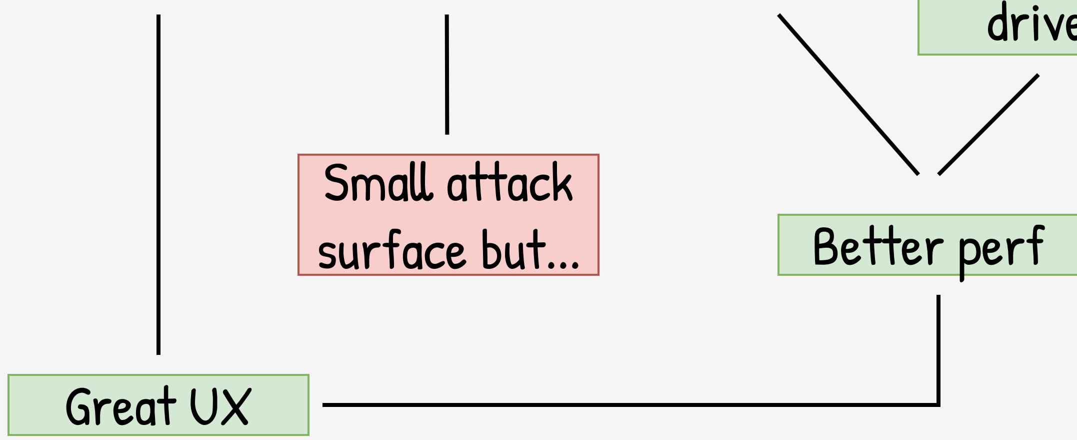


OS
Priv instr. without hypercalls (almost)
OS
Mem
App 1
CPU
Boot
Net, Disk, ...
Net
Net Driver
FS
Disk Driver
HyperVisor
No emulation + fewer hypercalls needed


CPU / Mem
Mother/...
Disk/Net/...
Guest OS
Full Virt. Mode
Emulated
Emulated
Emulated
Unmodified
Slow
Bad UX
Nobody will use it for sec
Complex
Likely with bugs
VM Escape
Good!


CPU / Mem
Mother/...
Disk/Net/...
Guest OS
Full Virt. Mode
Emulated
Emulated
Emulated
Unmodified
Hardware Virt. (HVM)
Virt
Emulated
Emulated
Unmodified
Near native performance.
Good UX
Complex
Likely with bugs
VM Escape
Good!

	CPU / Mem	Mother/...	Disk/Net/...	Guest OS
Full Virt. Mode	Emulated	Emulated	Emulated	Unmodified
Hardware Virt. (HVM)	Virt	Emulated	Emulated	Unmodified
HVM + Drivers Paravirtualized	Virt	Emulated	Virt	Unmodified, but custom drivers



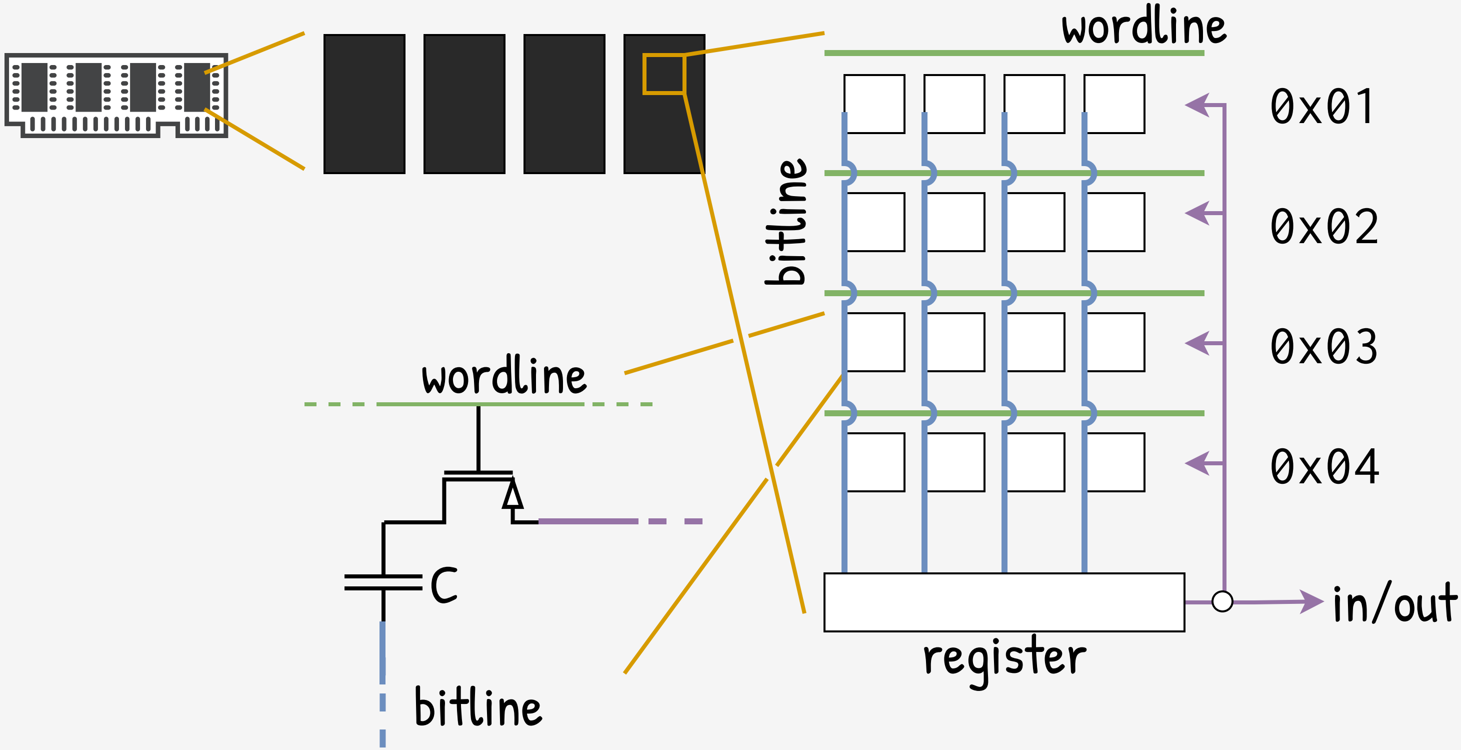
	CPU / Mem	Mother/...	Disk/Net/...	Guest OS
Full Virt. Mode	Emulated	Emulated	Emulated	Unmodified
Hardware Virt. (HVM)	Virt	Emulated	Emulated	Unmodified
HVM + Drivers Paravirtualized	Virt	Emulated	Virt	Unmodified, but custom drivers
HVM + Kernel PV (PVHVM / PVH)	Virt	Virt	Virt	Modified

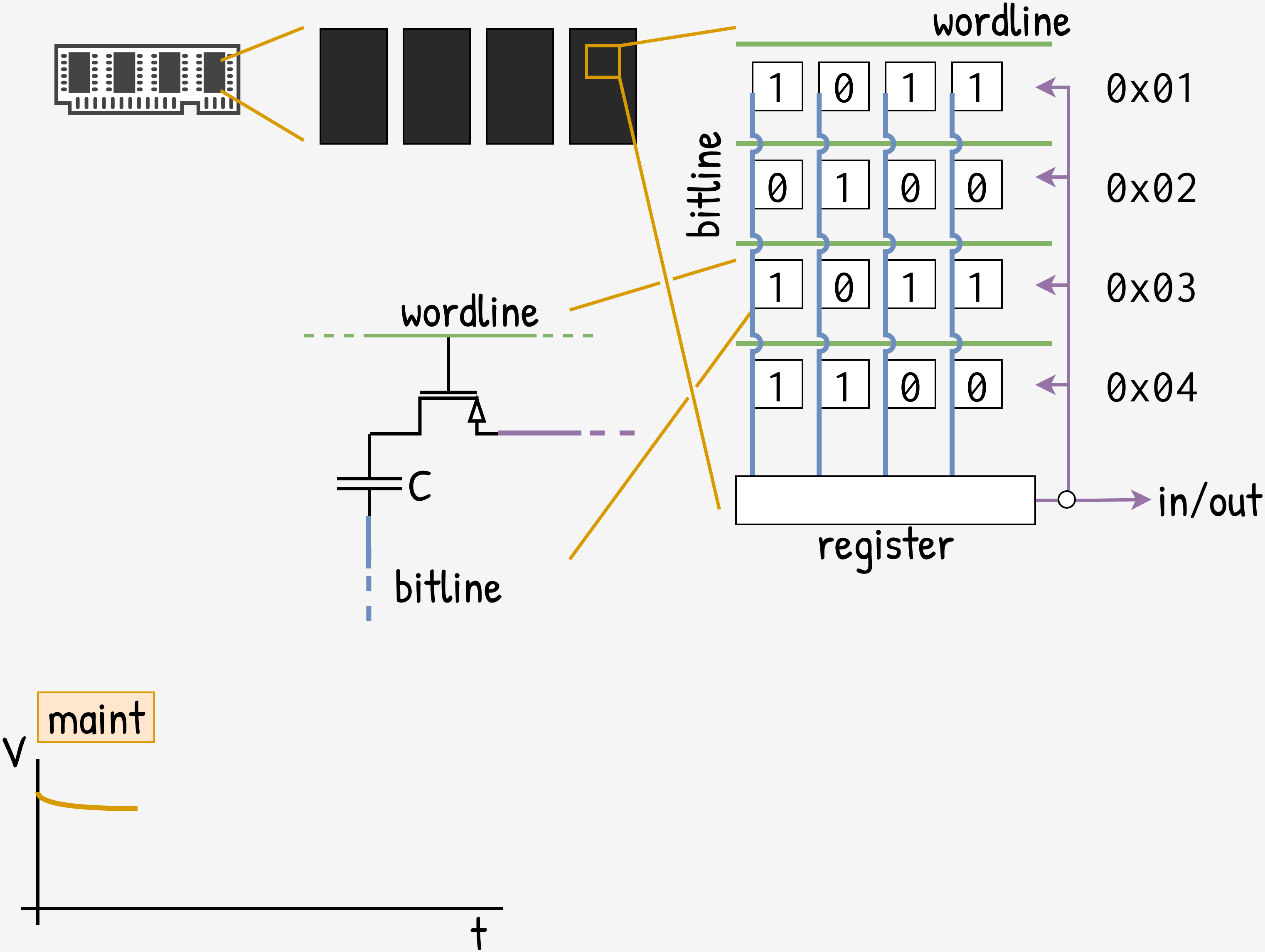
Native performance

Minimal exposition

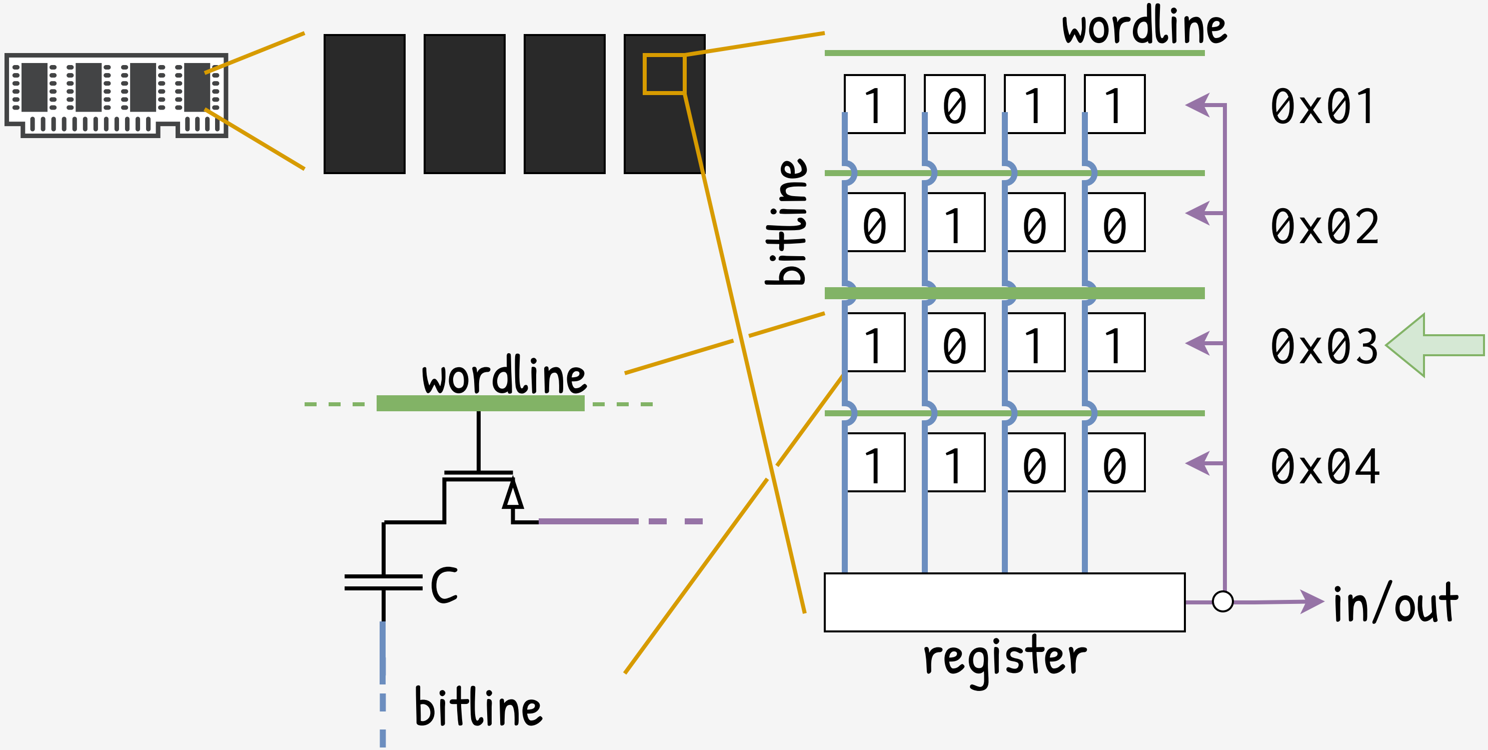
Require guest OS support

	CPU / Mem	Mother/...	Disk/Net/...	Guest OS
Full Virt. Mode	Emulated	Emulated	Emulated	Unmodified
Hardware Virt. (HVM)	Virt	Emulated	Emulated	Unmodified
HVM + Drivers Paravirtualized	Virt	Emulated	Virt	Unmodified, but custom drivers
HVM + Kernel PV (PVHVM / PVH)	Virt	Virt	Virt	Modified
Kernel PV + Emu (PVM)	Emulated	Virt	Virt	Modified
	Poor performance	Minimal exposition		Require guest OS support

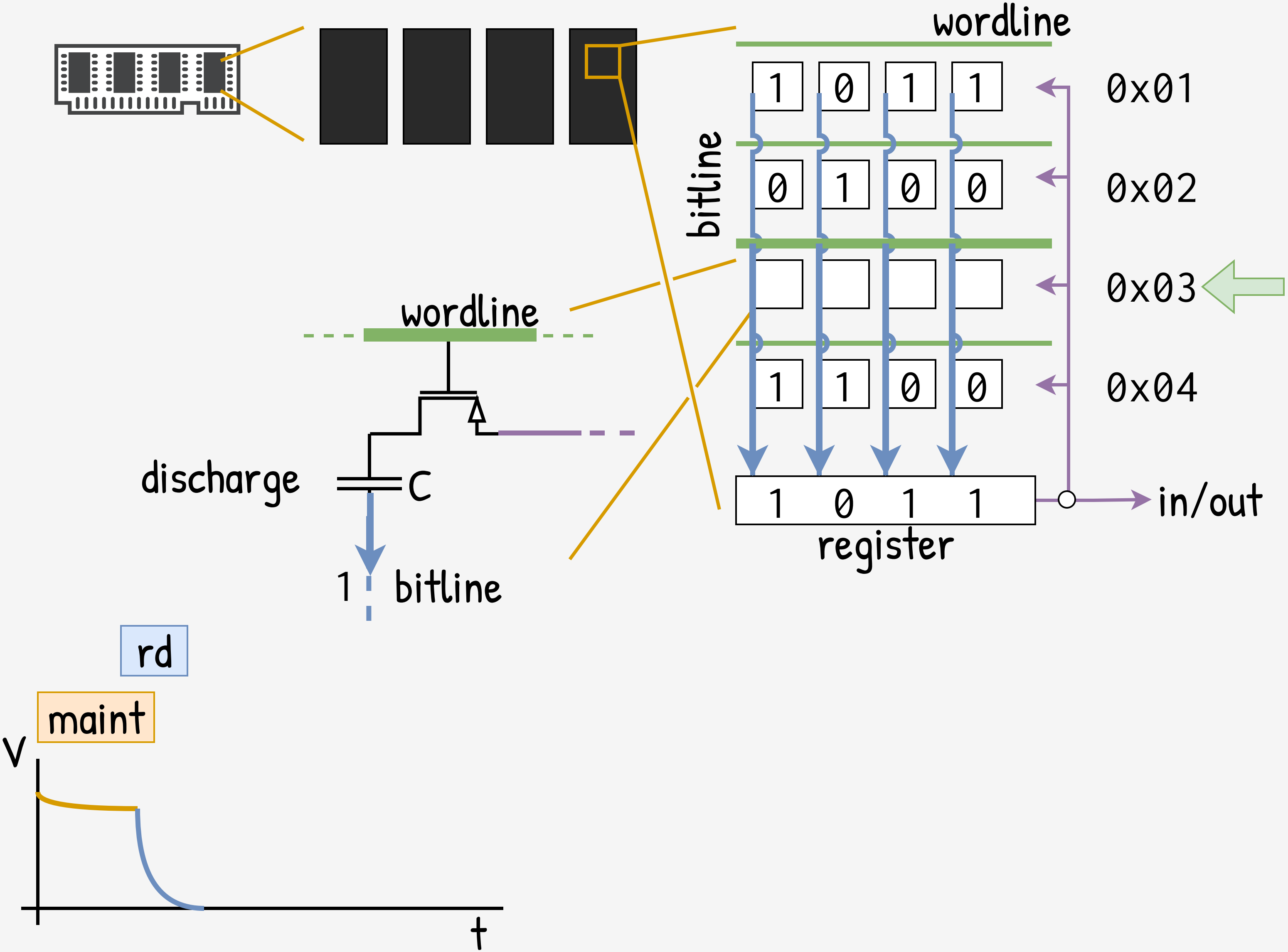

wordline
0x01
0x02
0x03
0x04
bitline
wordline
C
bitline
in/out
register

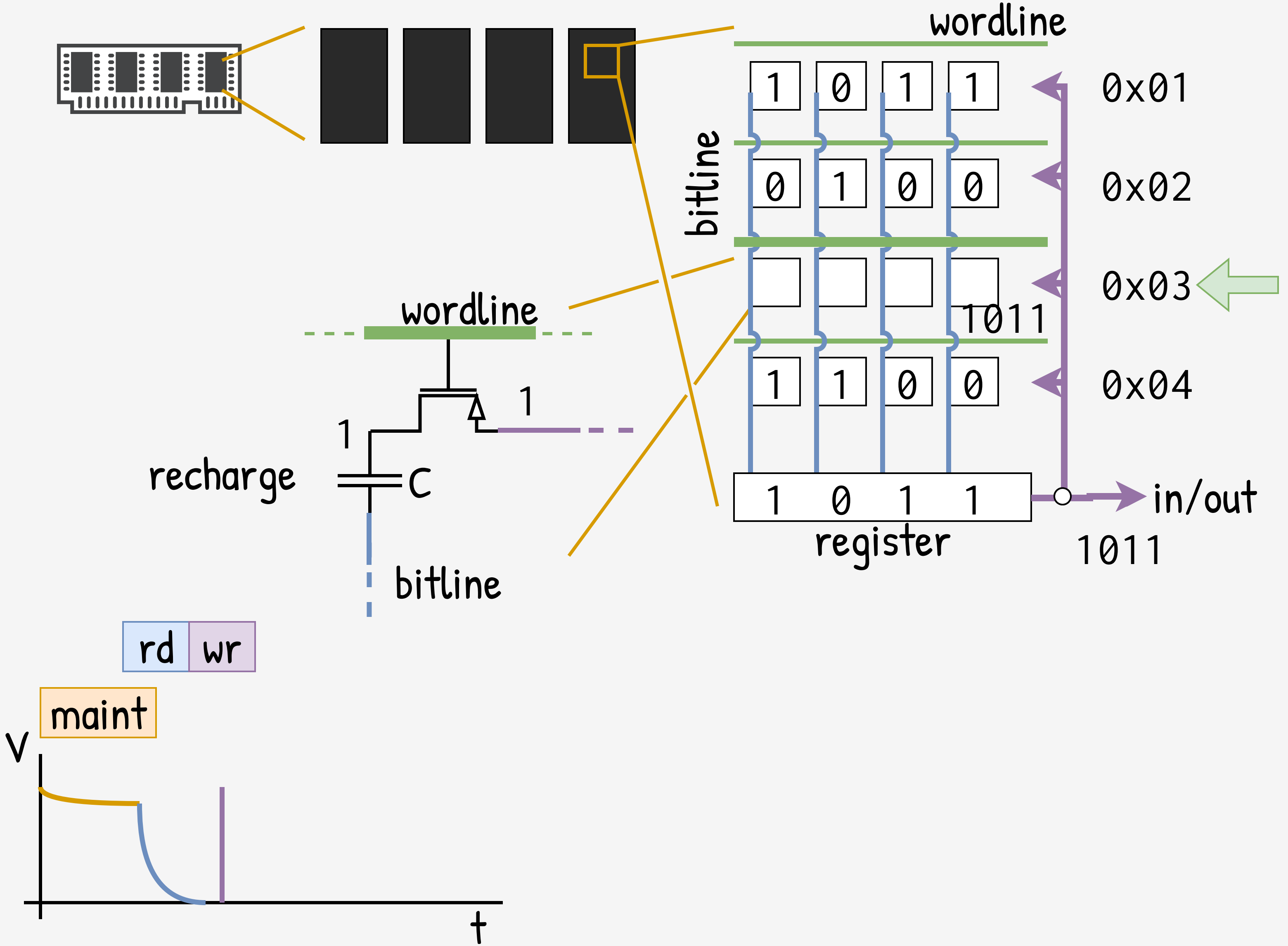


wordline
1 0 1 1
0 1 0 0
1 0 1 1
1 1 0 0
0x01
0x02
0x03
0x04
bitline
wordline
C
bitline
register
in/out
maint
V
t

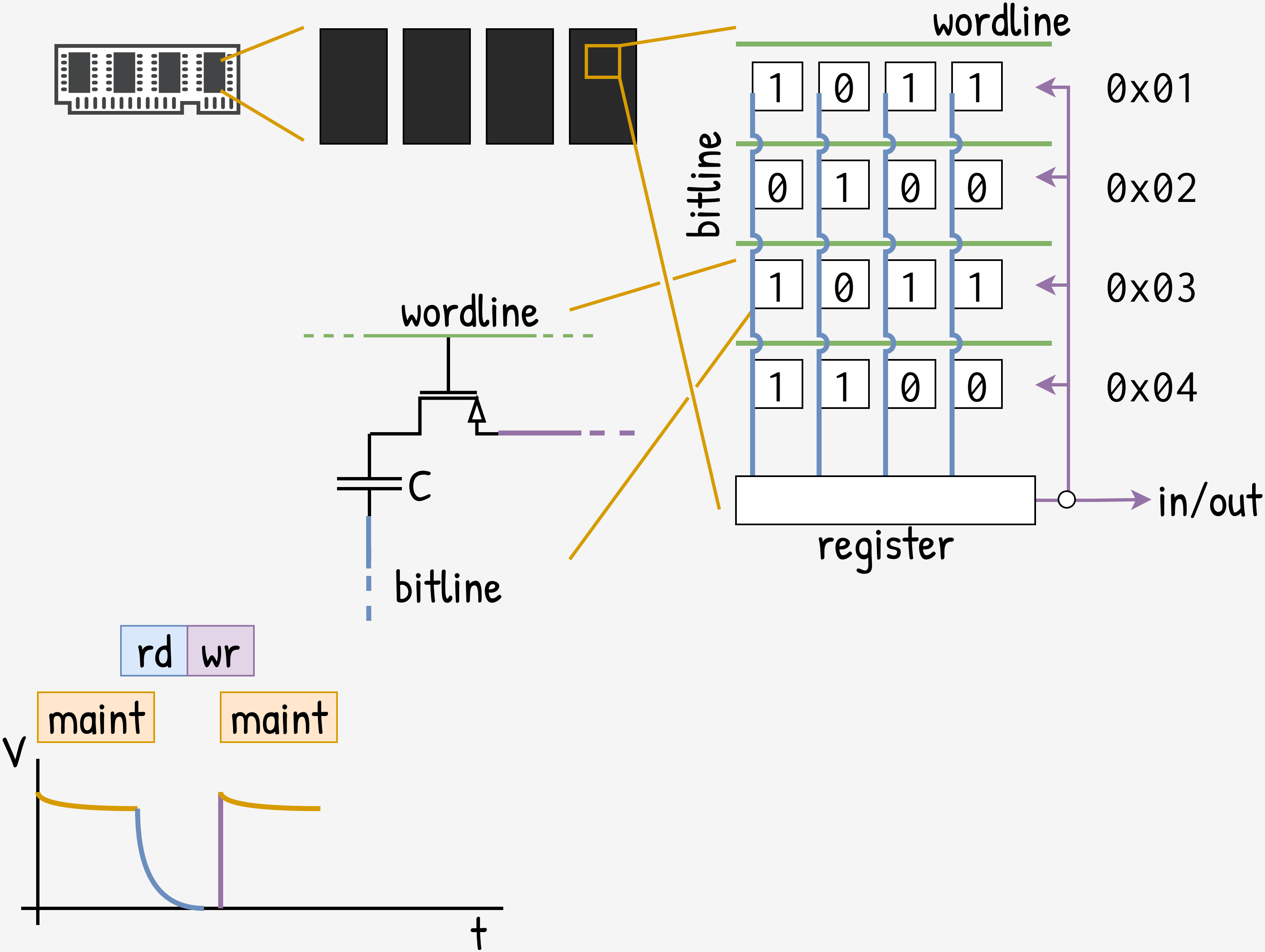

wordline
1 0 1 1
0x01
bitline
0 1 0 0
0x02
1 0 1 1
0x03
1 1 0 0
0x04
wordline
C
bitline
in/out
register


maint
V
t

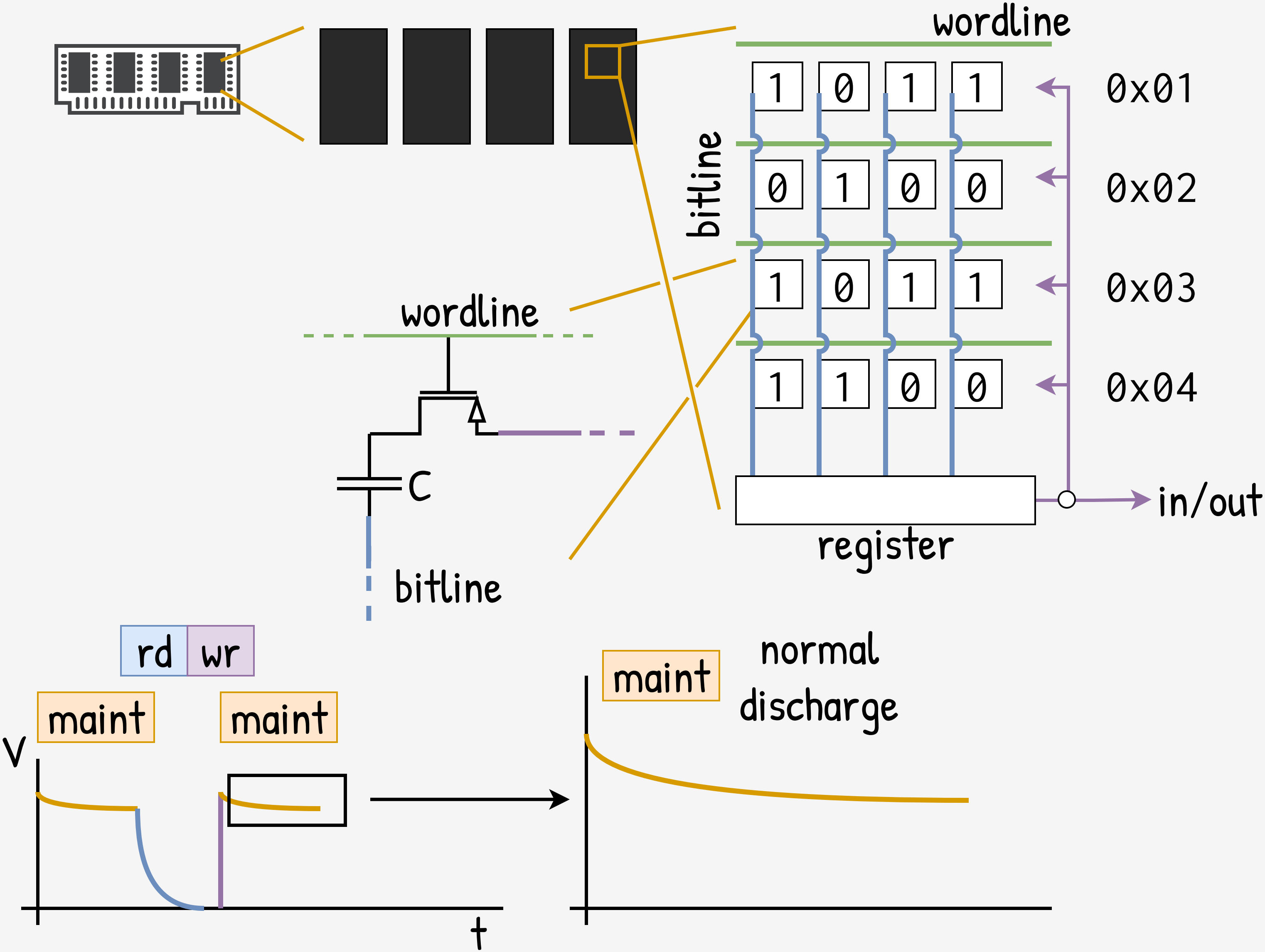

wordline
1 0 1 1
0x01
bitline
0 1 0 0
0x02
0x03
1 1 0 0
0x04
1 0 1 1
in/out
register
wordline
discharge
C
1
bitline
rd
maint
V
t



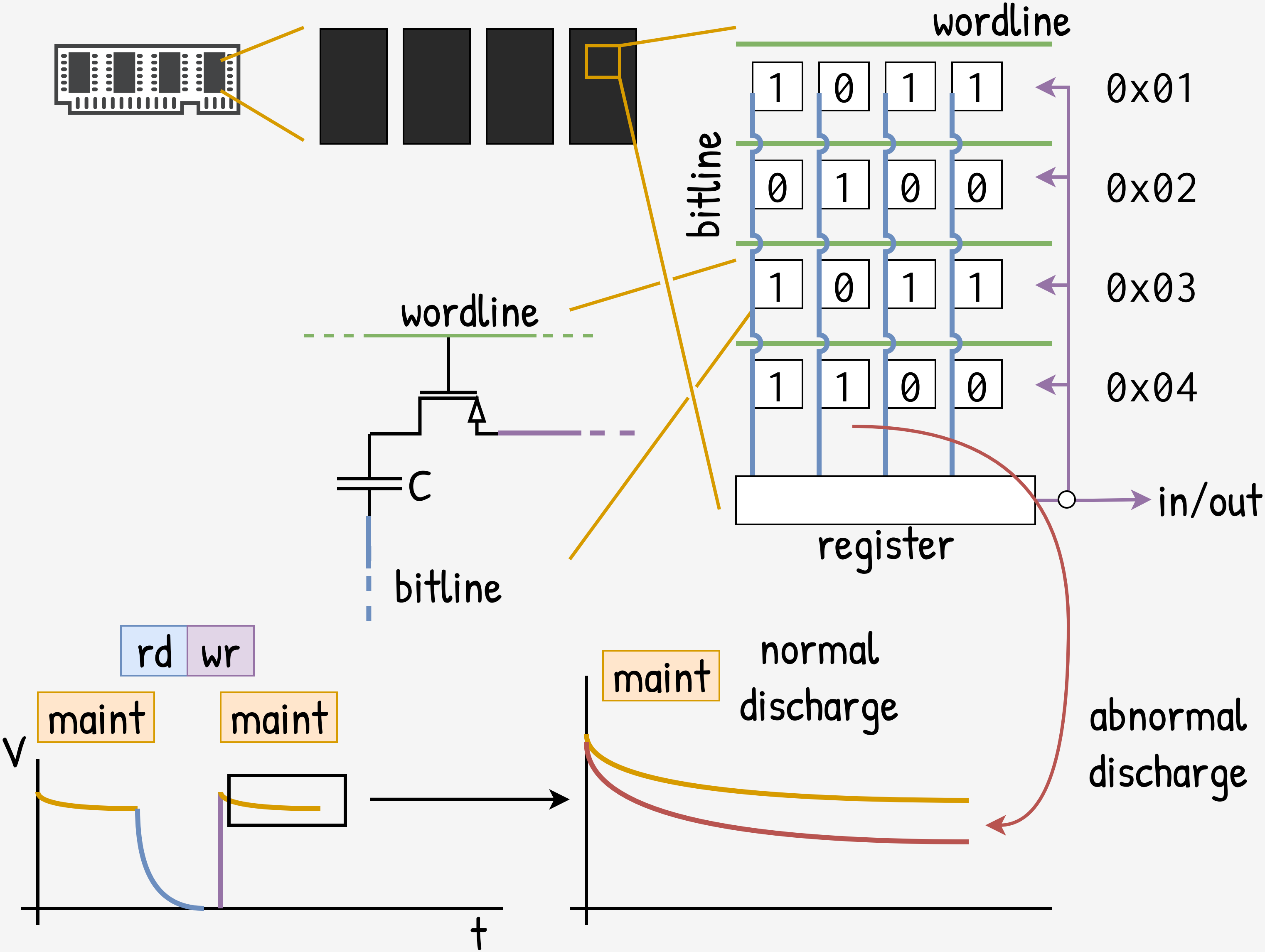
wordline
1 0 1 1
0x01
bitline
0 1 0 0
0x02
0x03
1011
1 1 0 0
0x04
wordline
1
1
recharge
C
1 0 1 1
register
in/out
1011
bitline
rd
wr
maint
V
t

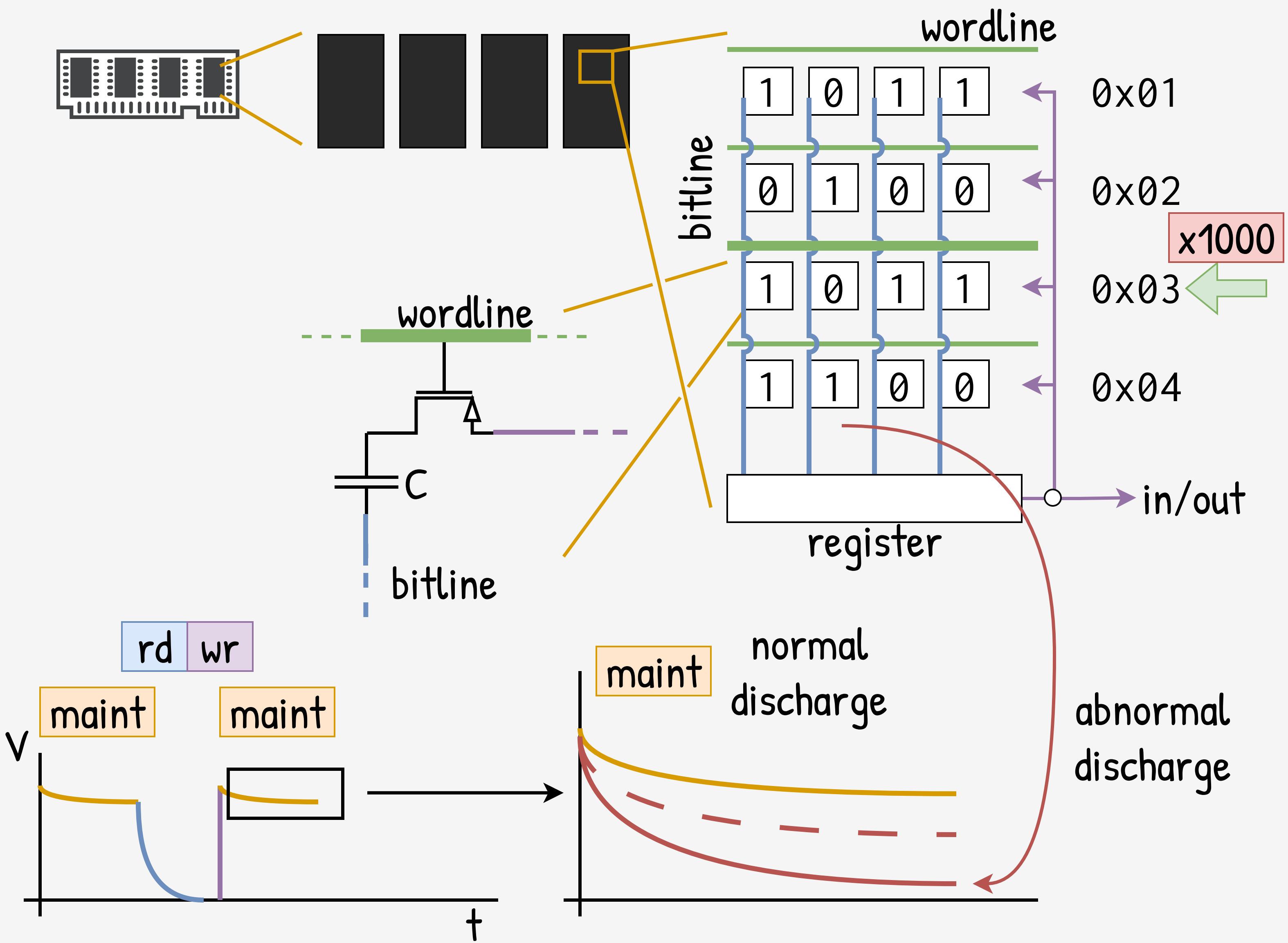


wordline
1 0 1 1
0x01
0 1 0 0
0x02
bitline
1 0 1 1
0x03
wordline
1 1 0 0
0x04
C
in/out
register
bitline
rd
wr
maint
maint
V
t



wordline
1 0 1 1
0 1 0 0
1 0 1 1
1 1 0 0
0x01
0x02
0x03
0x04
bitline
wordline
C
bitline
register
in/out
rd
wr
maint
maint
V
t
maint
normal
discharge


wordline
1 0 1 1
0x01
0 1 0 0
0x02
1 0 1 1
0x03
1 1 0 0
0x04
bitline
register
in/out
wordline
C
bitline
rd
wr
maint
maint
V
t
maint
normal
discharge
abnormal
discharge



wordline
1 0 1 1
0x01
bitline
0 1 0 0
0x02
x1000
1 0 1 1
0x03
1 1 0 0
0x04
wordline
C
in/out
register
bitline
rd
wr
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maint
V
t
maint
normal
discharge
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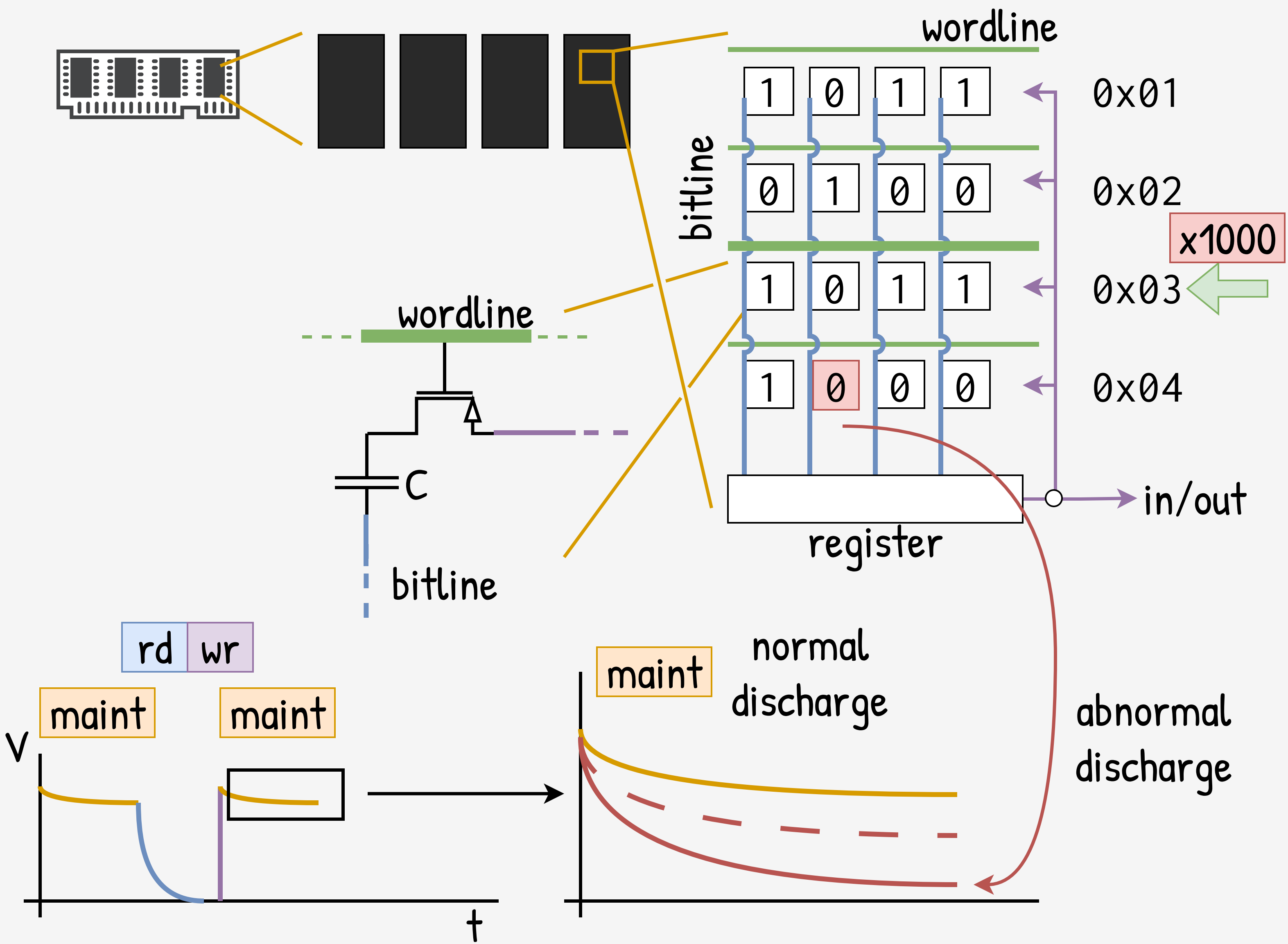
discharge



wordline
1 0 1 1
0x01
0 1 0 0
0x02
x1000
1 0 1 1
0x03
1 0 0 0
0x04
bitline
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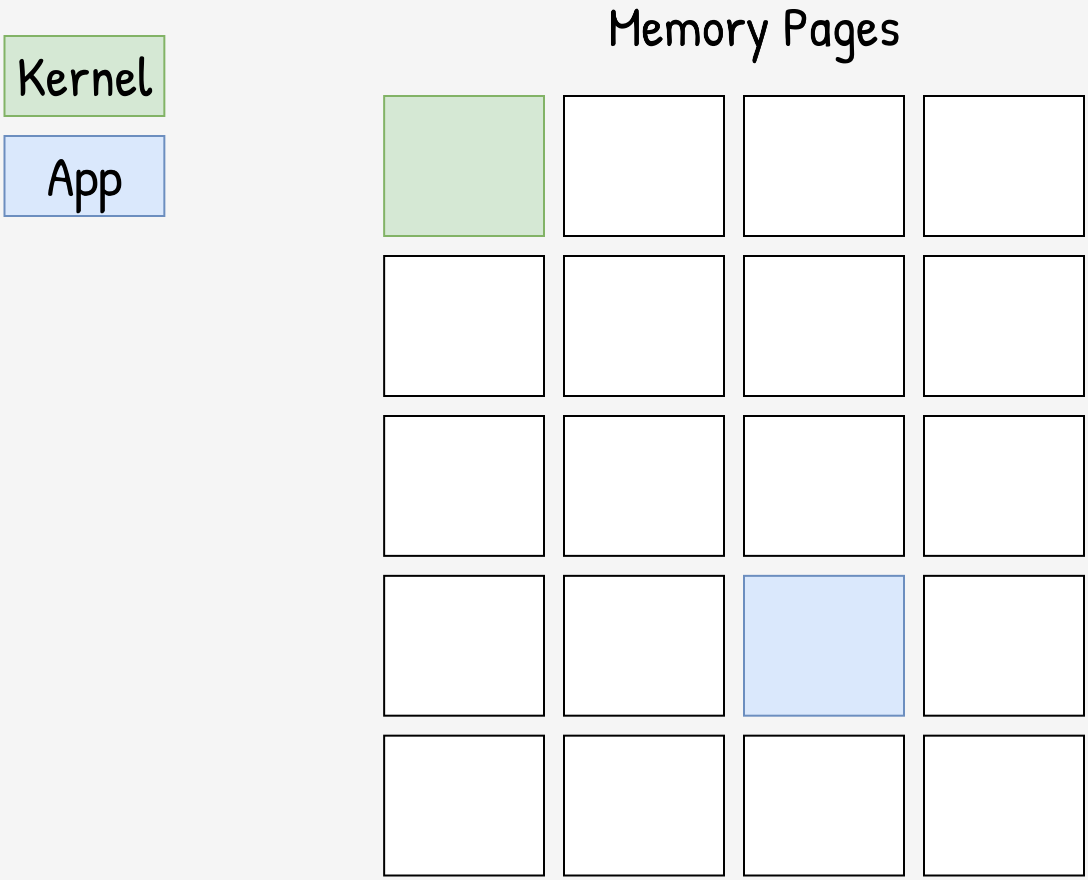
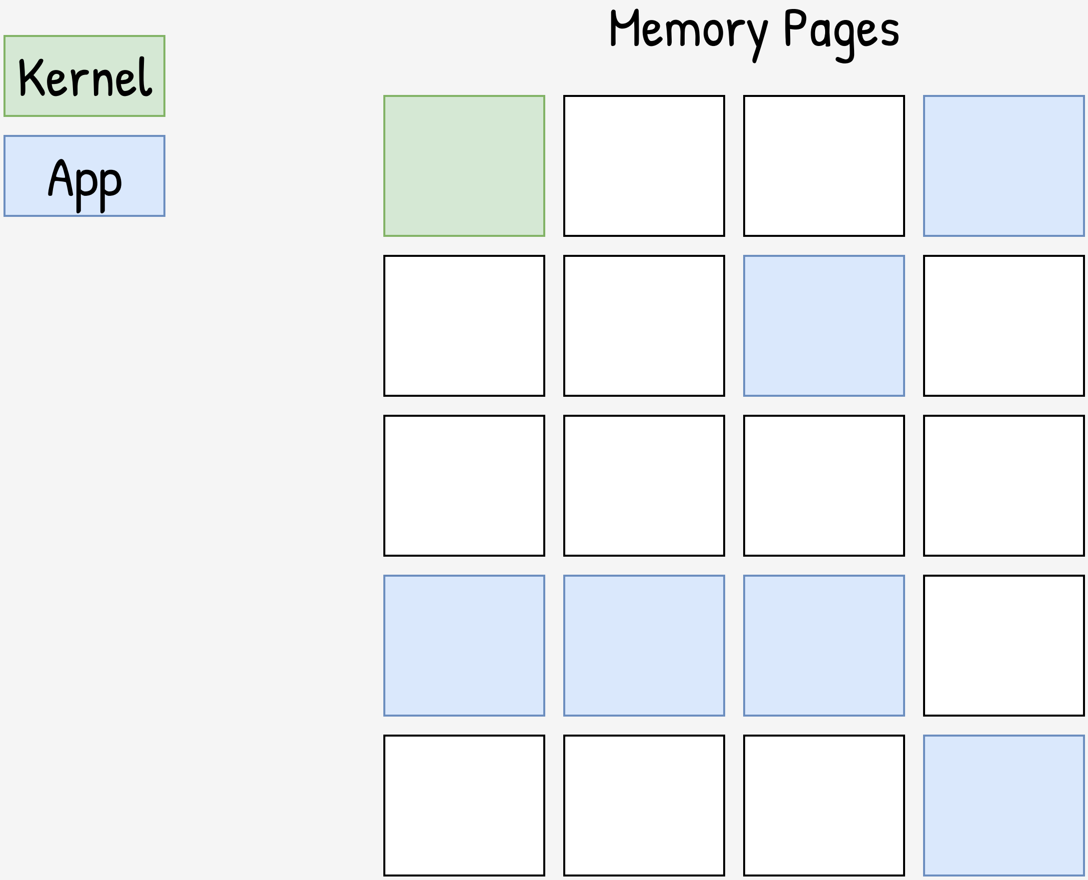
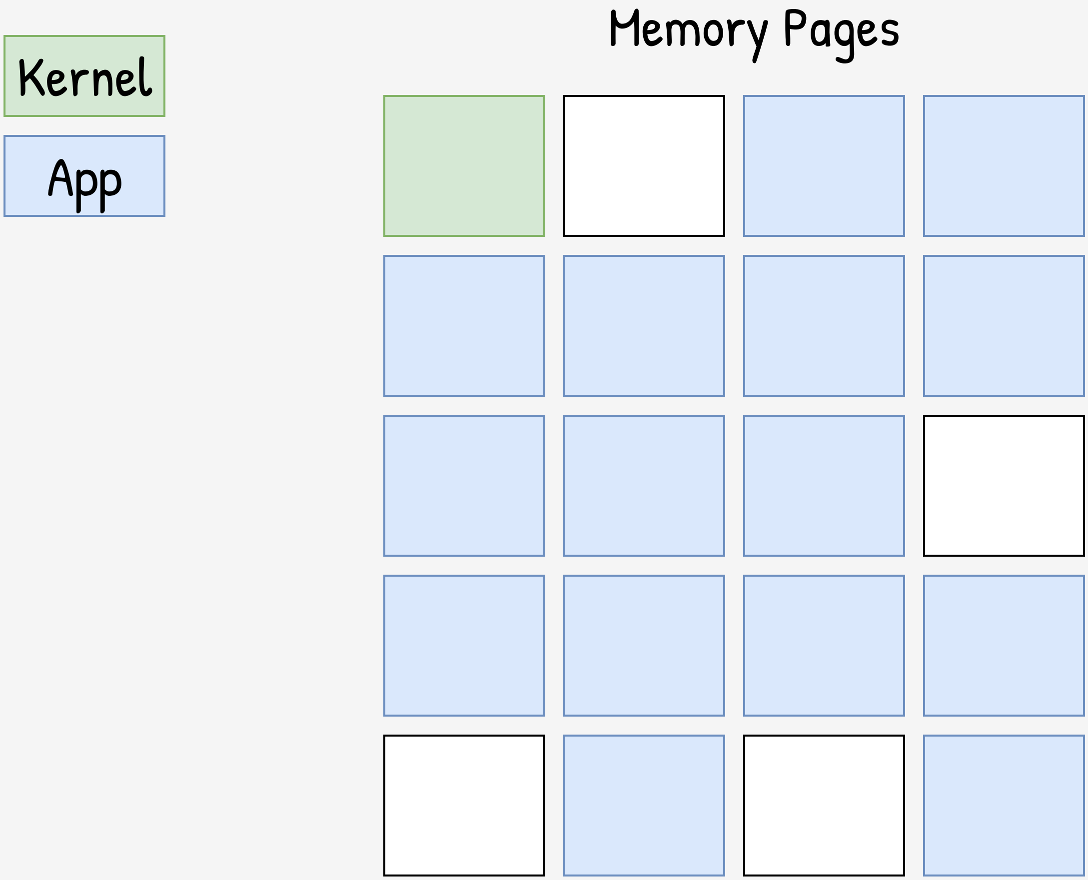
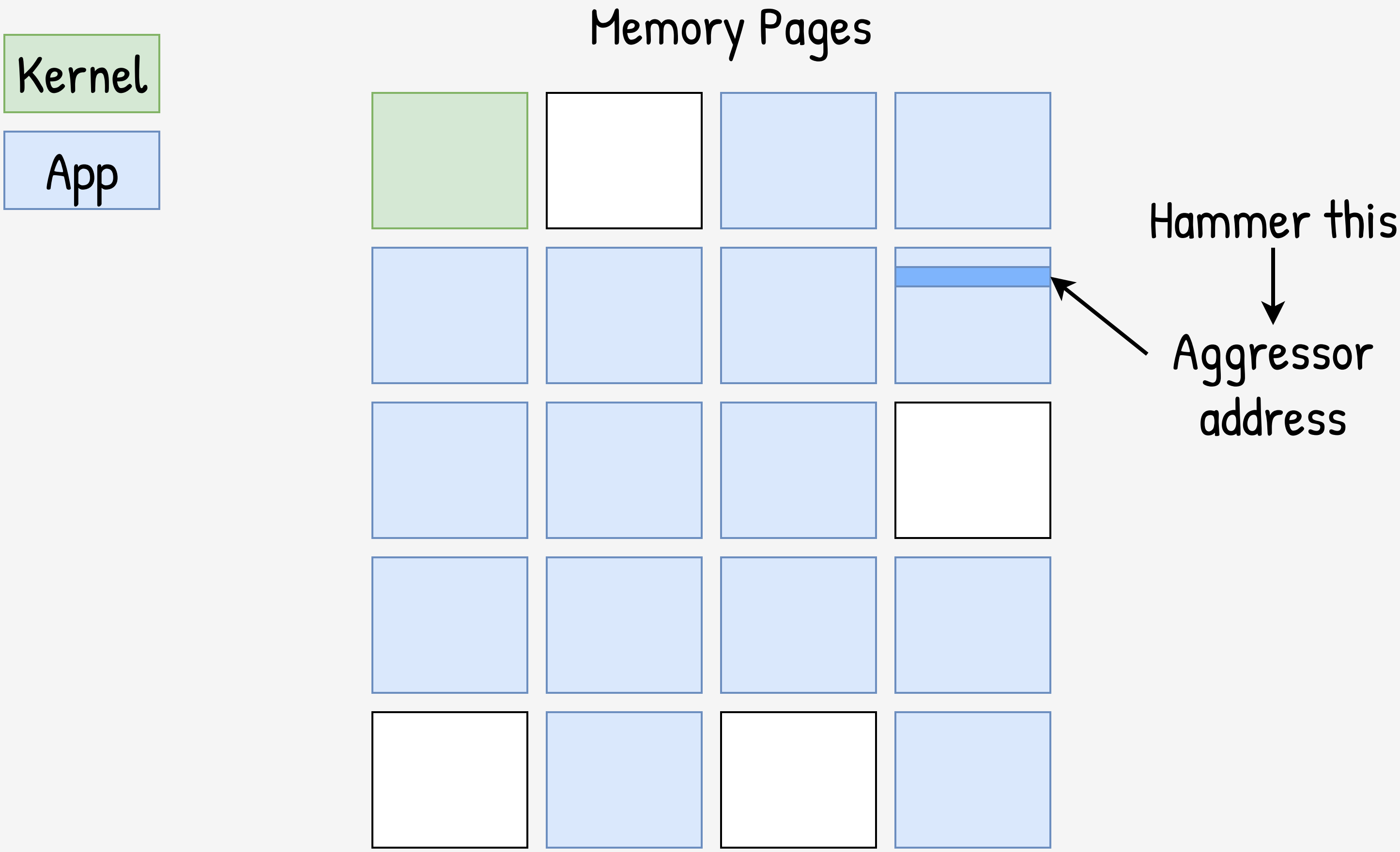
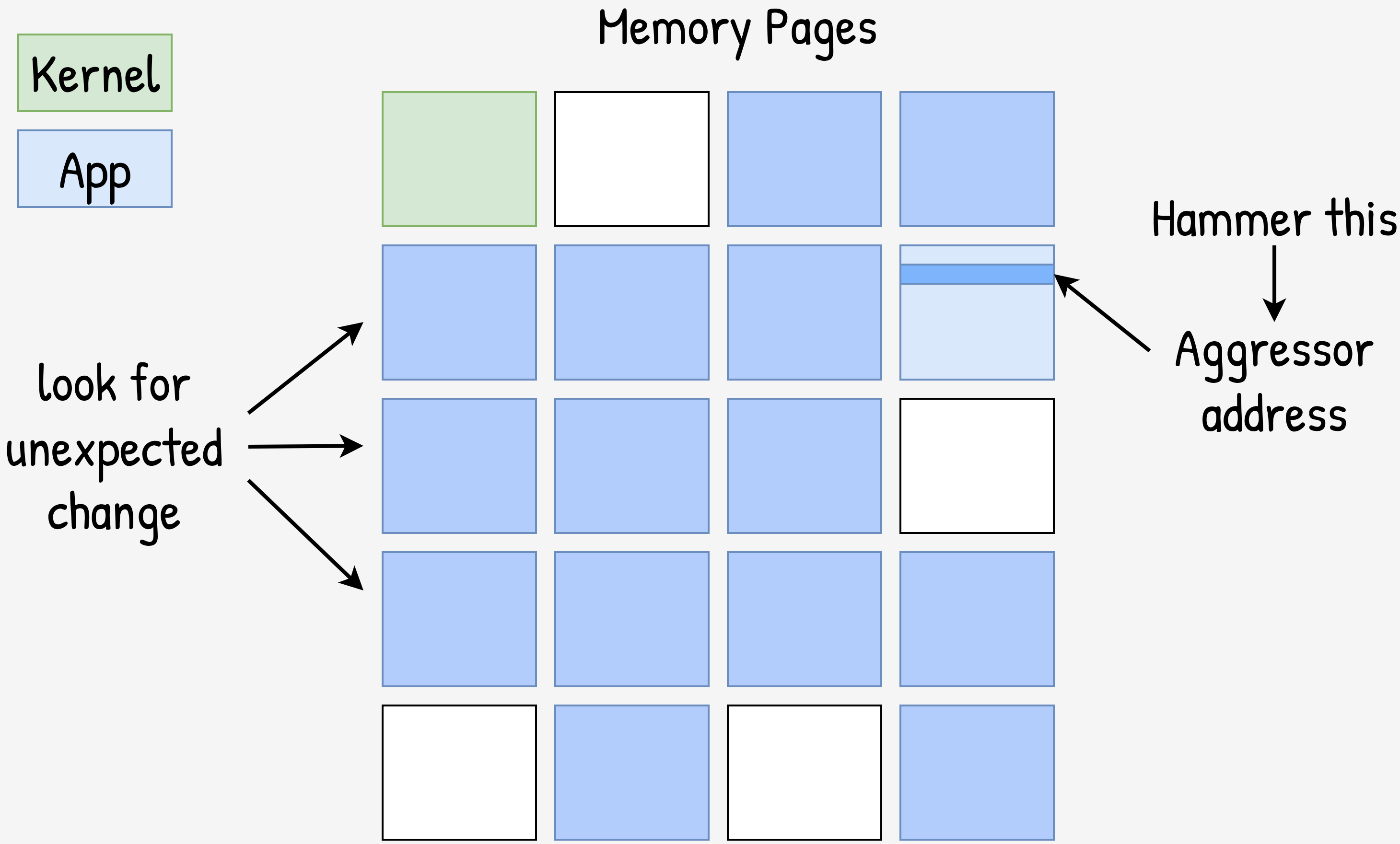
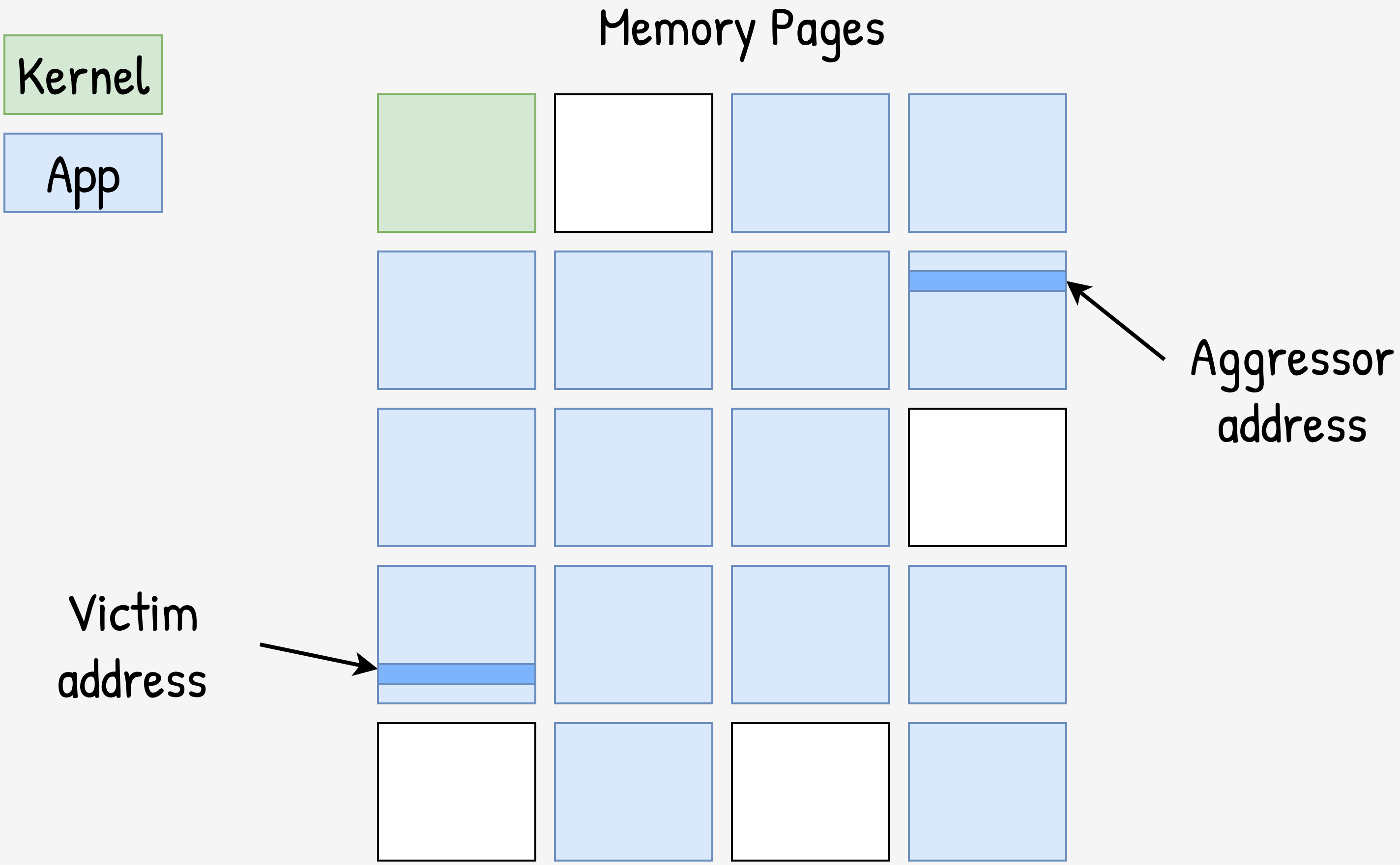

Kernel
App
Memory Pages


Memory Pages
Kernel
App

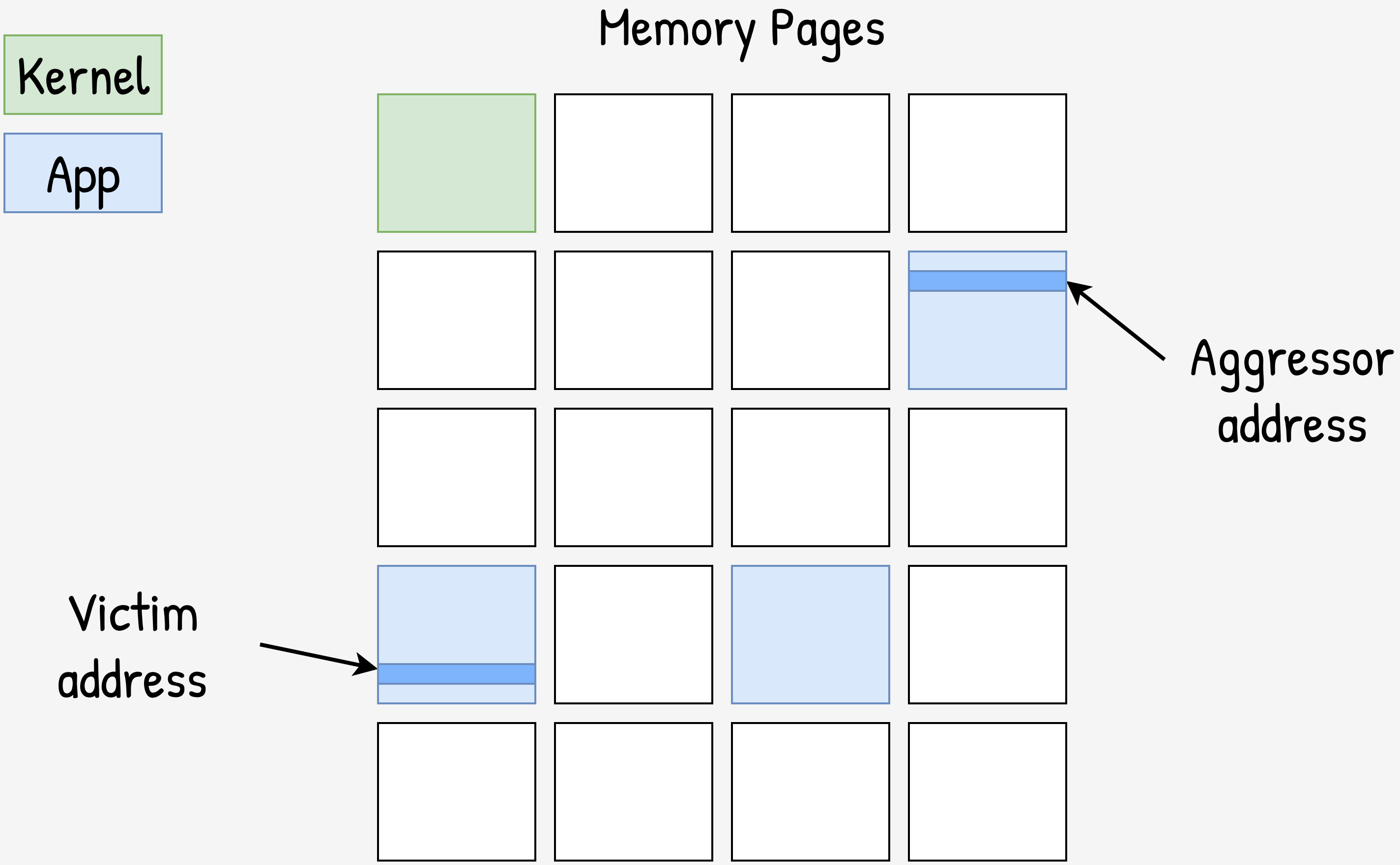

Memory Pages
Kernel
App

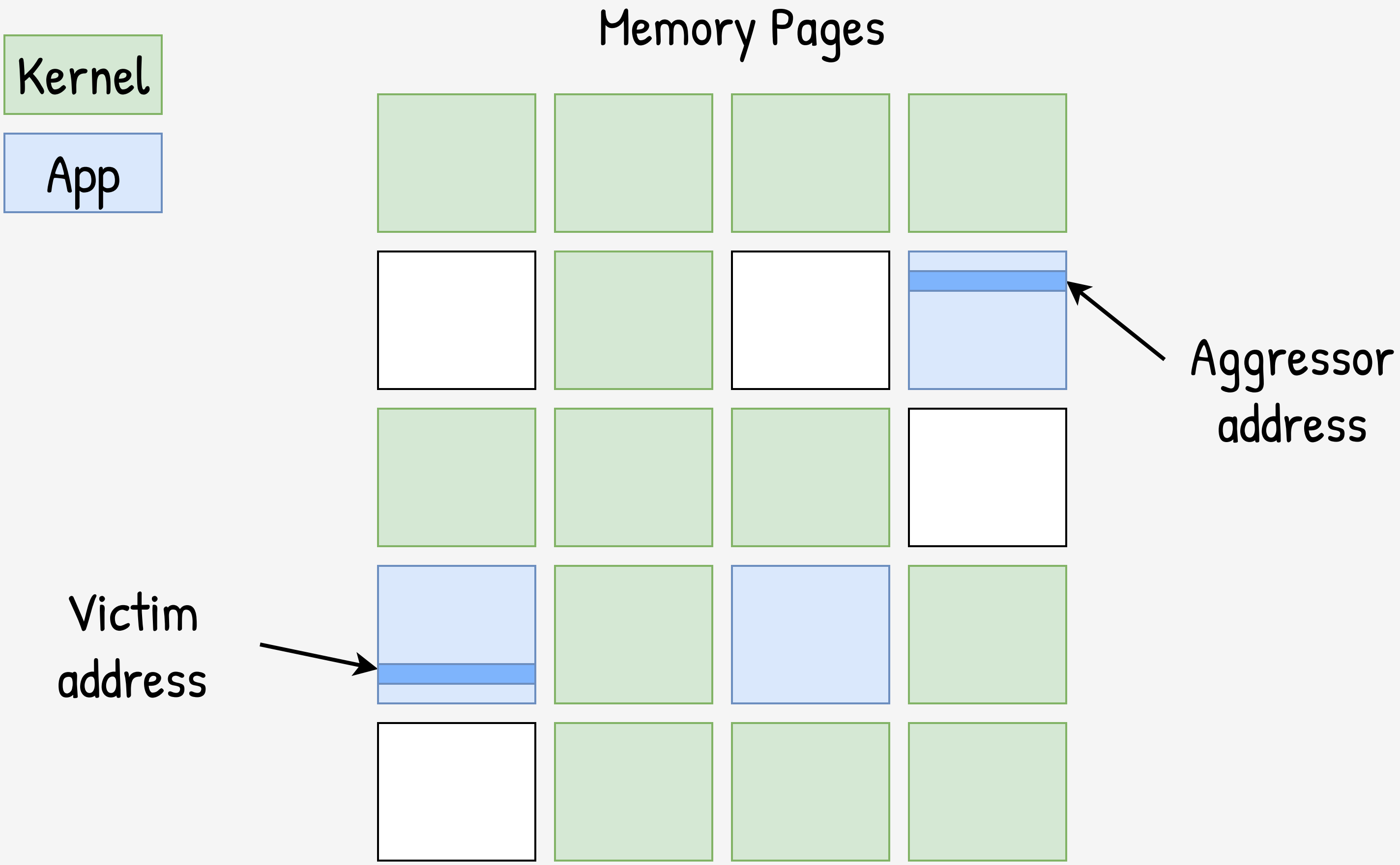

Memory Pages
Kernel
App
Hammer this
Aggressor address

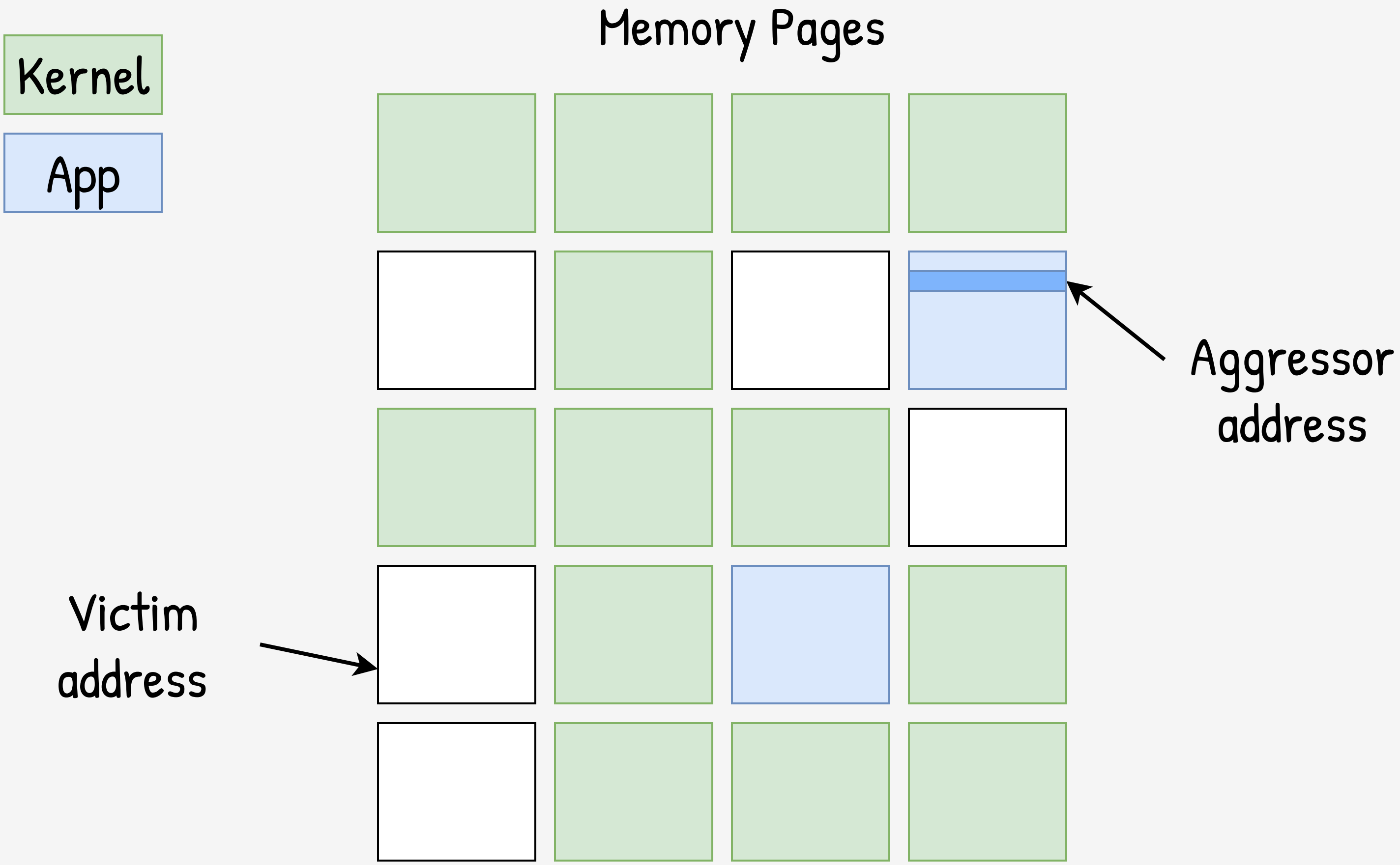

Memory Pages
Kernel
App
Hammer this
Aggressor address
look for unexpected change

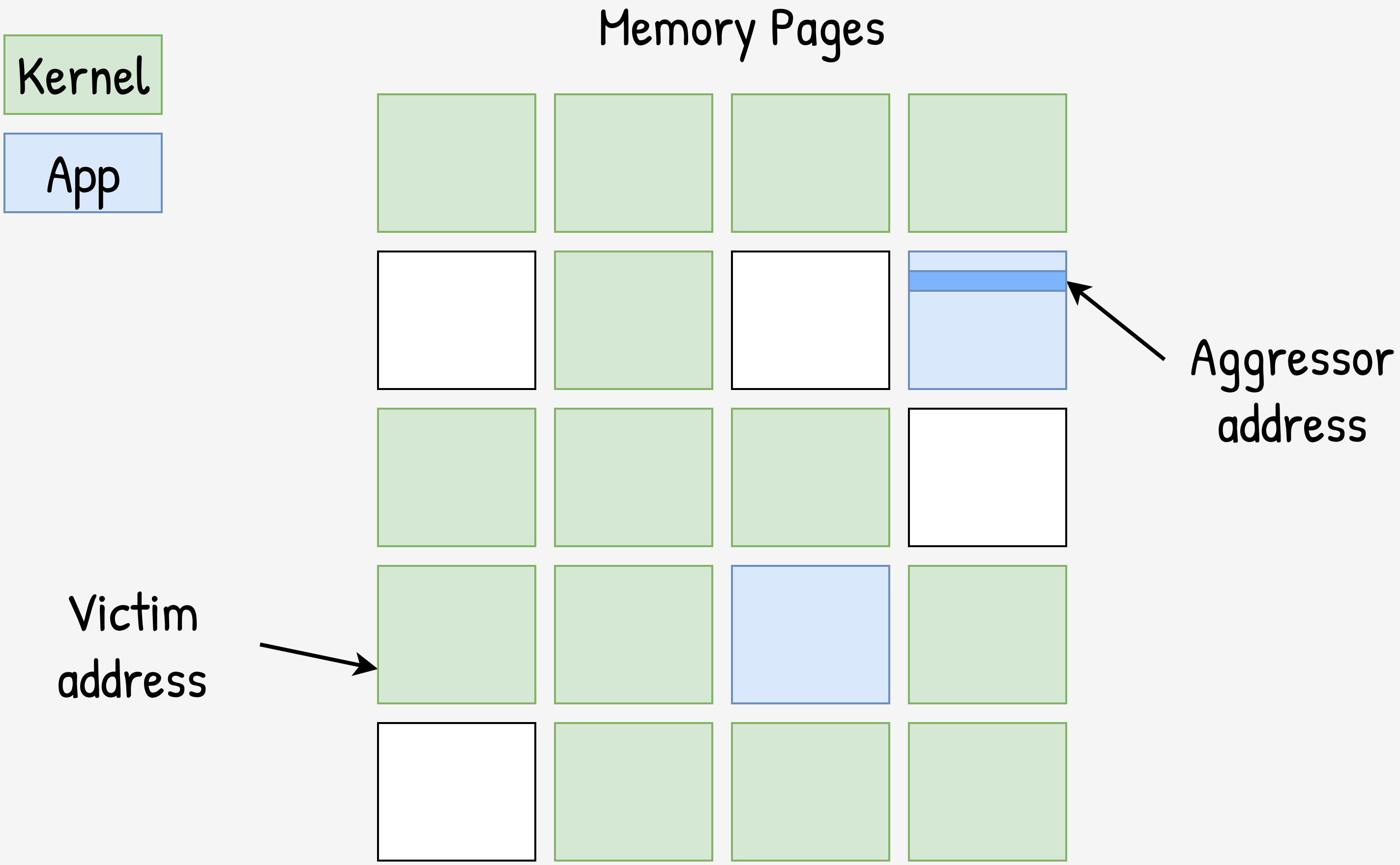


Memory Pages
Kernel
App
Aggressor address
Victim address

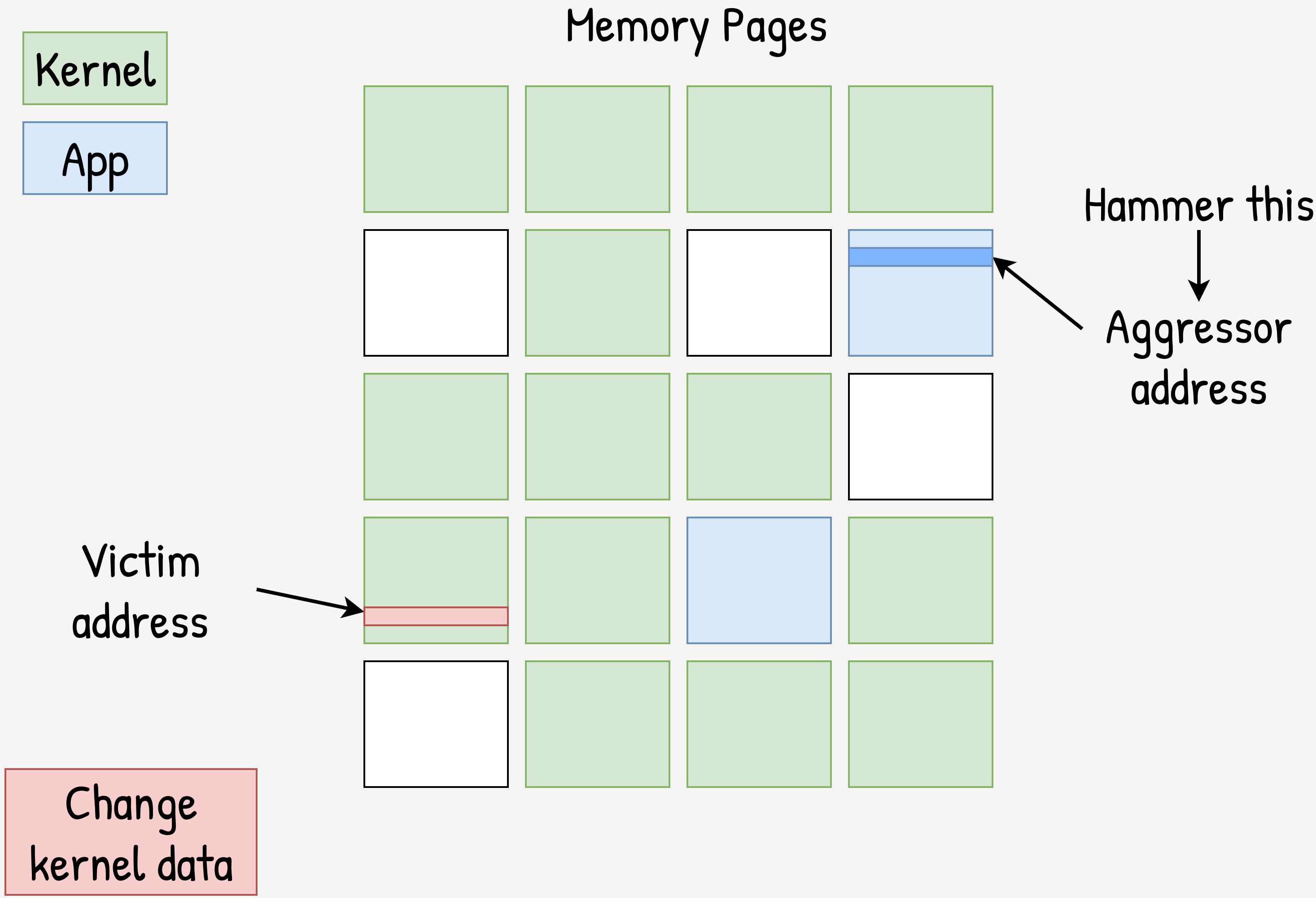

Memory Pages
Kernel
App
Aggressor address
Victim address


Memory Pages
Kernel
App
Aggressor address
Victim address

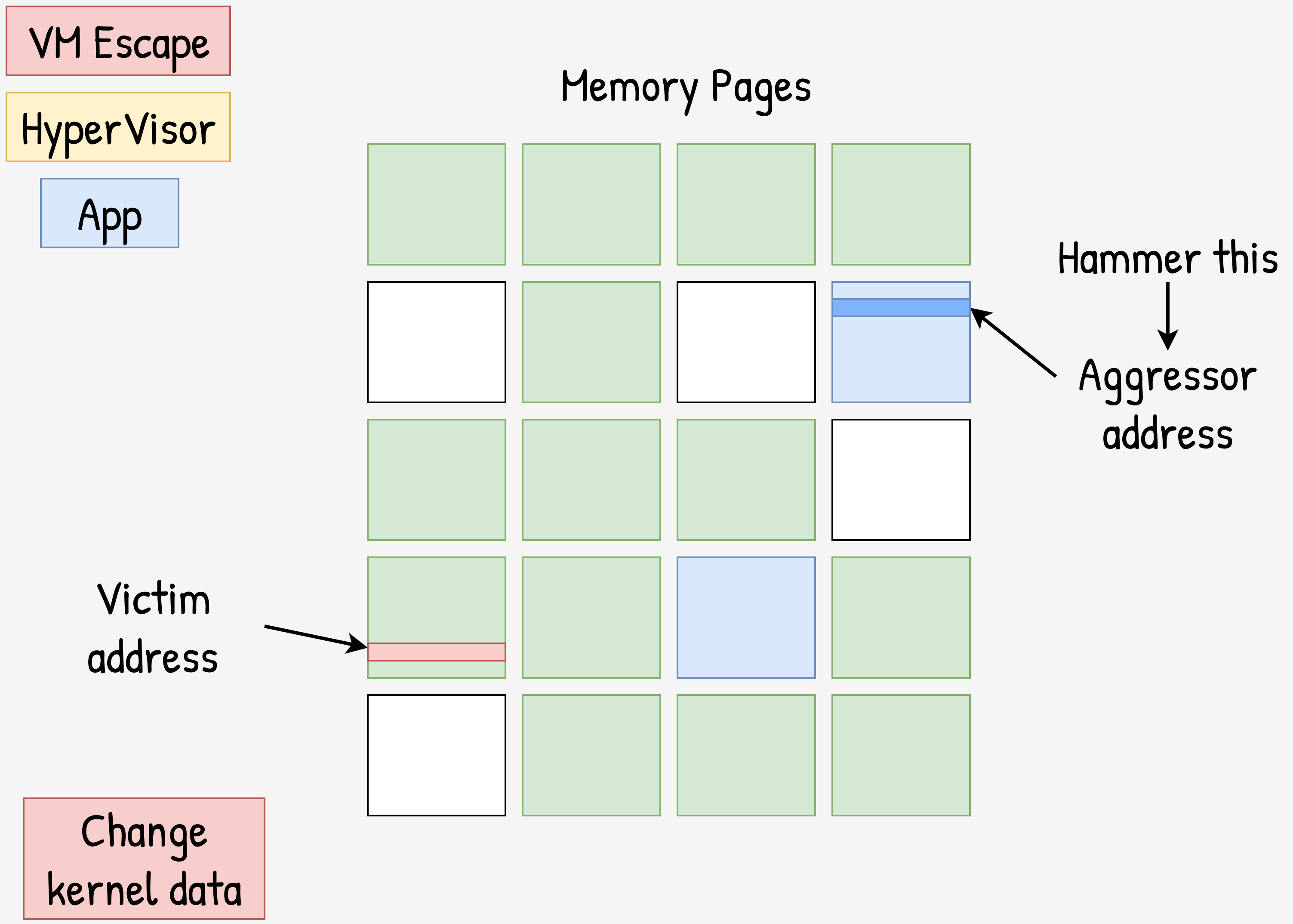

Memory Pages
Kernel
App
Aggressor address
Victim address



Kernel
App
Memory Pages
Aggressor address
Victim address



Memory Pages
Kernel
App
Hammer this
Aggressor address
Victim address
Change kernel data



VM Escape
HyperVisor
App
Memory Pages
Hammer this
Aggressor address
Victim address
Change kernel data

OS Isolation
(Virtual Machines)

Minimize privileged
code (hypervisor)

Near-native
performance / UX

Goals

Takeaways

OS Isolation (Virtual Machines)

Minimize privileged code (hypervisor)

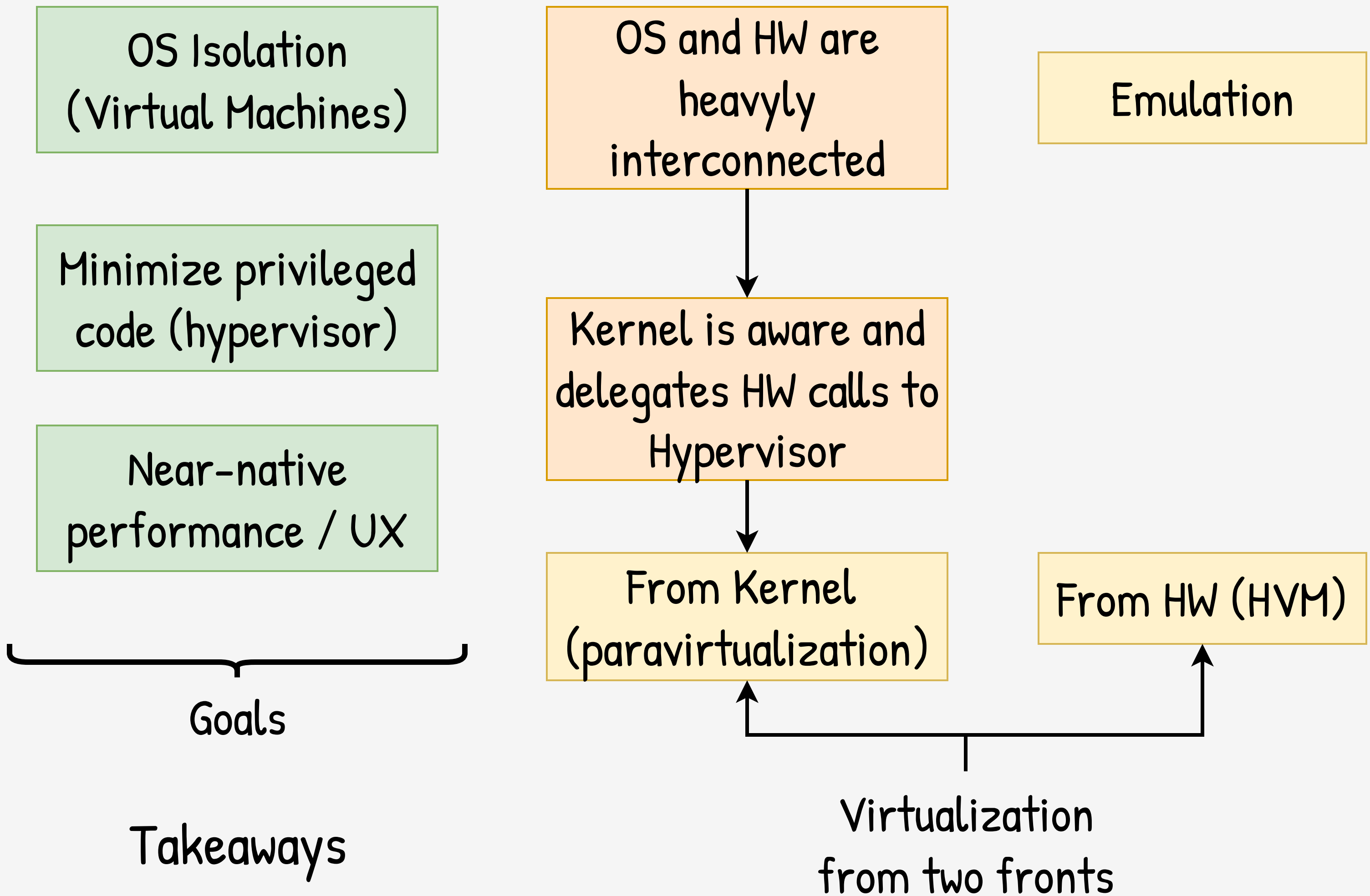
Near-native performance / UX

Goals

Takeaways

OS and HW are heavyly interconnected

Kernel is aware and delegates HW calls to Hypervisor


OS Isolation (Virtual Machines)
Minimize privileged code (hypervisor)
Near-native performance / UX
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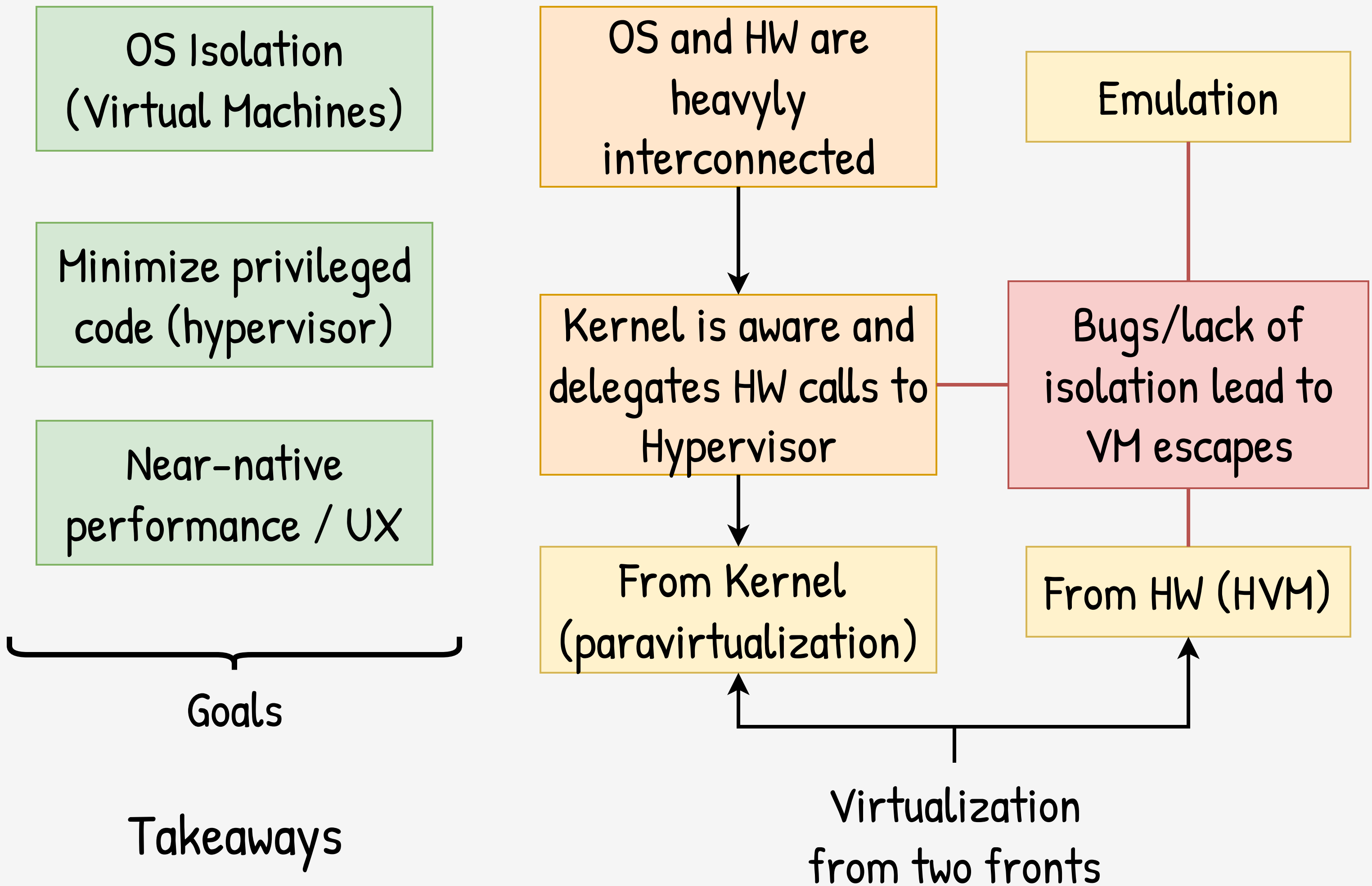
OS and HW are heavyly interconnected
Kernel is aware and delegates HW calls to Hypervisor
From Kernel (paravirtualization)
Emulation
From HW (HVM)
Virtualization from two fronts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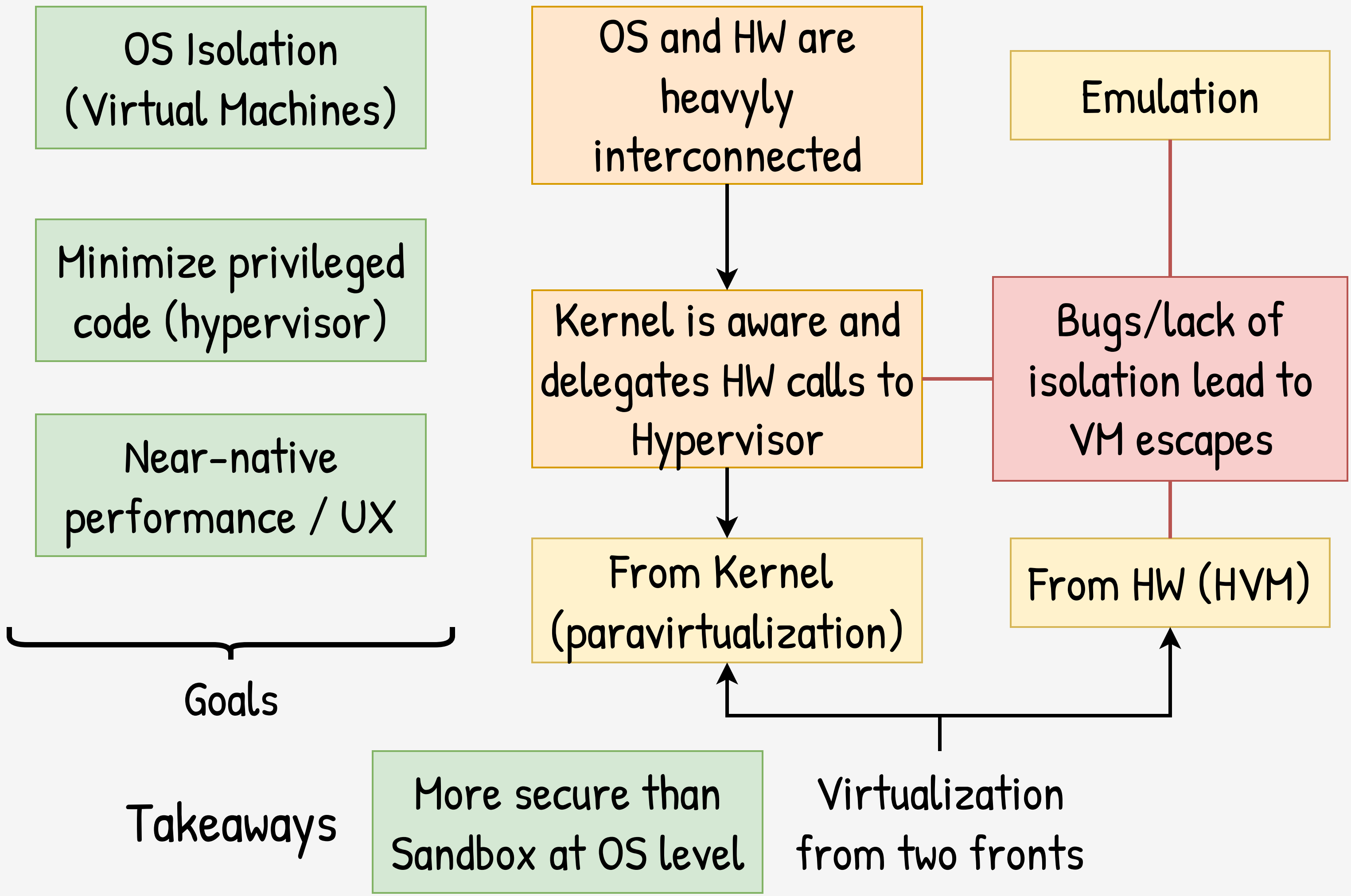
Virtualization from two fronts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More secure than Sandbox at OS level
Virtualization from two fronts